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CEN 302-Software Engineering

Easy Ticket APP

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1) Executive Summary

a) Project Overview

The Easy Ticket Albania (ETA) system is a comprehensive web-based solution designed to modernize the way transportation companies handle ticket reservations, passenger check-ins, trip scheduling, and overall transportation management operations. Currently, many transportation companies still rely on manual booking procedures, physical tickets, and traditional passenger verification methods. These processes are often time-consuming, inefficient, and prone to human errors, while also lacking the accessibility and flexibility required by modern transportation services and passengers.

Easy Ticket Albania aims to address these challenges by providing a centralized digital platform that streamlines and automates various transportation management processes, including ticket reservations, trip scheduling, passenger management, QR-based ticket validation, and administrative operations. By leveraging modern web technologies, the system offers a user-friendly interface accessible from any device with an internet connection, enabling seamless interaction between passengers, conductors, and administrators.

The platform improves operational efficiency, reduces manual workload, minimizes booking and verification errors, and enhances the overall passenger experience. Through digital ticketing and real-time transportation management, the system ensures a faster, more reliable, and hassle-free transportation service.

b) Purpose and Scope of this Specification

The purpose of this project is to develop a web application that facilitates the operations of transportation companies by digitalizing and automating ticket reservation and transportation management processes. This application aims to replace traditional paper-based ticketing and manual passenger management systems, allowing for easier access, improved management, and enhanced security of transportation data.

The system enables passengers to search available trips, reserve transportation tickets online, and receive digital QR-based tickets for validation during travel. Additionally, the application simplifies operational tasks for conductors and administrators by providing dedicated dashboards for ticket verification, route management, schedule administration, and passenger monitoring.

This documentation provides detailed information about the functionalities, features, workflows, software requirements, and technical implementation of the Easy Ticket Albania platform. The document is intended for all system users, including passengers, busmans, administrators, developers, and evaluators.

2) Product/Service Description

a) Product Context

Our software is designed for transportation companies and bus operators nationwide, serving as a comprehensive and standalone system accessible to administrators, busmans, and passengers. The platform streamlines transportation operations, enhances passenger experience, and optimizes ticket reservation,passenger verification,and schedule management processes.

The system ensures seamless coordination between different operational roles, including administrators responsible for managing routes and schedules, conductors responsible for passenger check-ins and ticket validation, and passengers who reserve and manage transportation tickets through the platform. By centralizing transportation management into a single digital solution, the application improves operational efficiency, reduces manual workload, and facilitates reliable and secure transportation services.

The platform also supports QR-based digital ticket verification and real-time reservation management, enabling transportation staff to manage passenger operations more effectively while providing passengers with a faster and more convenient booking experience.

b) User Characteristics

- 1. <u>Admin</u>:
- Generate detailed reports regarding ticket reservations, passenger activity, transportation schedules, and system usage.
- Manage and update user information, including creating, modifying, or removing conductor accounts and administrative records.
- Maintain and update transportation routes, schedules, stations, and ticket pricing, ensuring accurate and up-to-date information for passengers.
- Monitor overall transportation operations and oversee reservation management through the administrative dashboard.
- Track passenger reservations, ticket validations, and transportation activity to ensure smooth system operations

2. <u>Busman</u>:

- View assigned transportation schedules and passenger lists.
- Validate passenger tickets using QR-based ticket verification.
- Perform passenger check-in operations during boarding.
- Monitor ticket validation status and identify invalid or duplicated tickets.
- Assist passengers during boarding and transportation verification processes.

3. Passenger:

- Search available transportation routes, schedules, and ticket prices.
- Reserve transportation tickets online.
- Receive and access digital QR-based tickets.
- View reservation details and travel history.
- Manage personal reservations and ticket information.
- Access the platform through desktop or mobile devices for easier reservation management.

c) Assumptions

• Administrators and conductors will receive proper training to use the application efficiently.

• Transportation companies will have the necessary IT infrastructure, including computers, mobile devices, QR scanners, and internet connectivity, to support the implementation and smooth operation of the system.

• Proper verification of passenger information and reservation data will be conducted to maintain data integrity and prevent invalid ticket registrations.

• Passengers will use the application to search transportation schedules, reserve tickets, and access digital QR-based tickets without requiring external assistance.

• Busmans are responsible for ensuring that passenger check-ins and ticket validations are processed correctly during transportation operations.

• Administrators are responsible for maintaining accurate transportation schedules, routes, stations, and ticket pricing within the system.

• The application server and database services will remain operational to support real-time reservation and ticket validation functionalities.

d) Constraints

- The Easy Ticket Albania system requires a stable internet connection to function properly, as it is a web-based application.
- The system's efficiency depends on the proper training and adoption of the platform by administrators and busmans.
- The performance and reliability of the application are dependent on the transportation company's IT infrastructure, including server availability, network stability, and device accessibility.

- The accuracy and completeness of passenger, schedule, and route data directly impact the effectiveness of reservation, ticket validation, and search functionalities.
- QR-based ticket validation and passenger check-in functionalities rely on proper integration between the frontend, backend services, and QR generation mechanisms.
- The application’s functionality and user experience may be constrained by the limitations of the web technologies, frameworks, and third-party libraries used during development.
- The effectiveness of reporting and dashboard analytics depends on accurate reservation records, passenger data, and ticket validation operations entered into the system.
- Real-time ticket verification and reservation synchronization may be affected by internet interruptions or server downtime.

e) Dependencies

- • The creation of new ticket reservations depends on the availability of transportation schedules, routes, and active passenger participation within the platform (US_01, US_02).
- • Passenger check-in and QR ticket validation processes rely on proper ticket generation and accurate passenger data handling by the system and busmans (US_07, US_08).
- • Ticket validation functionalities depend on the QR generation and verification mechanisms implemented within the application.
- • Route and schedule management functionalities depend on administrators maintaining accurate and updated transportation information in the system (US_09, US_10).
- • The effectiveness of dashboard analytics and reporting depends on accurate reservation records, ticket validations, and passenger activity stored in the database.
- • Passenger access to digital tickets depends on successful reservation processing and authentication mechanisms.
- • Real-time reservation updates and synchronization depend on stable server availability and database connectivity.
- • Authentication and authorization functionalities depend on secure user credential handling and role-based access control mechanisms.

3) Requirements

a) Functional Requirements

Req#	**Requirement**	**Comments**	**Priority**	**Date Rvwd**	**SME Reviewed / Approved**
Req#	Requirement	Comment	Priority	Date	Reviewed/ Approved
FR_01	The application should provide separate dashboards and functionalities for different user roles.	Each role (Administrator, Busman, Passenger) will have access only to authorized functionalities.	1	05/05/20 26	Bora Tollkuci/Vikena Xhemali

Req#	**Requirement**	**Comments**	**Priority**	**Date Rvwd**	**SME Reviewed / Approved**
FR_02	The system shall implement secure user authentication and authorization mechanisms.	Authentication will use JWT-based authorization and encrypted password storage	1	06/05/20 26	Bora Tollkuci/Vikena Xhemali
FR_03	Users shall be able to register and log into the system securely.	The system validates user credentials before granting access.	1	07/05/20 26	Steisi Kurti/Paula Dollani
FR_04	Passengers shall be able to search available transportation routes and schedules.	The search process includes departure, destination, and travel date filtering.	1	08/05/20 26	Bora Tollkuci/Vi kena Xhemali
FR_05	Passengers shall be able to reserve transportation tickets online.	The reservation process stores passenger and trip information in the system database.	1	09/05/20 26	Bora Tollkuci/Vikena Xhemali
FR_06	The system shall generate digital QR-based tickets after successful reservations.	QR codes will be used for passenger verification and check-in operations.	1	10/05/20 26	Bora Tollkuci/Vikena Xhemali

FR_07	Passengers shall be able to view and manage their reservations.	Users can access reservation details and ticket information through their dashboard.	1	11/05/20 26	Bora Tollkuci/Vi kena Xhemali
FR_08	Busmans shall be able to validate passenger tickets using QR verification.	Ticket validation ensures ticket authenticity and prevents duplication.	1	12/05/20 26	Bora Tollkuci/Vikena Xhemali
FR_09	Busmans shall be able to perform passenger check-in operations.	Passenger boarding status will be updated after successful validation.	1	13/05/20 26	Bora Tollkuci/Vi kena Xhemali
FR_10	Administrators shall be able to create, update, and manage transportation routes.	Route management includes departure and destination information.	1	14/05/20 26	Ornela Senja/Trev is Koxha/ Rebeka Cekini
FR_11	Administrators shall be able to create and manage transportation schedules.	Schedules include departure time, arrival time, and assigned routes.	1	15/05/20 26	Ornela Senja/Trev is Koxha/ Rebeka Cekini
FR_12	Administrators shall be able to monitor transportation operations and reservation activities.	The system stores and updates station-related information.	2	16/05/20 26	Ornela Senja/Trev is Koxha/ Rebeka Cekini
FR_13	Administrators shall be able to monitor ticket reservations and transportation operations.	Dashboard analytics will provide operational insights and reservation statistics.	2	17/05/20 26	Ornela Senja/Trev is Koxha/ Rebeka Cekini
FR_14	The application shall implement QR-based ticket validation functionality.	QR validation must support real-time ticket verification.	1	18/05/20 26	Bora Tollkuci/Vi kena Xhemali
FR_15	The system shall validate all user inputs before processing data.	Input validation protects the system from invalid or malicious data.	1	19/05/20 26	Bora Tollkuci/Vi kena Xhemali

FR_16	The application shall be responsive and accessible from multiple devices.	Users can access the platform from desktop, tablet, and mobile devices.	1	20/05/2026	Steisi Kurti/ Paula Dollani
FR_17	The system shall maintain reservation and ticket history for authenticated users.	Reservation records are stored for tracking and monitoring purposes.	2	21/05/2026	Ornela Senja/Trevis Koxha/ Rebeka Cekini
FR_18	The system shall allow administrators to manage user accounts and permissions.	Role-based access control ensures secure system usage.	1	22/05/2026	Steisi Kurti/ Paula Dollani
FR_19	The application shall automatically update reservation and ticket information after booking operations.	Reservation and ticket data should update dynamically across the platform.	2	23/05/2026	Steisi Kurti/ Paula Dollani
FR_20	The system shall generate operational reports and dashboard statistics.	Reports help administrators monitor transportation operations and ticket activity.	2	24/05/2026	Steisi Kurti/ Paula Dollani

b) Non-Functional Requirements

i) Product Requirements

(1) User Interface Requirements

The user interface for the Easy Ticket Albania platform should be compatible with modern web browsers in order for users to access the application from desktop and mobile devices. The user interface is divided into several main sections:

- ***Home interface***

- A navigation bar containing the application name and options such as Home, Routes, Schedules Login, and Register.

- The main body will display transportation information, available routes, schedules, and ticket reservation options.
- A ticket reservation/search section allowing passengers to search for available trips.
- A footer containing contact details, system information, and navigation links.

- ***Login/Register interface:***

- The registration interface will contain fields required for user registration, including personal information and authentication credentials.
- The login interface will contain username/email and password fields.
- All user inputs will be validated and error messages will be displayed for invalid data.

- ***Passenger interface:***

- Passengers will be able to search routes and schedules.
- Passengers can reserve transportation tickets online.
- The dashboard will display reservation history and QR-based digital tickets.
- Users can view ticket details and travel information

- ***Bussman interface:***

- Busmans will access a dashboard containing assigned trips and passenger lists.
- Busmans can validate passenger tickets using QR verification.
- Passenger check-in operations can be performed through the platform.
- The interface will display ticket validation status and passenger boarding information.

- ***Administrator interface:***

- Administrators will access dashboards containing route, schedule, and station management functionalities.
- The system will display reservation statistics and operational reports.
- Administrators can manage users, schedules, and transportation information.
- Dashboard analytics and operational monitoring tools will be available

(2) Learnability

The Easy Ticket Albania system shall be designed to ensure that users can quickly and efficiently learn how to use the platform with minimal training. Key learnability features include:

1. Intuitive and user-friendly interface.
2. Role-based dashboards that display only relevant functionalities.
3. Consistent navigation structure throughout the application.
4. Familiar design patterns for reservation and ticket management processes.
5. Minimal training dependency for passengers and transportation staff

(3) Accessibility

The platform shall ensure accessibility and usability across multiple devices and user environments:

- The application will follow responsive web design principles.
- The system will be accessible through desktop, tablet, and mobile devices.
- The interface will support keyboard navigation and readable layouts.
- Text and interface elements will remain usable across different screen sizes.

(4) Efficiency

The system shall optimize task completion time and operational resource usage:

- Quick Actions: Frequent operations such as ticket reservations, passenger check-ins,and
- QR Validation will be streamlined for faster execution.
- Performance: Pages and reservation operations should load within a few seconds under normal network
- conditions.
- Search&Filters:Users will be able to quickly locate schedules ,routes,rezervations and passenger
- Information using dynamic search and filtering functionalities.
- Automation: Repetitive operations such as QR ticket generation, reservation updates and dashboard synchronization will be automated whenever possible.

(5) Dependability/Memorability

The Easy Ticket Albania system shall ensure that users can easily recall how to perform operations after periods of interactivity**:**

- Preventiv Consistent Layout: Core functionalities such as reservations, , schedules, dashboards, and QR validation will maintain consistent placement throughout the application.
- Visual Cues: Icons and labels will remain standardized across the platform to improve recognition and usability.
- Saved Preferences: User session information and dashboard settings will persist between authenticated sessions whenever applicable.
- Task History: Recently accessed reservations and transportation information will be easily accessible for users.

- Error Recovery: Clear instructions and validation messages will guide users during invalid operations or incorrect input formats.

(6) Errors

The system shall minimize user errors and provide clear recovery paths:

- Preventive Design: Input validation will prevent invalid reservation, authentication, and schedule data submissions.
- Descriptive Messages: Error messages will explain issues clearly and provide corrective suggestions.
- Confirmation Actions: Critical operations such as reservation cancellation or schedule modification will include confirmation dialogs.
- Logging: System and user-facing errors will be logged for troubleshooting and maintenance purposes without exposing sensitive data.
- Graceful Degradation: If a functionality such as reservation synchronization or QR validation fails, the system will preserve user progress and provide alternative recovery steps.

3.2.1.7 Satisfaction

The Easy Ticket Albania platform is designed to provide a simple, efficient, and user-friendly transportation management experience for passengers, busmans, and administrators.The application simplifies transportation reservation and ticket verification processes while reducing manual operations and operational delays. The responsive interface and digital ticketing functionalities improve accessibility and usability across multiple devices. Overall, the system aims to improve transportation management efficiency and provide passengers with a reliable and modern ticket reservation experience.

3.2.1.8 Capacity

The Easy Ticket Albania system is designed to support simultaneous access by multiple passengers, busmans, and administrators. The application operates in real time, ensuring that reservation updates, ticket validations, and transportation information remain synchronized across the platform. All users interact with the same centralized database.

The system is hosted on a web server architecture and is designed to maintain efficient performance while handling reservation operations, QR validation requests, and transportation management processes.

(7) Security

The data stored within the Easy Ticket Albania system is considered sensitive and must be protected from unauthorized access. Security measures implemented within the system include:

- Password encryption using hashing algorithms.
- JWT-based authentication and role-based authorization
- Restricted access to functionalities based on user roles.
- Validation of all user inputs.
- Secure handling of reservation and passenger information.
- Protection against unauthorized ticket validation and duplicated ticket usage.

ii) Organizational Requirements

Requirements which are a consequence of organisational policies and procedures e.g. process stand ards used, implementation requirements, etc

(1) Availability

The system should remain available continuously to support transportation reservation and ticket validation operations.

(2) Latency

Critical operations such as reservations, authentication, and QR validation should complete within a few seconds under normal network conditions.

(3) Maintenance

Regular system updates and maintenance procedures should be performed to improve security, stability, and application performance.

(4) Operations

The system is designed to provide an intuitive workflow for passengers and transportation staff,ensuring ease of use and operational efficiency.

(5) Standarts Compilance

The application shall follow modern web application development standards and secure authentication practices.

(6) Portabiliy

The platform is web-based and accessible from modern operating systems and browsers without requiring additional software installation.

iii) External Requirements

– Requirements which arise from factors which are external to the system and its development process e.g. interoperability requirements, legislative requirements, etc.

(1) Protection

To protect the Easy Ticket Albania system from unauthorized access, malicious modifications, disclosure, destruction, or misuse, the following measures will be implemented:

Encrypting sensitive user data, such as passwords, using hashing algorithms to maintain privacy and security.Logging and auditing user activities to ensure accountability and assist in identifying unauthorized or suspicious actions.

Conductors and administrators are responsible for ensuring the accuracy of passenger and transportation information entered into the system.

Validating all user input data to prevent invalid data submissions and potential cyberattacks.

Requiring user confirmation through dialogs or pop-up windows for critical operations such as reservation cancellation or schedule modifications.

Each passenger will only be able to access their personal reservation and ticket information.

Administrators and conductors will have limited access according to their assigned system roles to ensure passenger data privacy and operational security.

(2) Authorization and Authentication

The Authorization and Authentication factors:

- User authentication will be implemented using username/email, password, and JWT-based authentication mechanisms to enhance security.
- Authorization will be role-based, ensuring that each user can access only the functionalities and information assigned to their role.
- Sessions and authentication tokens will be used to manage authenticated users within the system.
- If the user tries to log in with the wrong credentials, an error message will be displayed.
- If a user attempts to log in using invalid credentials, the system will display an authentication error message.

(3) Legislative Requirements

The system shall follow general software security and data protection principles regarding passenger information and reservation management. The application shall ensure that sensitive passenger and transportation information remains protected and accessible only to authorized users.

Operational logs and reservation records may be maintained for monitoring, troubleshooting, and auditing purposes. The system shall maintain activity tracking for important operations such as reservation creation, ticket validation, and user authentication in order to improve accountability and operational monitoring.

The platform shall support secure access control and operational tracking functionalities to facilitate reliable transportation and reservation management services.

c) Domain Requirements

This Web Application operates in the field of Transportation Management and Online Ticket Reservation.
The main purpose of the platform is to digitalize and automate transportation reservation processes, passenger ticket management, and transportation operations.
However, data security and access control are of utmost importance. Sensitive information such as passenger details, reservation records, QR-based tickets, and user authentication data must be protected and accessible only to authorized users. The application ensures role-based access control, allowing each user to interact only with the functionalities relevant to their role.

This system is intended for transportation companies and bus operators and does not require direct integration with external transportation systems. The web application will be responsive and accessible across differen t devices while implementing modern authentication, authorization, and QR-based validation mechanisms to ensure secure and reliable transportation management operations.

4) User Scenarios/Use Cases

4.1 Requirements Analysis

4.1.1 User Scenarios List

Nr	Name	Description
US_01	User Registration	Guest creates a new account in the system by providing personal information such as name, email, and password
US_02	User Login	Passenger, Busman, and Admin log into the system
US_03	Manage profile	Passenger/Admin/Busman: The user updates personal information such as name, password, profile image, or contact details.
US_04	Search routes	Guest/Passenger: The user searches for available transportation routes by selecting departure and destination locations.
US_05	View schedules	Guest/Passenger: The user views available transportation schedules including departure times and seat availability.
US_06	Purchase ticket	Passenger: The user selects a transportation schedule and reserves a ticket.
US_07	Select seat	Passenger: The user selects a preferred seat from available options during reservation.
US_08	Confirm reservation	Passenger: The user confirms the reservation using the cash payment option.
US_09	Generate QR ticket	System: The system generates a QR-based digital ticket after successful reservation
US_10	View bookings	Passenger: The user views current and previous reservations/tickets.
US_11	Cancel Reservation	Passenger cancels existing reservation
US_12	Manage routes	Admin: The administrator creates, updates, or removes transportation routes.
US_13	Manage schedules	Admin: The administrator creates and manages

		transportation schedules and departure timings.
US_14	Manage busman	Admin: The administrator creates, updates, or removes busman
US_15	View dashboard statistics	Admin: The administrator views reservation statistics and transportation activity.
US_16	Scan ticket	Busman: The user scans and validates the passenger QR ticket.
US_17	Mark ticket as used	Busman: The user marks the ticket as used after successful passenger verification.
US_18	Authenticate user roles	System: The system validates user roles and grants role-based access permissions.
US_19	Logout user	Passenger/Admin/Busman: The authenticated user securely logs out from the platform.

4.1.2 User Scenarios Extended

US_01: User Registration

1. Guest navigates to the registration page.
2. Guest enters personal information.
3. Guest enters email and password.
4. System validates user input.
5. System checks if the account already exists.
6. System creates a new user account.
7. System stores user information securely.
8. System confirms successful registration.

US_02: User Login

1. User navigates to the login page.
2. User enters email/username and password.
3. User submits login request.
4. System validates credentials.
5. System authenticates the user.
6. System identifies assigned role.
7. System redirects the user to the appropriate dashboard.

US_03: Manage Profile

1. User logs into the system.
2. User navigates to the profile section.
3. System displays current profile information.
4. User updates personal information.

5. System validates updated data.
6. System stores updated information.
7. System confirms successful profile update.

US_04: Search Routes

1. Guest or Passenger accesses the search section.
2. User selects departure location.
3. User selects destination location.
4. User selects travel date.
5. User submits search request.
6. System retrieves matching transportation routes.
7. System displays available routes.

US_05: View Schedules

1. User selects transportation route.
2. System retrieves available schedules.
3. System displays departure times and seat availability.
4. User reviews available schedules.

US_06: Purchase Ticket

1. Passenger selects transportation schedule.
2. System displays reservation information.
3. Passenger proceeds with reservation.
4. System stores reservation details.
5. System confirms reservation process.

US_07: Select Seat

1. Passenger views available seats.
2. System displays seat layout.
3. Passenger selects preferred seat.
4. System reserves selected seat.
5. System updates seat availability.

US_08: Confirm Reservation

1. Passenger reviews reservation details.
2. Passenger confirms reservation using cash payment option.
3. System validates reservation information.
4. System stores reservation data.
5. System confirms successful reservation.

US_09: Generate QR Ticket

1. Reservation process completes successfully.
2. System generates QR-based digital ticket.
3. System links QR ticket with reservation.
4. System stores QR ticket information.
5. Passenger receives QR ticket confirmation.

US_10: View Bookings

1. Passenger logs into dashboard.
2. Passenger navigates to booking section.
3. System retrieves reservation history.
4. System displays current and previous bookings.

US_11: Cancel Reservation

1. Passenger selects reservation to cancel.
2. System displays reservation information.
3. Passenger confirms cancellation request.
4. System updates reservation status.
5. System confirms successful cancellation.

US_12: Manage Routes

1. Admin logs into dashboard.
2. Admin accesses route management section.
3. Admin creates, updates, or removes transportation routes.
4. System validates route information.
5. System stores updated route data.

US_13: Manage Schedules

1. Admin accesses schedule management section.
2. Admin creates or updates transportation schedules.
3. Admin selects departure time and assigned route.
4. System validates schedule information.
5. System stores updated schedule data.

US_14: Manage Busman

1. Admin accesses busman management section.
2. Admin creates, updates, or removes busman accounts.

3. System validates user information.
4. System stores updated account data.
5. System confirms successful operation.

US_15: View Dashboard Statistics

1. Admin accesses dashboard analytics section.
2. System retrieves reservation statistics and operational data.
3. System displays charts and transportation activity information.
4. Admin reviews dashboard analytics.

US_16: Scan Ticket

1. Busman opens QR scanner section.
2. Passenger presents QR ticket.
3. Busman scans QR code.
4. System retrieves ticket information.
5. System validates QR ticket.
6. System displays validation result.

US_17: Mark Ticket as Used

1. Busman validates passenger ticket.
2. System confirms successful verification.
3. Busman marks ticket as used.
4. System updates ticket status.
5. System stores updated validation information.

US_18: Authenticate User Roles

1. User attempts to access the system.
2. System validates authentication credentials.
3. System identifies assigned user role.
4. System grants role-based access permissions.
5. System restricts unauthorized functionalities.

US_19: Logout User

1. User clicks logout button.
2. System terminates user session.
3. System removes authentication token.
4. System redirects user to login or home page.

4.1.3 User Cases

UC_01: User Register

UC Name	User register
Dependency	None
Actors	Passenger
Preconditions	1. Passenger is not authenticated. 2. Registration page is accessible.
Description of Main Sequence	1. Passenger accesses the registration page. 2. Passenger enters personal information such as full name, email, and password. 3. System validates entered information. 4. System checks if the email already exists. 5. System encrypts passenger password. 6. System creates a new passenger account. 7. System stores passenger information in the database. 8. System confirms successful registration.
Description of Alternative Sequence	1. Invalid input data: a. System displays validation errors. b. Passenger corrects invalid fields. 2. Existing email: a. System displays duplicate account error message.
Non-functional requirements	1. Passwords must be encrypted. 2. Registration should complete within a few seconds. 3. User information must be stored securely.
Postconditions	1. Passenger account is created successfully. 2. Passenger can access login functionality.

UC_02: Log In

UC Name	User logs in
Dependency	UC_01 (User Registers)
Actors	Passenger, Admin, Busman
Preconditions	1. User account exists in the database. 2. Login page is accessible.

Description of Main Sequence	1. User accesses the login page. 2. User enters email and password. 3. User submits login request. 4. System validates credentials. 5. System authenticates user account. 6. System identifies user role. 7. System generates authentication token/session. 8. System redirects user to appropriate dashboard.

Description of Alternative Sequence	1. Invalid credentials: a. System displays authentication error message. 2. Missing credentials: a. System requests required login information.
Non-functional requirements	1. Authentication must be secure. 2. JWT authorization must be implemented. 3. Login process should complete quickly.
Postconditions	1. User becomes authenticated. 2. Role-based access is granted.

UC_03: Manage profile

UC Name	Manage profile
Dependency	UC_02(User Login)
Actors	Passenger, Admin, Busman
Preconditions	1.User is authenticated. 2.User profile information exists.
Description of Main Sequence	1. User accesses profile management section. 2. System displays current profile information. 3. User updates personal information. 4. User submits updated profile information. 5. System validates entered information. 6. System stores updated profile data. 7. System confirms successful profile update
Description of Alternative Sequence	1. Invalid information: a. System displays validation errors. 2. Upload failure: a. System displays profile image upload error.

Non-functional requirements	1. Profile information must be stored securely. 2. System should validate uploaded files. 3. Update process should complete efficiently.
Postconditions	1. User profile information is updated successfully. 2. Updated information becomes visible in the system.

UC_04: Search routes

UC Name	Search routes
Dependency	None
Actors	Passenger
Preconditions	1. Transportation routes exist in the system. 2. Search functionality is available.
Description of Main Sequence	1. Passenger accesses route search section. 2. Passenger selects departure location. 3. Passenger selects destination location. 4. Passenger selects travel date. 5. Passenger submits search request. 6. System retrieves matching transportation routes. 7. System displays available transportation routes.
Description of Alternative Sequence	1. No matching routes: a. System displays no routes found message. 2. Missing search criteria: a. System does not performs search action.
Non-functional requirements	1. Route search should return results quickly. 2. Search results must be accurate. 3. System should support multiple simultaneous searches.
Postconditions	1. Matching routes are displayed successfully. 2. User can continue to schedule selection.

UC_05: View schedules

UC Name	View schedules
Dependency	UC_04(Search Routes)

Actors	Passenger
Preconditions	1. Transportation schedules exist for selected route. 2. Passenger has selected a route.
Description of Main Sequence	1. Passenger selects transportation route. 2. System retrieves transportation schedules. 3. System retrieves departure and arrival times. 4. System retrieves available seat information. 5. System displays transportation schedules and seat availability. 6. Passenger reviews schedule information.
Description of Alternative Sequence	1. No schedules available: a. System displays unavailable schedule message.
Non-functional requirements	1. Schedule information must be accurate. 2. Seat availability should update in real time. 3. Response time should be minimal.
Postconditions	1. Passenger views transportation schedules successfully. 2. Passenger can continue reservation process.

UC_07: Select Seat

UC Name	Seat Select
Dependency	UC_05 (View schedules)
Actors	Passenger
Preconditions	1. Passenger is authenticated. 2. Transportation schedule is selected. 3. Seats are available.
Description of Main Sequence	1. Passenger accesses seat selection section. 2. System displays available seat layout. 3. Passenger reviews available seats. 4. Passenger selects preferred seat. 5. System validates seat availability. 6. System temporarily reserves selected seat. 7. System updates seat availability information.
Description of Alternative Sequence	1. Seat already reserved: a. System displays unavailable seat message. 2. Invalid seat selection: a. System requests another seat selection.
Non-functional requirements	1. Seat availability must update in real time. 2. System must prevent double booking. 3. Seat selection should be responsive and user-friendly.

Postconditions	1. Passenger seat is selected successfully. 2. Passenger can continue to payment confirmation.

UC_07: Make Payment

UC Name	Make Payment
Dependency	UC_06 (Select Seat)
Actors	Passenger
Preconditions	1. Passenger selected transportation schedule. 2. Available seats exist and selected successfully. 3.Rezervation information is valid.
Description of Main Sequence	1. Passenger reviews reservation details. 2. Passenger selects cash payment option. 3. Passenger confirms payment process. 4. System validates reservation information. 5. System stores payment confirmation data. 6. System updates reservation status. 7. System confirms successful payment confirmation.
Description of Alternative Sequence	1. Reservation validation failure: a. System displays reservation error message. 2. Reservation cancellation: a. System cancels reservation process.
Non-functional requirements	1. Reservation confirmation must be secure. 2. Reservation processing should complete quickly. 3. System must maintain data consistency.
Postconditions	1. Reservation payment is confirmed successfully. 2. Passenger can have generated transportation ticket.

UC_08: Generate Ticket

UC Name	Generate Ticket
Dependency	US_07 (Make Paayment)
Actors	Passenger

Preconditions	1. Reservation payment is confirmed successfully. 2. Reservation information exists in the system.
Description of Main Sequence	1. Passenger requests ticket generation. 2. System retrieves reservation details. 3. System generates unique QR-based transportation ticket. 4. System links generated ticket with reservation. 5. System stores ticket information in the database. 6. System displays generated transportation ticket.
Description of Alternative Sequence	1. QR generation failure: a. System displays ticket generation error. 2. Database storage failure: a. System retries ticket generation process.
Non-functional requirements	1. QR code must be unique. 2. Ticket generation should complete quickly. 3. Ticket information must remain secure.
Postconditions	1. Transportation ticket is generated successfully. 2. Passenger can download transportation ticket.

UC_09: Purchase Ticket

UC Name	Purchase Ticket
Dependency	UC_06 (Select Seat,) UC_07 (Make Payment), UC_08 (Generate Ticket)
Actors	Passenger
Preconditions	1. Passenger selected transportation schedule. 2. Reservation process is active.
Description of Main Sequence	1. Passenger selects transportation schedule. 2. Passenger selects preferred seat. 3. Passenger confirms cash payment. 4. System validates reservation information. 5. System generates QR transportation ticket. 6. System stores purchase information. 7. System confirms successful transportation ticket purchase.

Description of Alternative Sequence	1. Reservation process failure: a. System displays purchase error message. 2. Seat unavailable during process: a. System requests another seat selection.
Non-functional requirements	1. Ticket purchase must be secure. 2. Reservation data must remain consistent. 3. Purchase process should complete efficiently.
Postconditions	1. Transportation ticket purchase is completed successfully. 2. Passenger receives valid QR transportation ticket.

UC_10: Downloaad Ticket

UC Name	Download Ticket
Dependency	US_09 (Purchase Ticket)
Actors	Pasenger
Preconditions	1. Transportation ticket is generated successfully. 2. Passenger is authenticated.
Description of Main Sequence	1. Passenger accesses generated ticket section. 2. System displays QR transportation ticket. 3. Passenger selects download option. 4. System prepares downloadable ticket file. 5. System downloads transportation ticket successfully.
Description of Alternative Sequence	1. Download failure: a. System displays download error message. 2. Ticket unavailable: a. System requests ticket regeneration.
Non-functional requirements	1. Download process should complete quickly. 2. Downloaded ticket must remain readable. 3. QR ticket information must remain accurate.
Postconditions	1. Passenger downloads transportation ticket successfully. 2. Passenger possesses valid QR transportation ticket.

UC_11: View bookings

UC Name	View bookings
Dependency	US_02 (User login)

Actors	Passenger
Preconditions	1. Passenger is authenticated. 2. Reservation history exists in the system.
Description of Main Sequence	1. Passenger accesses booking section. 2. System retrieves current reservations. 3. System retrieves previous reservations. 4. System retrieves ticket details. 5. System displays booking history and transportation tickets.
Description of Alternative Sequence	1. No bookings found: a. System displays empty booking history message. 2. Retrieval failure: a. System displays temporary system error message.
Non-functional requirements	1. Booking information must load quickly. 2. Reservation history must remain secure. 3. System must display accurate booking information.
Postconditions	1. Passenger views booking history successfully. 2. Passenger can access reservation details.

US_12: Manage Routes

UC Name	Manage Routes
Dependency	US_02 (User logs in)
Actors	Admin
Preconditions	1. Admin is authenticated. 2. Route management permissions exist.
Description of Main Sequence	1. Admin accesses route management section. 2. System displays existing transportation routes. 3. Admin creates, updates, or removes transportation routes. 4. System validates route information. 5. System stores updated route data. 6. System confirms successful route management operation.
Description of Alternative Sequence	1. Invalid route information: a. System displays validation errors. 2. Database update failure: a. System displays route management error message.

Non-functional requirements	1. Route information must remain accurate. 2. Route updates should synchronize quickly. 3. System must maintain route consistency.
Postconditions	1. Transportation routes are updated successfully. 2. Updated routes become visible in the system.

UC_13: Manage Schedules

UC Name	Manage Schedules
Dependency	US_12 (Manage Routes)
Actors	Admin
Preconditions	1. Transportation routes exist. 2. Admin is authenticated.
Description of Main Sequence	1. Admin accesses schedule management section. 2. System displays existing schedules. 3. Admin creates or updates transportation schedules. 4. Admin assigns route and departure time. 5. System validates schedule information. 6. System stores updated schedule data. 7. System confirms successful schedule operation.
Description of Alternative Sequence	1. Invalid schedule information: a. System displays validation errors. 2. Scheduling conflict: a. System displays schedule conflict message.
Non-functional requirements	1.Schedule information must update in real time. 2. Schedule data must remain consistent. 3. System should handle multiple schedule updates efficiently.
Postconditions	1. Transportation schedules are updated successfully. 2. Updated schedules become accessible to passengers.

UC_14: Manage Tickets

UC Name	Manage tickets
Dependency	US_02 (User logs in)
Actors	Admin
Preconditions	1. Ticket information exists in the system. 2. Admin is authenticated.
Description of Main Sequence	1. Admin accesses ticket management section. 2. System retrieves transportation ticket information. 3. Admin reviews ticket status and reservation details. 4. Admin updates or manages ticket information. 5. System stores updated ticket data. 6. System confirms successful ticket management operation.
Description of Alternative Sequence	1. Ticket retrieval failure: a. System displays retrieval error message. 2. Invalid ticket information: a. System displays validation errors.
Non-functional requirements	1. Ticket data must remain secure. 2. Ticket management operations should complete quickly. 3. System must maintain ticket consistency.
Postconditions	1. Transportation ticket information is updated successfully. 2. Updated ticket data becomes available in the system.

UC_15: Manage Busman

UC Name	Manage Busman
Dependency	US_02 (User logs in)
Actors	Admin
Preconditions	1. Busman is authenticated 2. Busman management permissions exist.
Description of Main Sequence	1. Admin accesses busman management section. 2. System displays existing busman accounts. 3. Admin creates, updates, or removes busman accounts. 4. System validates entered busman information. 5. System stores updated busman data. 6. System confirms successful management operation.
Description of Alternative Sequence	1. Invalid busman information: a. System displays validation errors. 2. Duplicate account detected: a. System displays duplicate account error message.

Non-functional requirements	1. Busman information must remain secure. 2. User roles must be validated correctly. 3. Management operations should complete efficiently.
Postconditions	1. Busman information is updated successfully. 2. Updated busman accounts become active in the system.

UC_16: Monitor Route Performance

UC Name	Monitor Route Performance
Dependency	UC_18 Generate Reports
Actors	Admin
Preconditions	1. Transportation activity data exists. 2. Admin is authenticated.

Description of Main Sequence	1. Admin accesses reporting section. 2. Admin selects route performance monitoring option. 3. System retrieves transportation route statistics. 4. System processes route activity information. 5. System generates route performance report. 6. System displays transportation route statistics and analytics.
Description of Alternative Sequence	1. No statistics available: a. System displays unavailable report message. 2. Data retrieval failure: a. System displays temporary reporting error.
Non-functional requirements	1. Reports must display accurate information. 2. Data processing should complete efficiently. 3. Statistical information must remain consistent.
Postconditions	1. Route performance statistics are displayed successfully. 2. Admin can review transportation analytics.

UC_17: Monitor Ticket Sales

UC Name	Monitor Ticket Sales

Dependency	UC_18 (Generate Reports)
Actors	Admin
Preconditions	1. Ticket purchase data exists. 2. Admin is authenticated.
Description of Main Sequence	1. Admin accesses reporting section. 2. Admin selects ticket sales monitoring option. 3. System retrieves transportation ticket sales data. 4. System processes reservation and sales statistics. 5. System generates ticket sales report. 6. System displays transportation ticket sales analytics.
Description of Alternative Sequence	1. No ticket sales data available: a. System displays unavailable report message. 2. Report generation failure: a. System displays reporting error message.

Non-functional requirements	1. Sales reports must remain accurate. 2. Statistical calculations should process efficiently. 3. Sales information must remain secure.
Postconditions	1. Transportation ticket sales analytics are displayed successfully. 2. Admin can review reservation and sales statistics.

UC_18: Generate Reports

UC Name	Generate Reports
Dependency	US_02(User logs in)
Actors	Admin
Preconditions	1. Admin is authenticated. 2. Reservation and transportation data exist in the system.
Description of Main Sequence	1. Admin accesses reporting section. 2. System retrieves transportation and reservation statistics. 3. Admin selects preferred report type. 4. System processes analytical information. 5. System generates requested report. 6. System displays generated report and statistics.

Description of Alternative Sequence	1. Report generation failure: a. System displays reporting error message. 2. Missing statistical data: a. System displays unavailable analytics information.
Non-functional requirements	1. Reports must generate efficiently. 2. Statistical information must remain accurate. 3. System should support multiple report requests stimuntaneously.
Postconditions	1. Reports must generate efficiently. 2. Statistical information must remain accurate. 3. System should support multiple report requests simultaneously.

UC_19: View Passenger Details

UC Name	View Passenger Details
Dependency	US_20 (View Schedules details)
Actors	Busman
Preconditions	1. Busman is authenticated. 2. Passenger reservation information exists.
Description of Main Sequence	1. Busman accesses transportation schedule details. 2. System retrieves passenger reservation information. 3. System retrieves ticket validation information. 4. System displays passenger details and reservation status. 5. Busman reviews passenger information.
Description of Alternative Sequence	1. Passenger information unavailable: a. System displays unavailable passenger information message. 2. Retrieval failure: a. System displays temporary system error message.
Non-functional requirements	1. Passenger information must remain secure. 2. Retrieval operations should complete efficiently. 3. Passenger data must remain accurate.
Postconditions	1. Passenger information is displayed successfully. 2. Busman can continue validation process.

UC_20: Schedules details

UC Name	View available room types
Dependency	UC_02 User Login
Actors	Busman

Preconditions	1. Busman is authenticated. 2. Transportation schedules exist in the system.
Description of Main Sequence	1. Busman accesses schedule details section. 2. System retrieves transportation schedules. 3. System retrieves passenger reservation information. 4. System displays schedule details and passenger list. 5. Busman reviews transportation schedule information.
Description of Alternative Sequence	1. Schedule unavailable: a. System displays unavailable schedule message. 2. Data retrieval failure: a. System displays temporary system error message.
Non-functional requirements	1. Schedule information must remain accurate. 2. Passenger information should load efficiently. 3. Schedule retrieval must remain reliable.
Postconditions	1. Transportation schedule details are displayed successfully. 2. Busman can review passenger information.

UC_21: Mark Ticket as Used

UC Name	Mark Ticket as Used
Dependency	UC_22 Scan Ticket
Actors	Busman
Preconditions	1. Transportation ticket is validated successfully. 2. Passenger possesses valid QR ticket.

Description of Main Sequence	1. Busman validates passenger ticket. 2. System confirms ticket authenticity. 3. Busman marks transportation ticket as used. 4. System updates ticket validation status. 5. System stores passenger boarding information. 6. System confirms successful ticket validation update.

Description of Alternative Sequence	1. Invalid ticket: a. System displays invalid ticket message. 2. Already used ticket: a. System displays duplicate validation warning.
Non-functional requirements	1. Ticket validation must complete quickly. 2. QR validation data must remain accurate. 3. System must prevent duplicate ticket usage.
Postconditions	1. 1. Transportation ticket status is updated successfully. 2. Passenger boarding is confirmed.

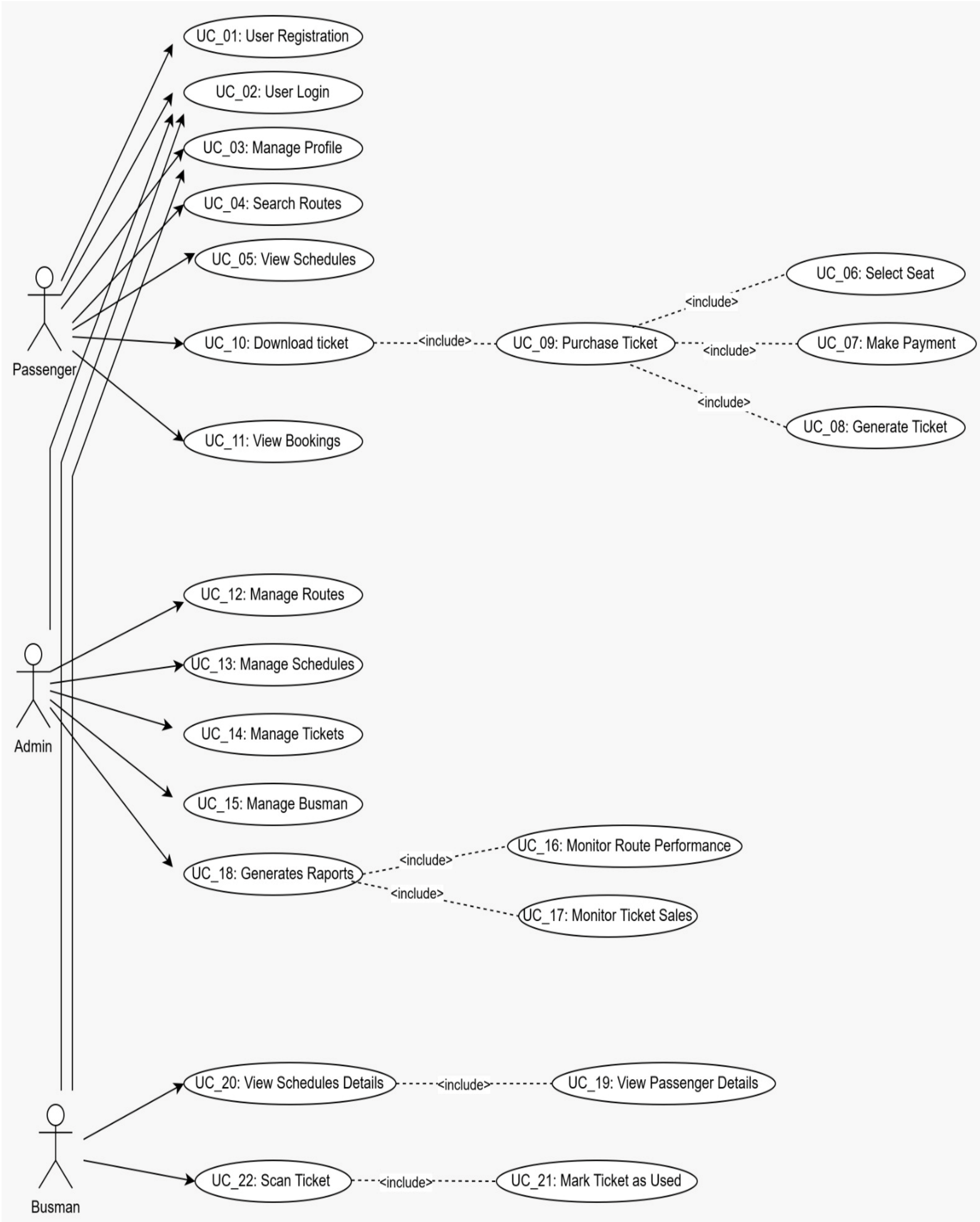
UC_22: Scan Ticket

UC Name	Scan Ticket
Dependency	UC_20 (View Schedule Details)
Actors	Busman
Preconditions	1. Busman is authenticated. 2. Passenger possesses QR transportation ticket.
Description of Main Sequence	1. Busman opens QR scanning section. 2. Passenger presents QR transportation ticket. 3. Busman scans QR code. 4. System retrieves ticket information. 5. System validates transportation ticket. 6. System displays validation result and passenger information. 7. Busman confirms ticket verification process.
Description of Alternative Sequence	1. Invalid QR code: a. System displays invalid ticket message. 2. QR scan failure: a. System requests another scan attempt.

Non-functional requirements	1. QR scanning should complete quickly. 2. Validation information must remain accurate. 3. System must support reliable QR recognition.
Postconditions	1. Transportation ticket is validated successfully. 2. Busman can mark ticket as used.

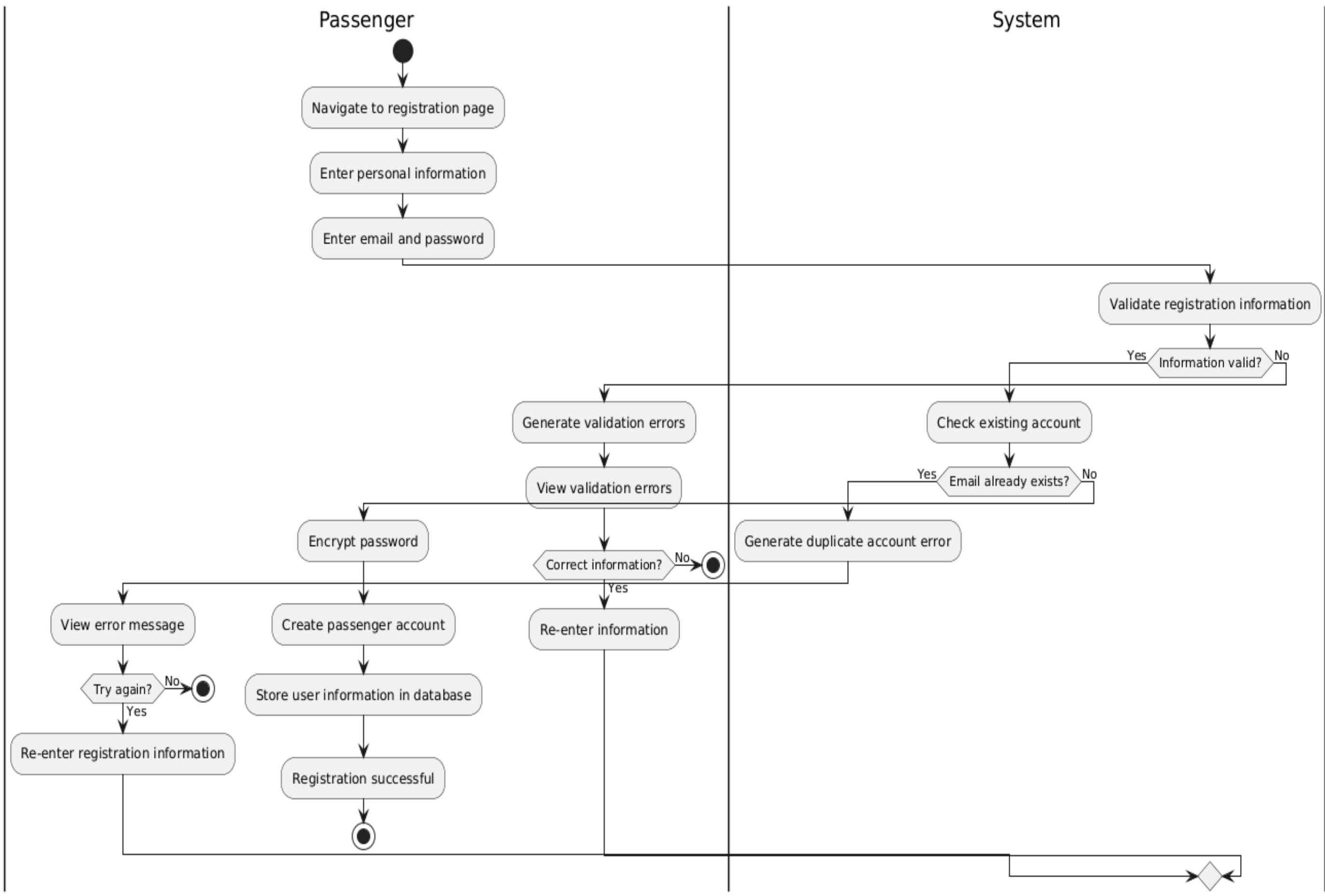
4.2 Behavioral Diagrams

4.2.1 Use Case Diagrams

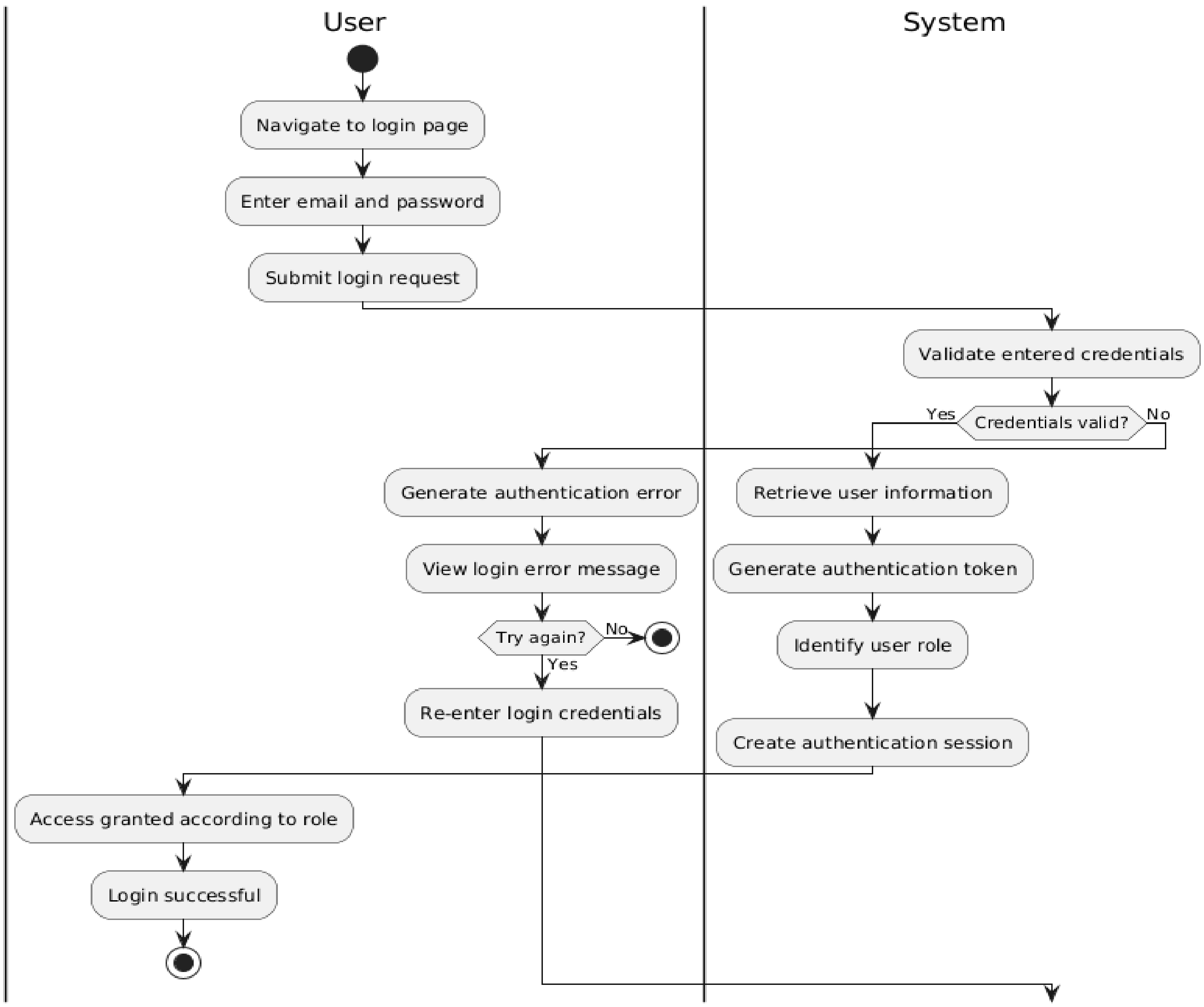

UC_01: User Registration
UC_02: User Login
UC_03: Manage Profile
UC_04: Search Routes
UC_05: View Schedules
UC_06: Select Seat
<include>
UC_10: Download ticket
<include>
UC_09: Purchase Ticket
<include>
UC_07: Make Payment
Passenger
<include>
UC_08: Generate Ticket
UC_11: View Bookings
UC_12: Manage Routes
UC_13: Manage Schedules
Admin
UC_14: Manage Tickets
UC_15: Manage Busman
<include>
UC_16: Monitor Route Performance
UC_18: Generates Raports
<include>
UC_17: Monitor Ticket Sales
UC_20: View Schedules Details
<include>
UC_19: View Passenger Details
UC_22: Scan Ticket
<include>
UC_21: Mark Ticket as Used
Busman

4.2.2 Activity Diagrams

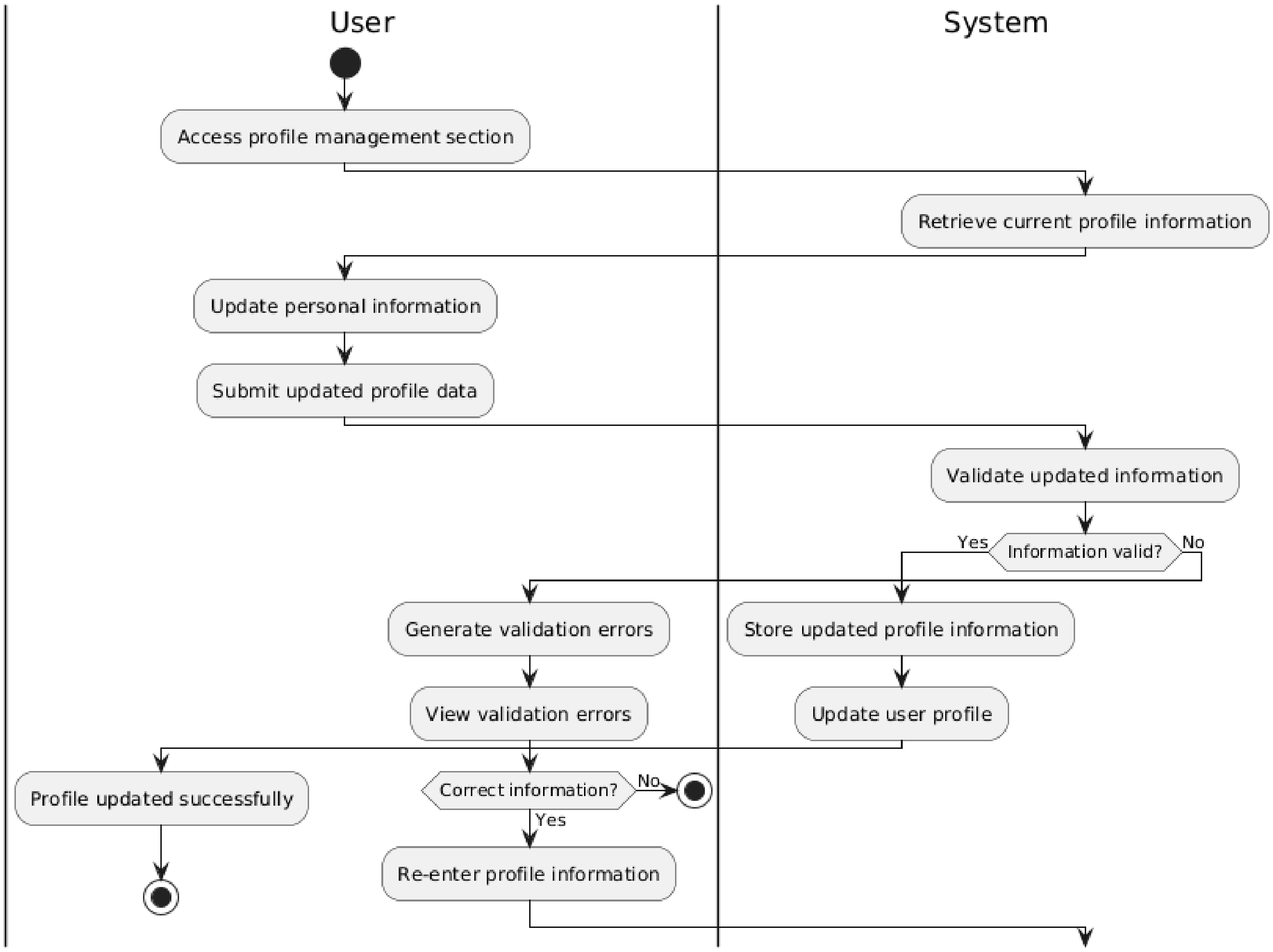
AD_01 User Registration



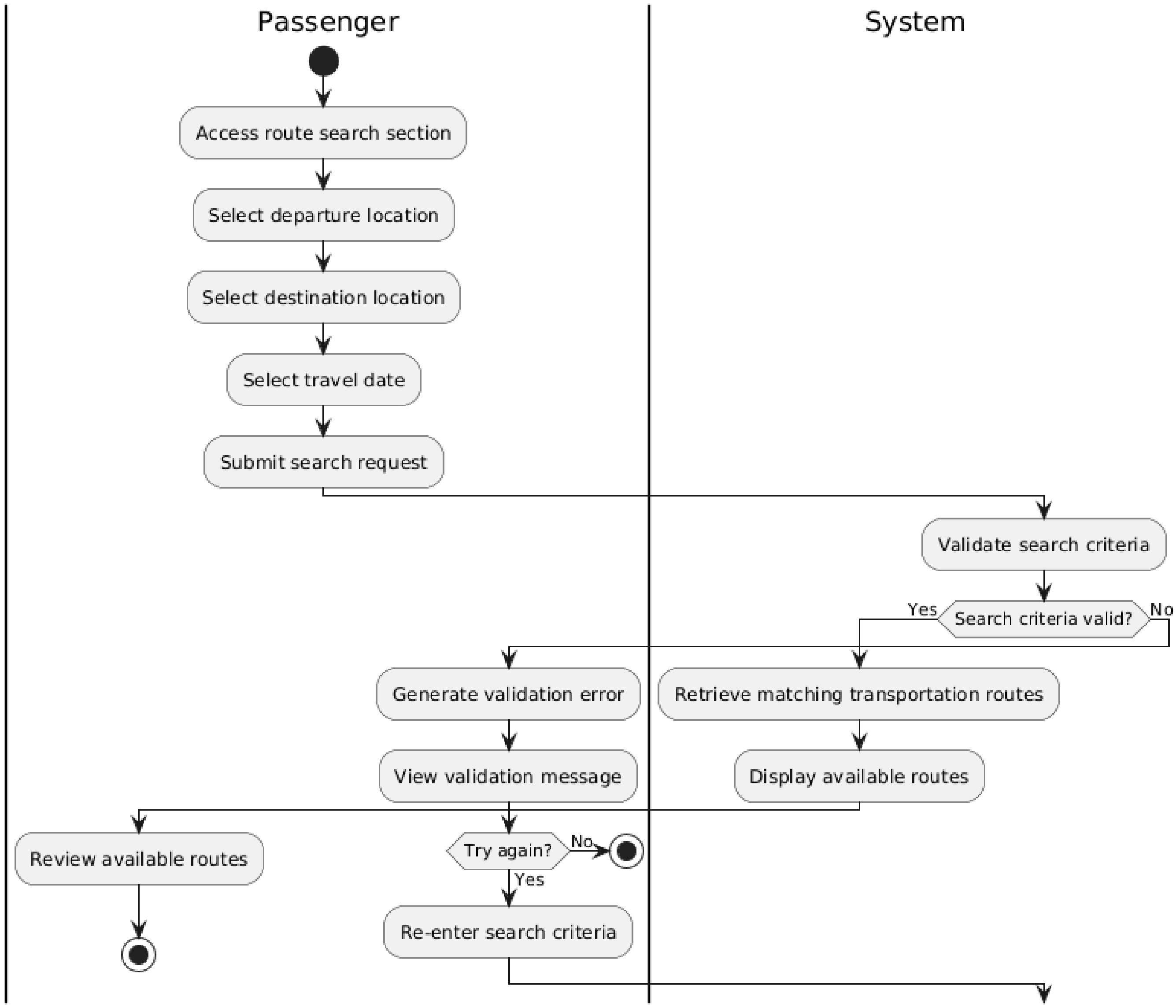
AD_02 Log In


User
System
Navigate to login page
Enter email and password
Submit login request
Validate entered credentials
Credentials valid?
Yes
No
Generate authentication error
Retrieve user information
View login error message
Generate authentication token
Try again?
No
Yes
Identify user role
Re-enter login credentials
Create authentication session
Access granted according to role
Login successful

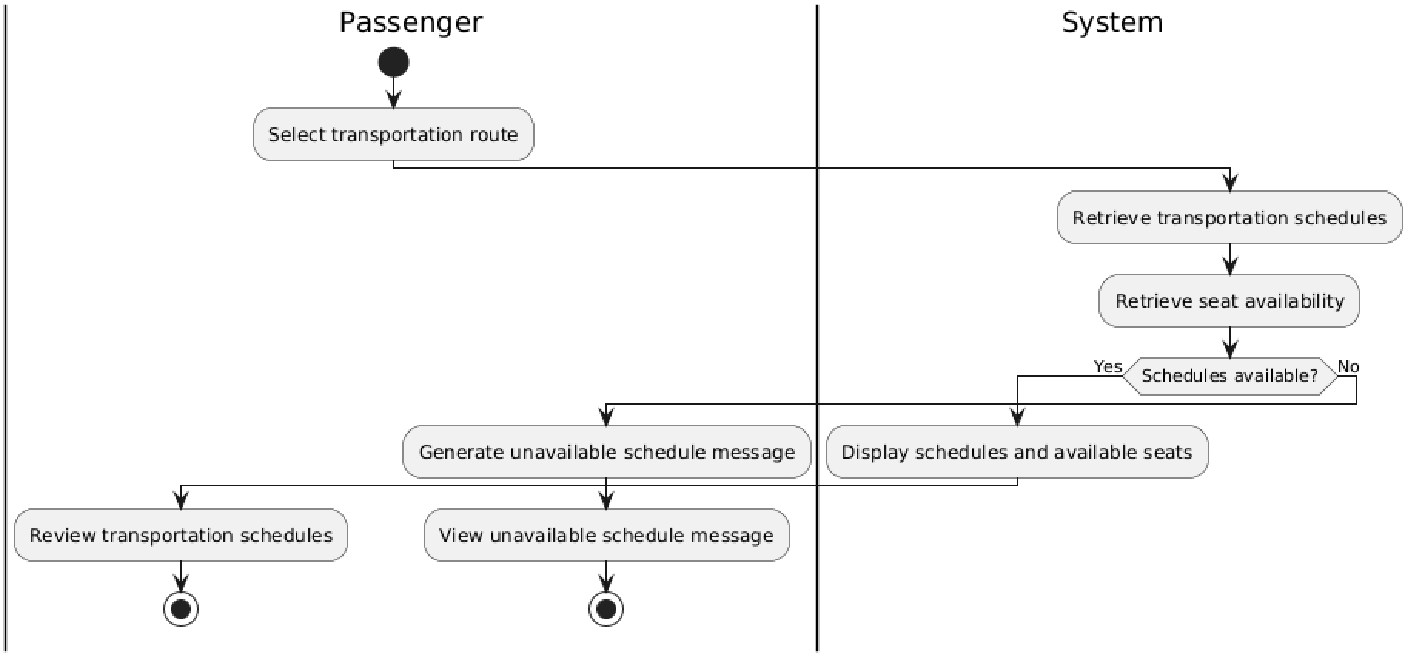
Ad_03_Manage Profile



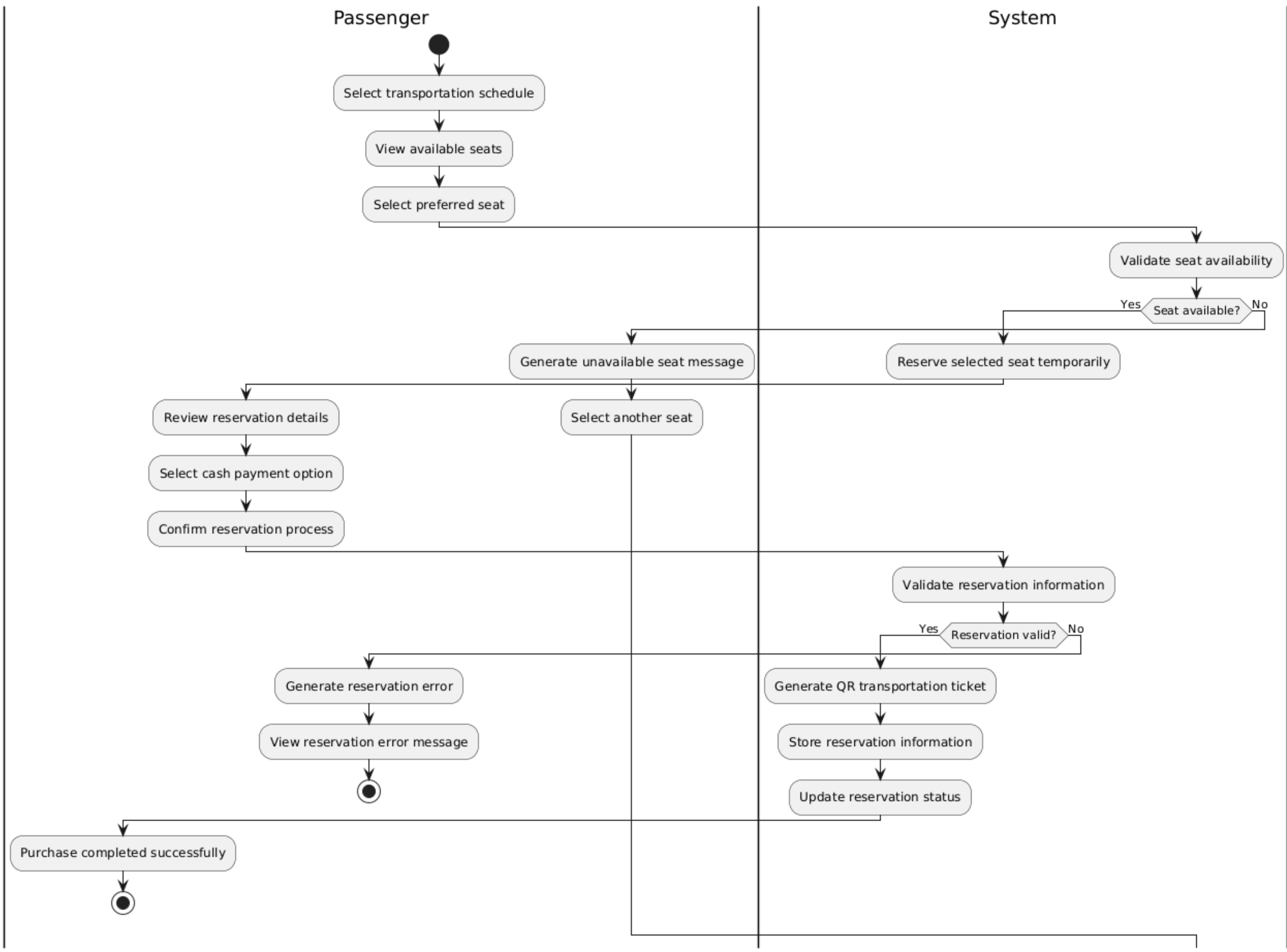
AD_04_ Search Routes



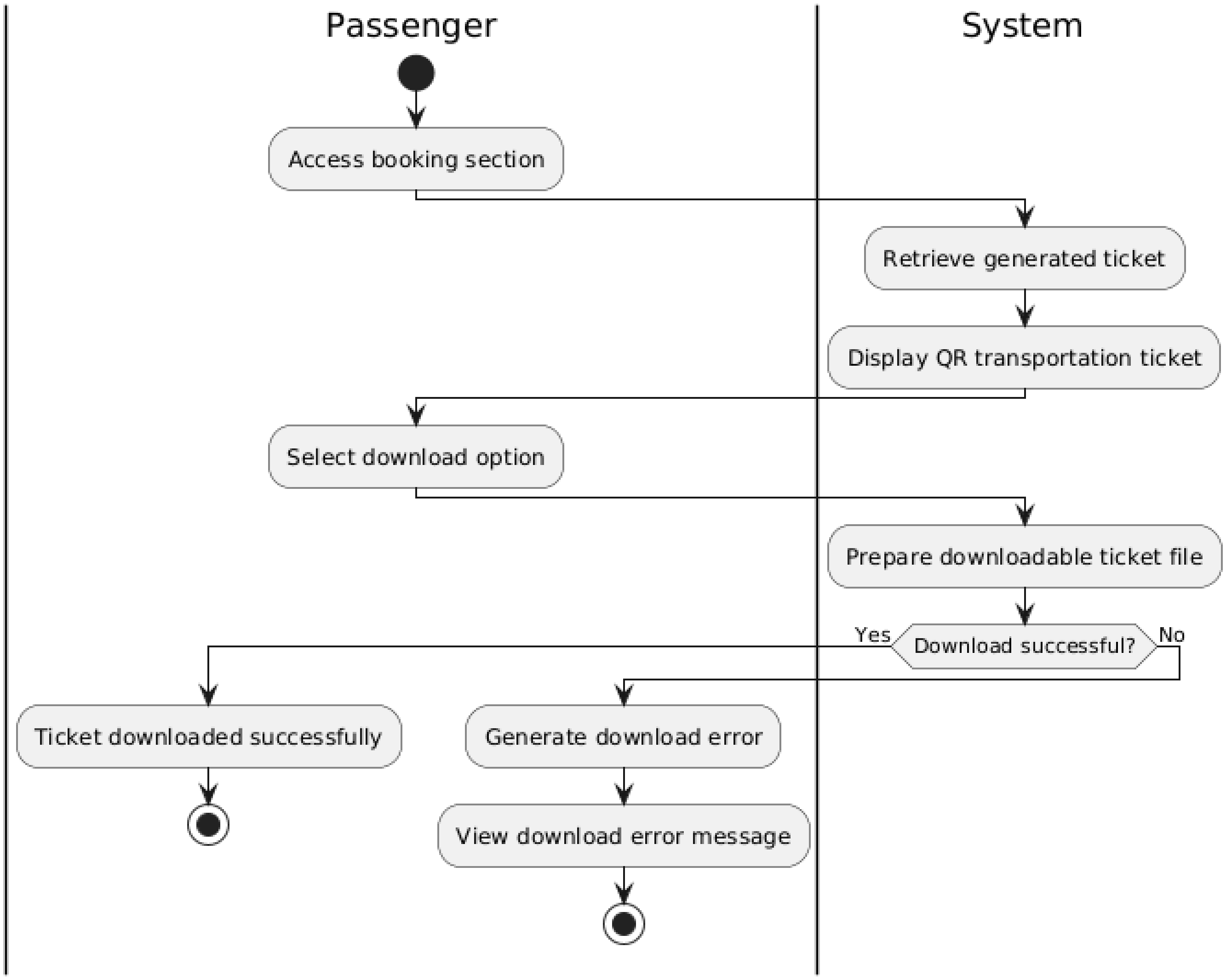
Ad_05_View Schedules



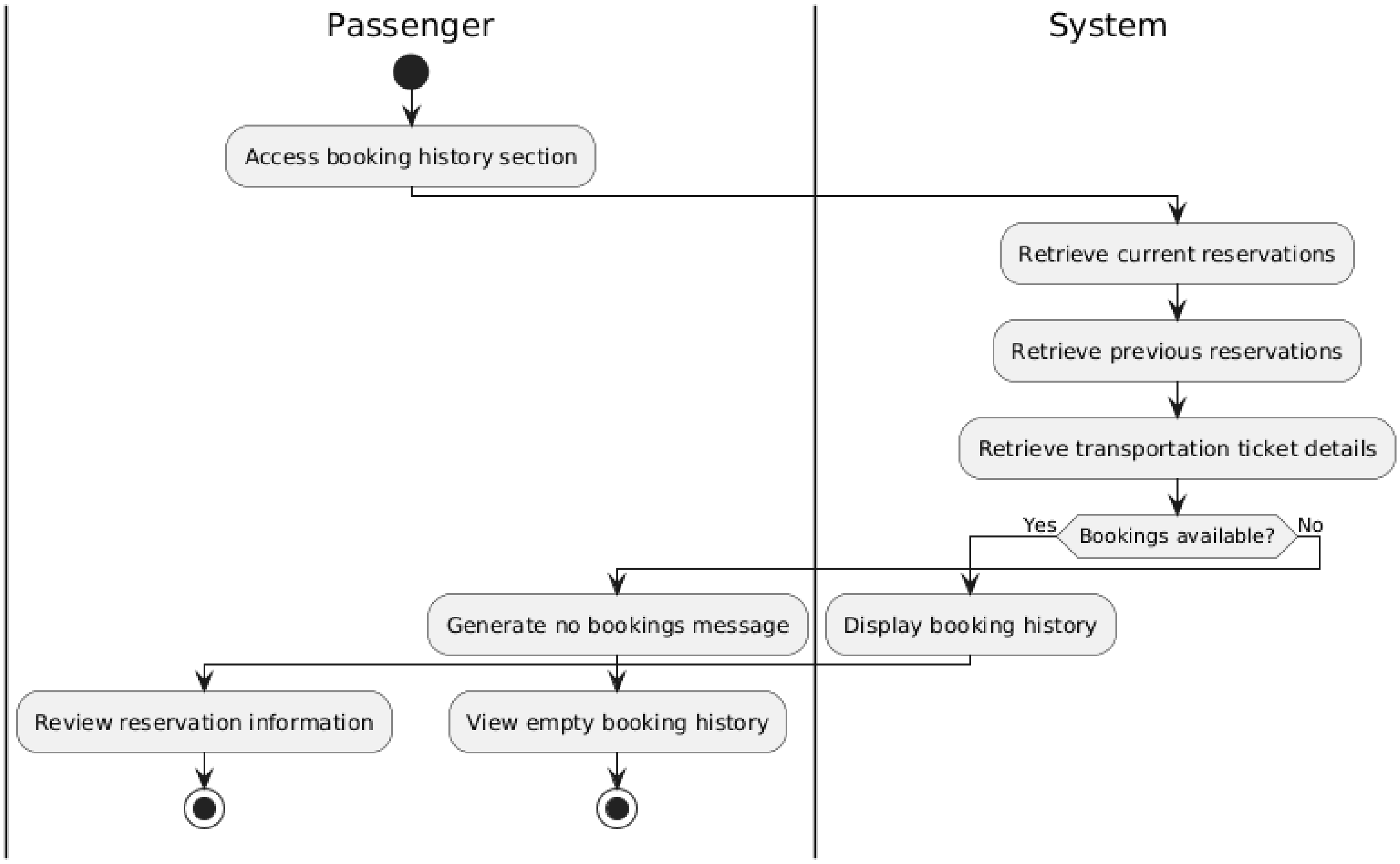
Ad_06_Purchase Ticket



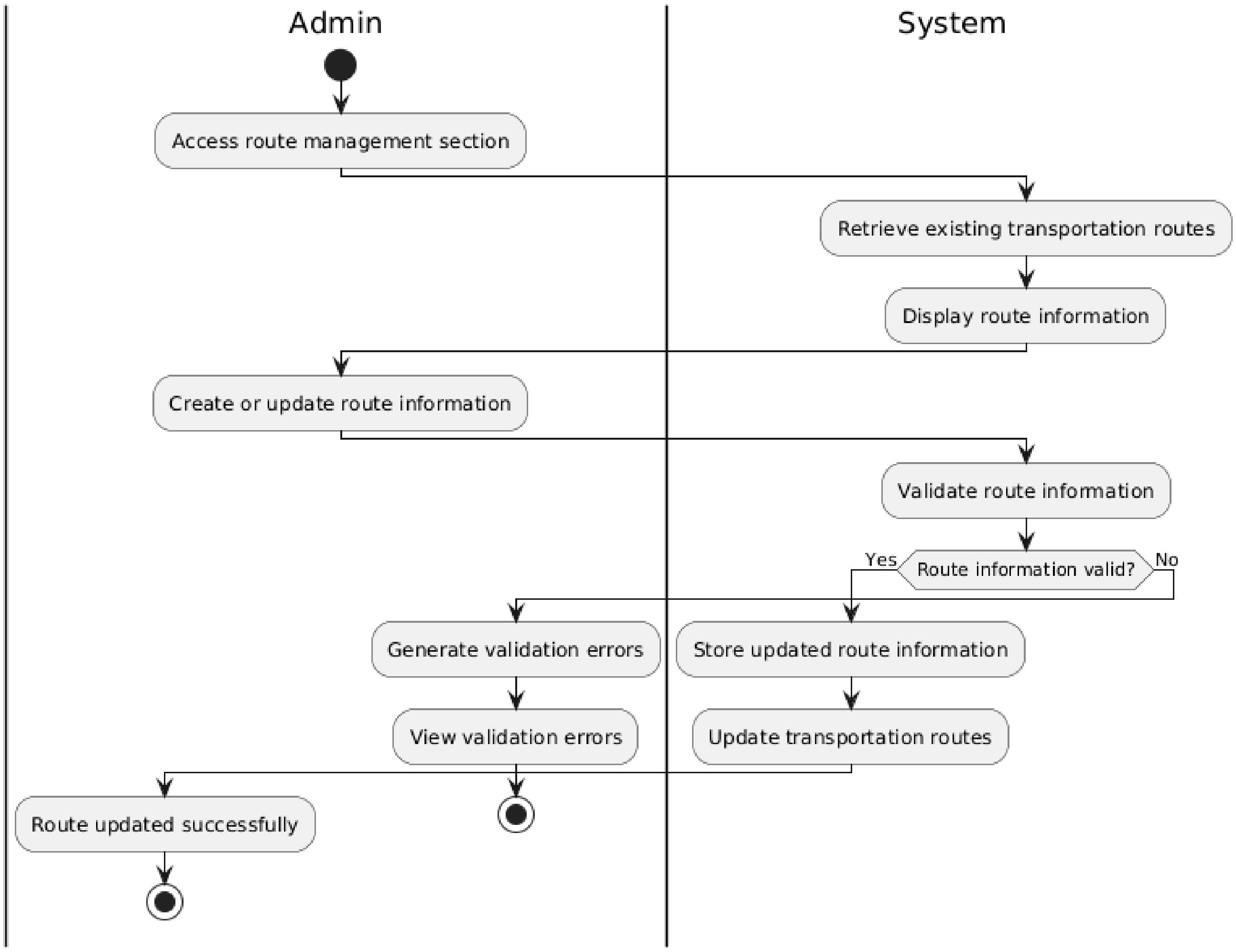
Ad_07_Download Ticket



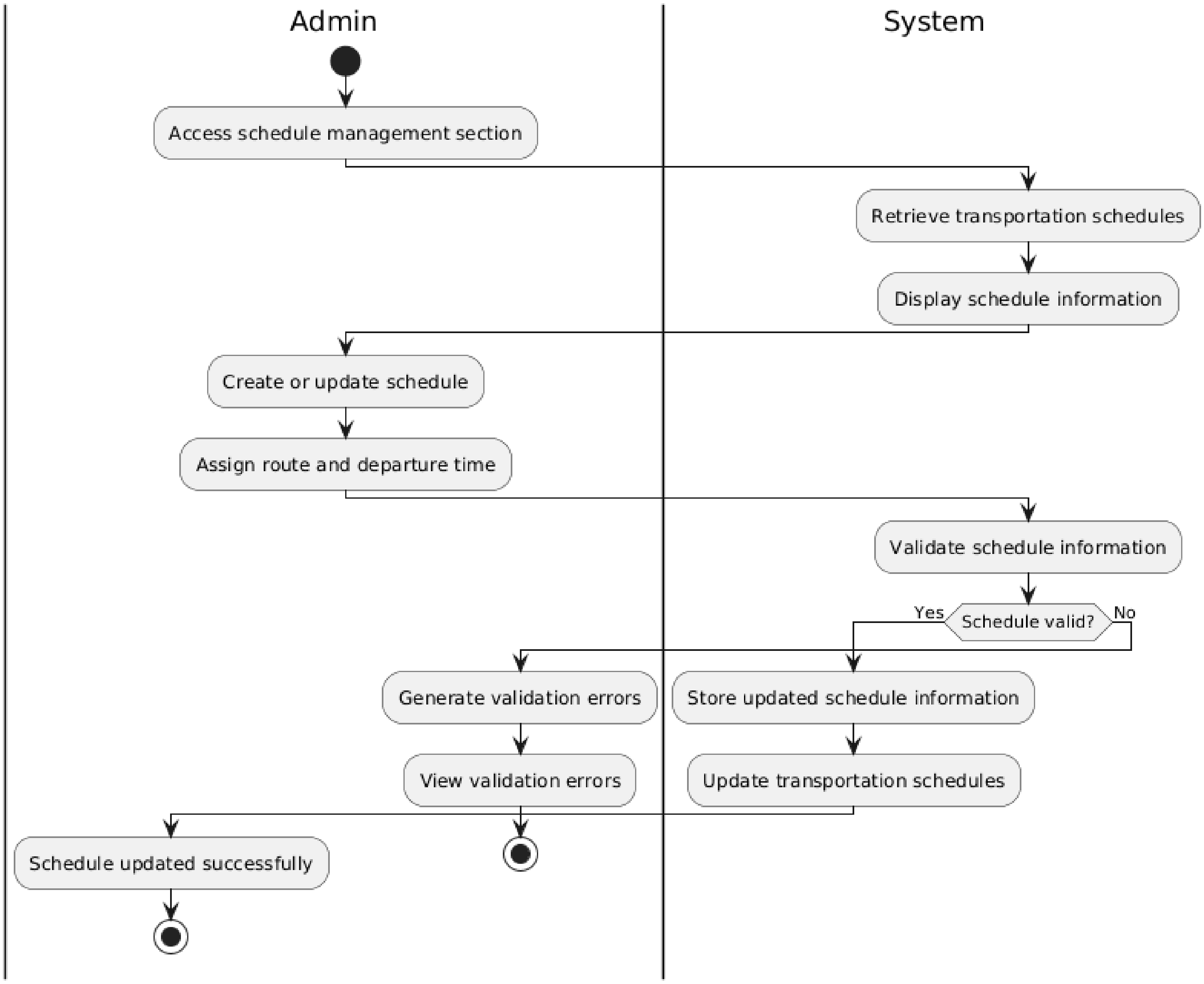
Ad_08_View Bookings



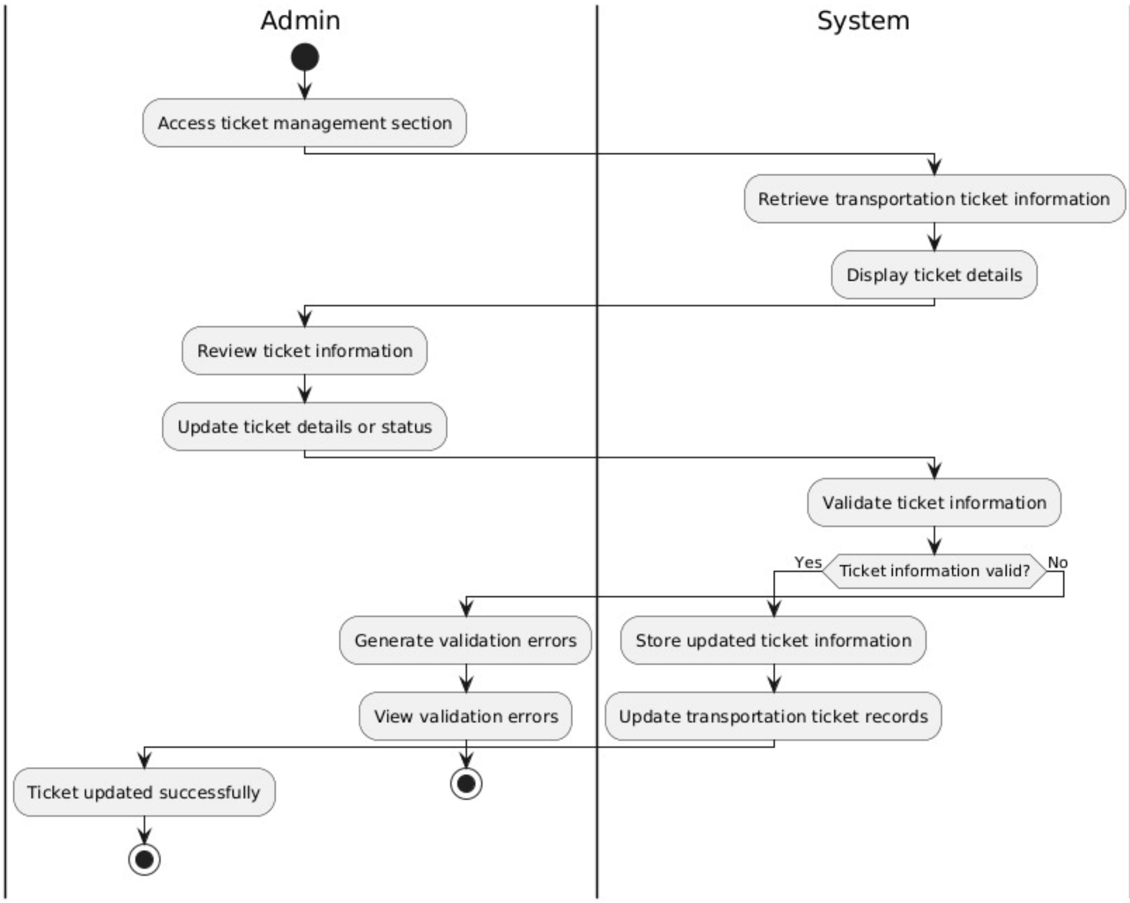
Ad_09_Manage Routes



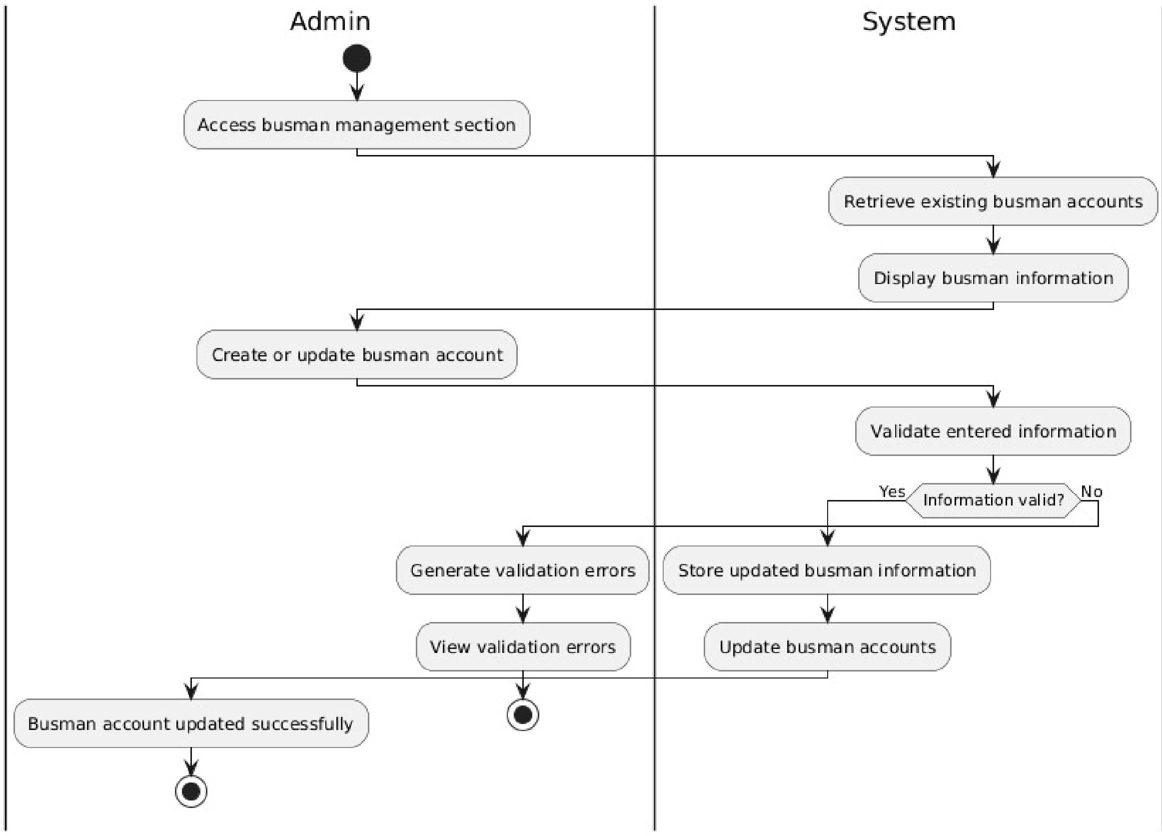
Ad_10_Manage Schedules



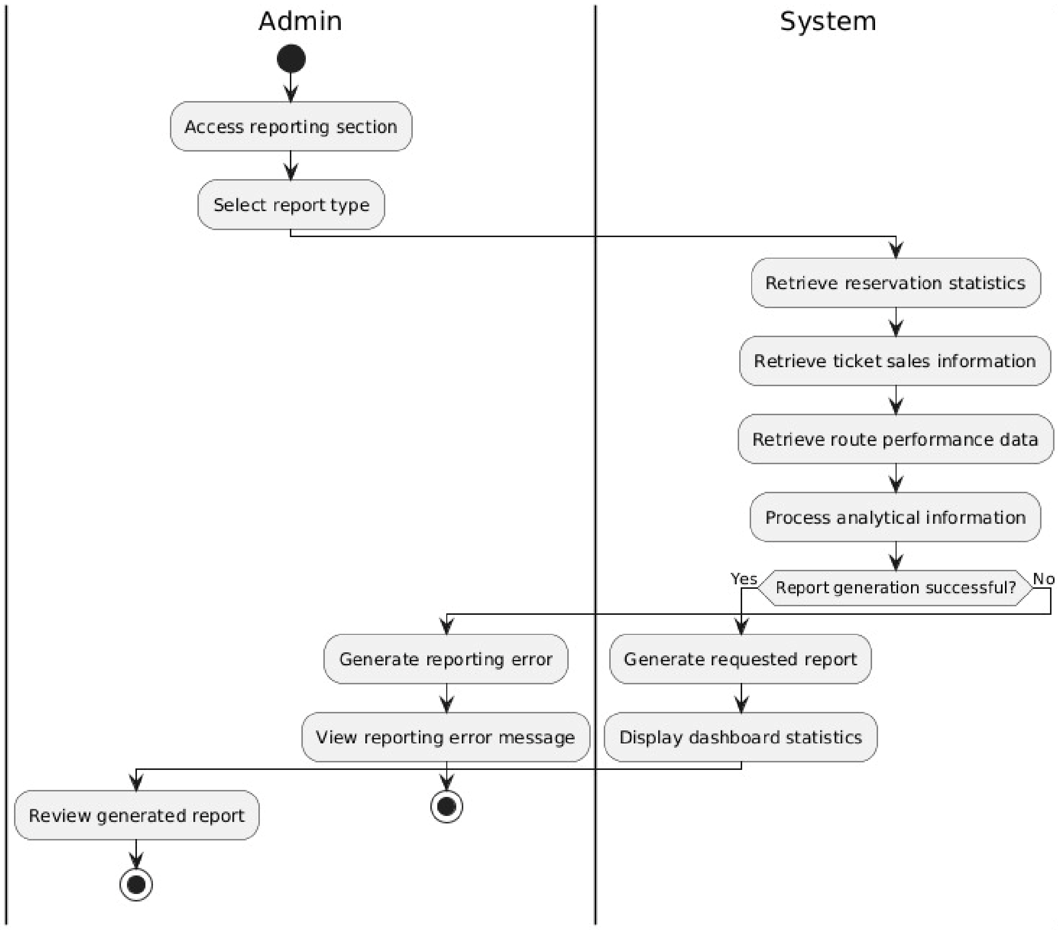
AD_11 _Manage Tickets



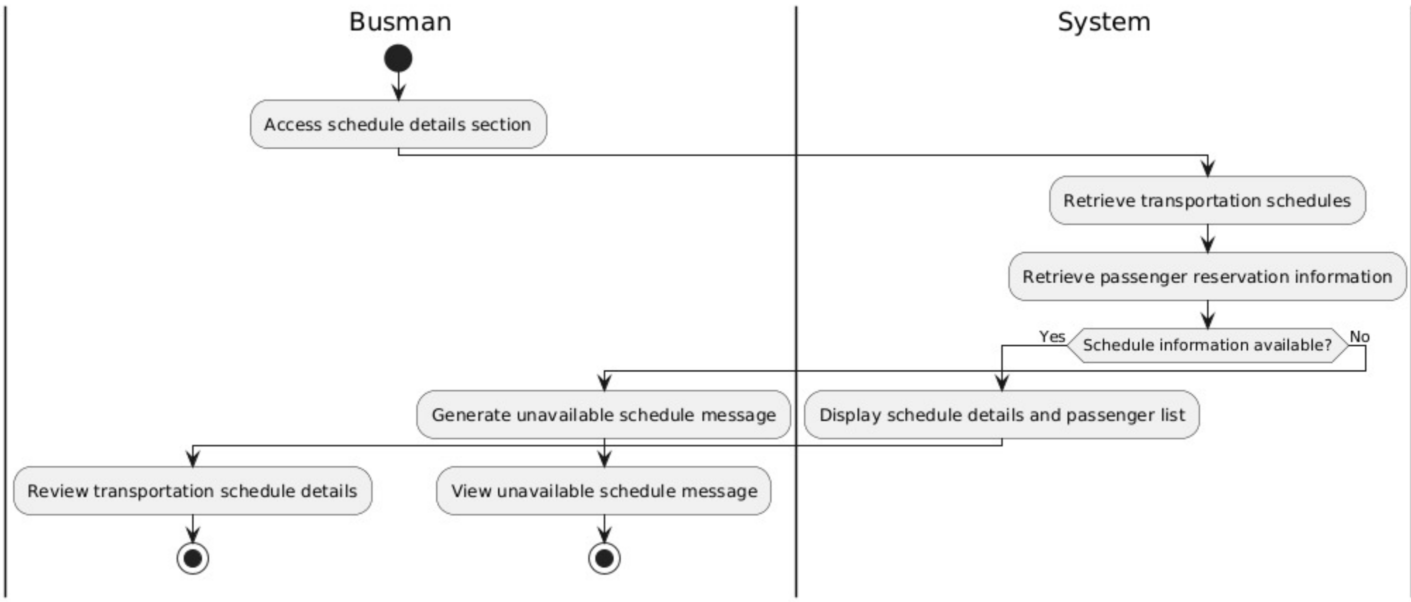
AD_12 _Manage Busman



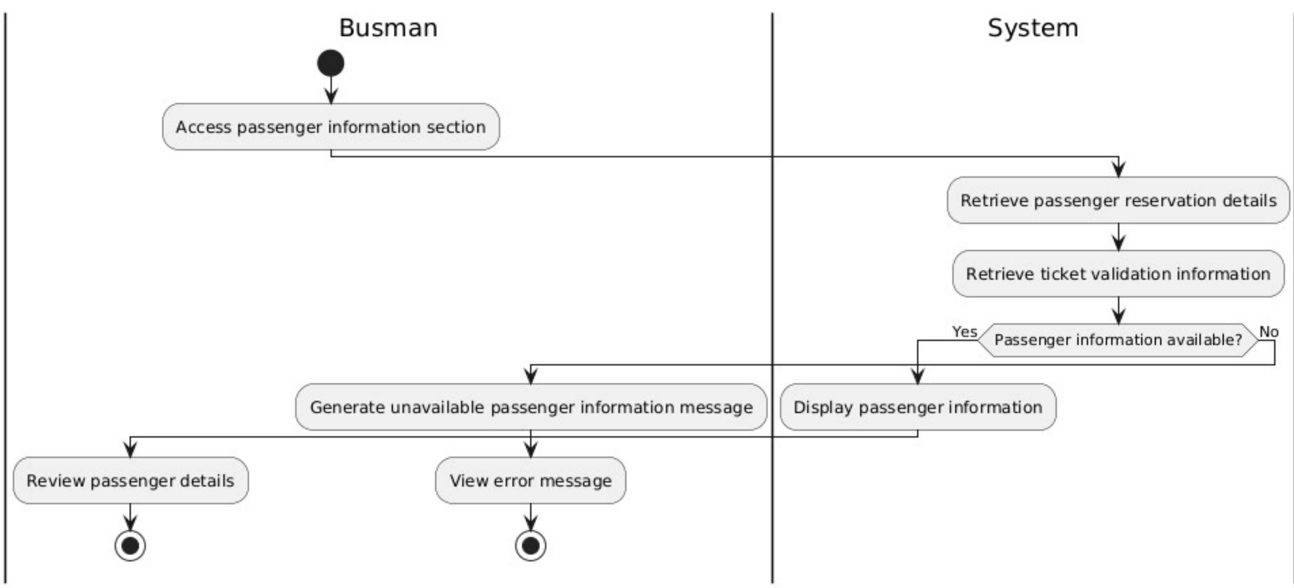
AD_13 _ Generate Reports



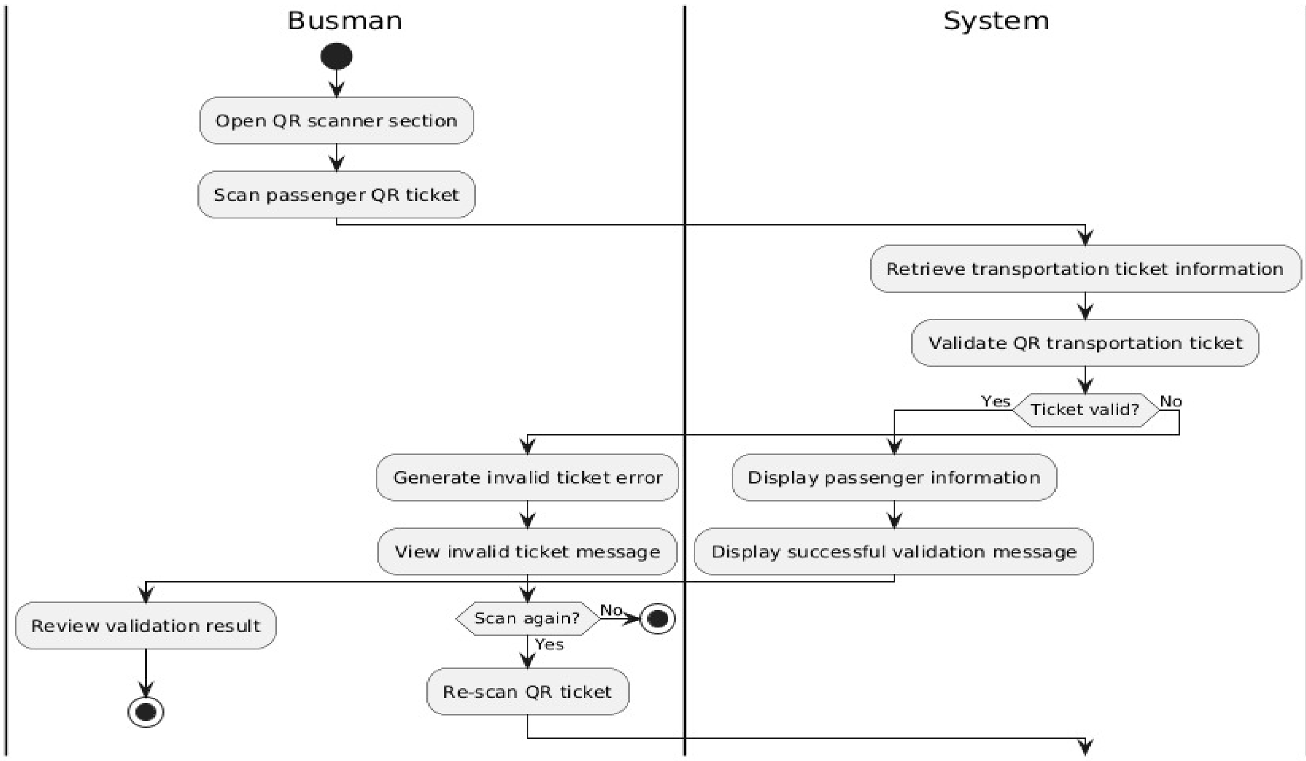
AD_14 _ View Schedule Details



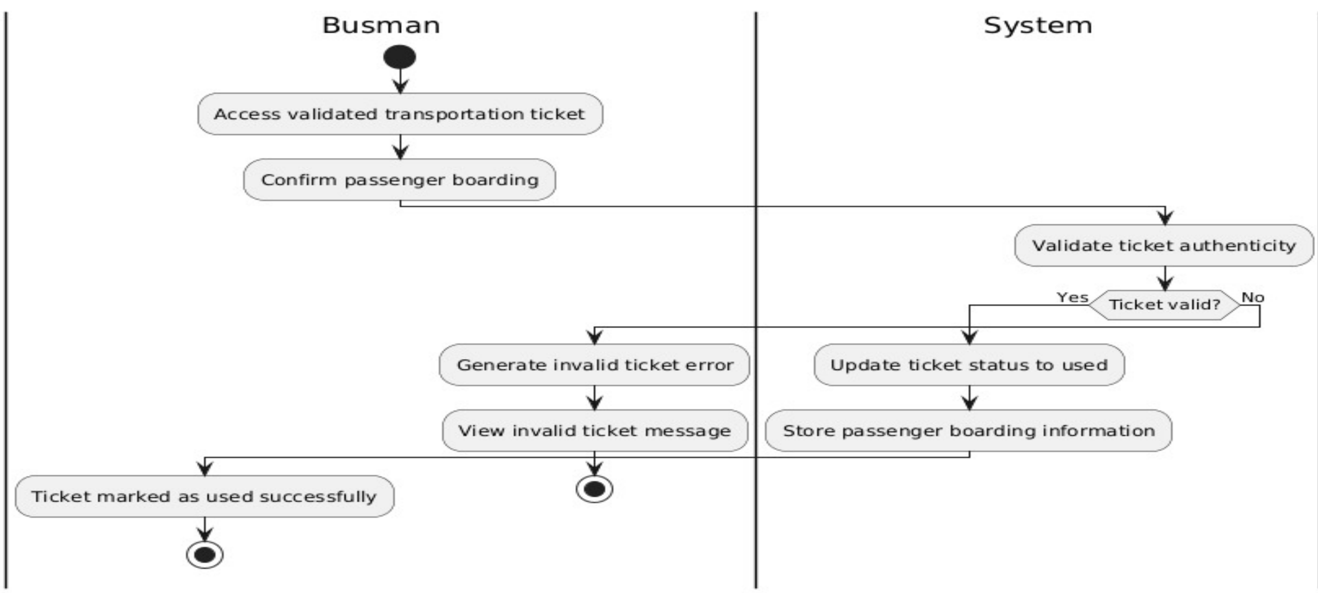
AD_15 _ View Passenger Details



AD_16 _ Scan Ticket

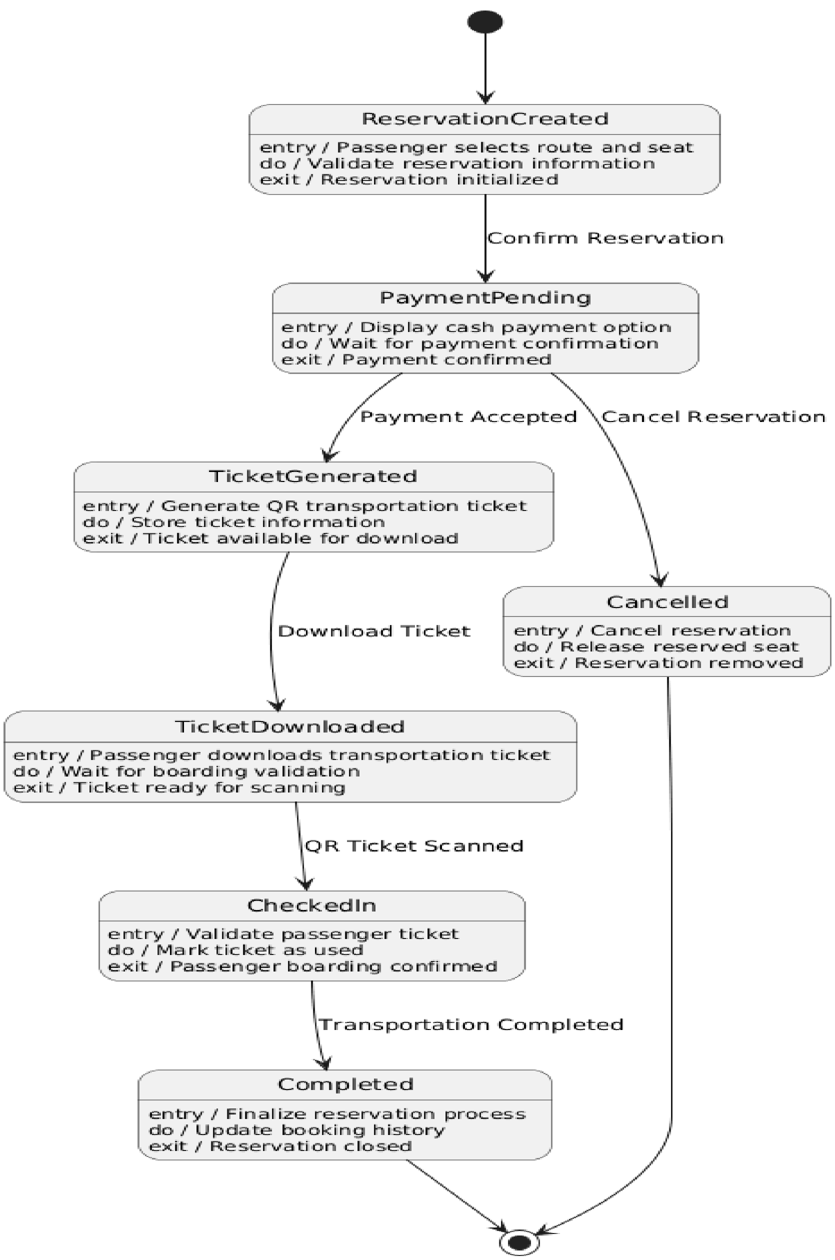


AD_17 —_Mark Ticket as Used

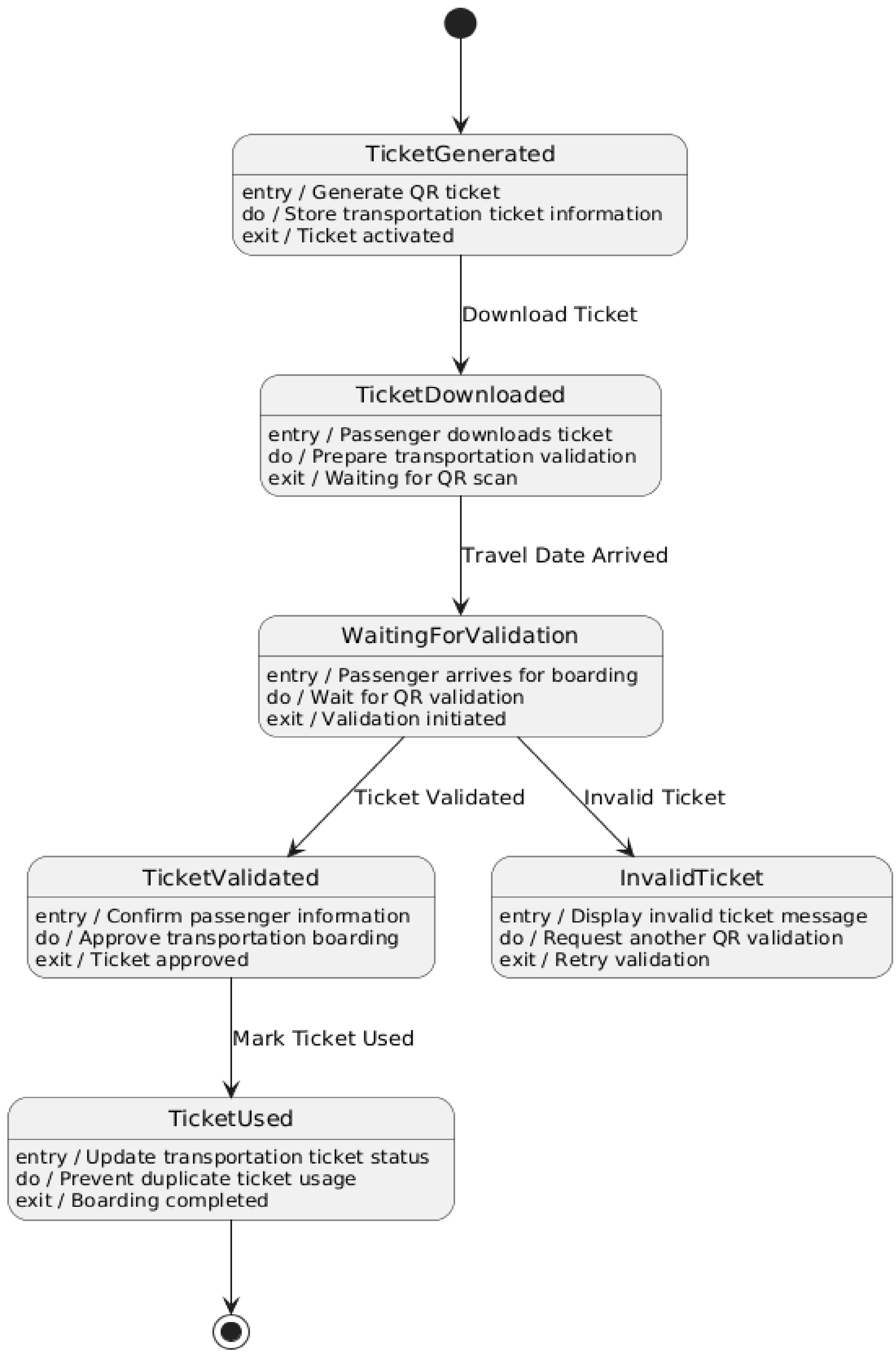


4.2.3 State Diagrams

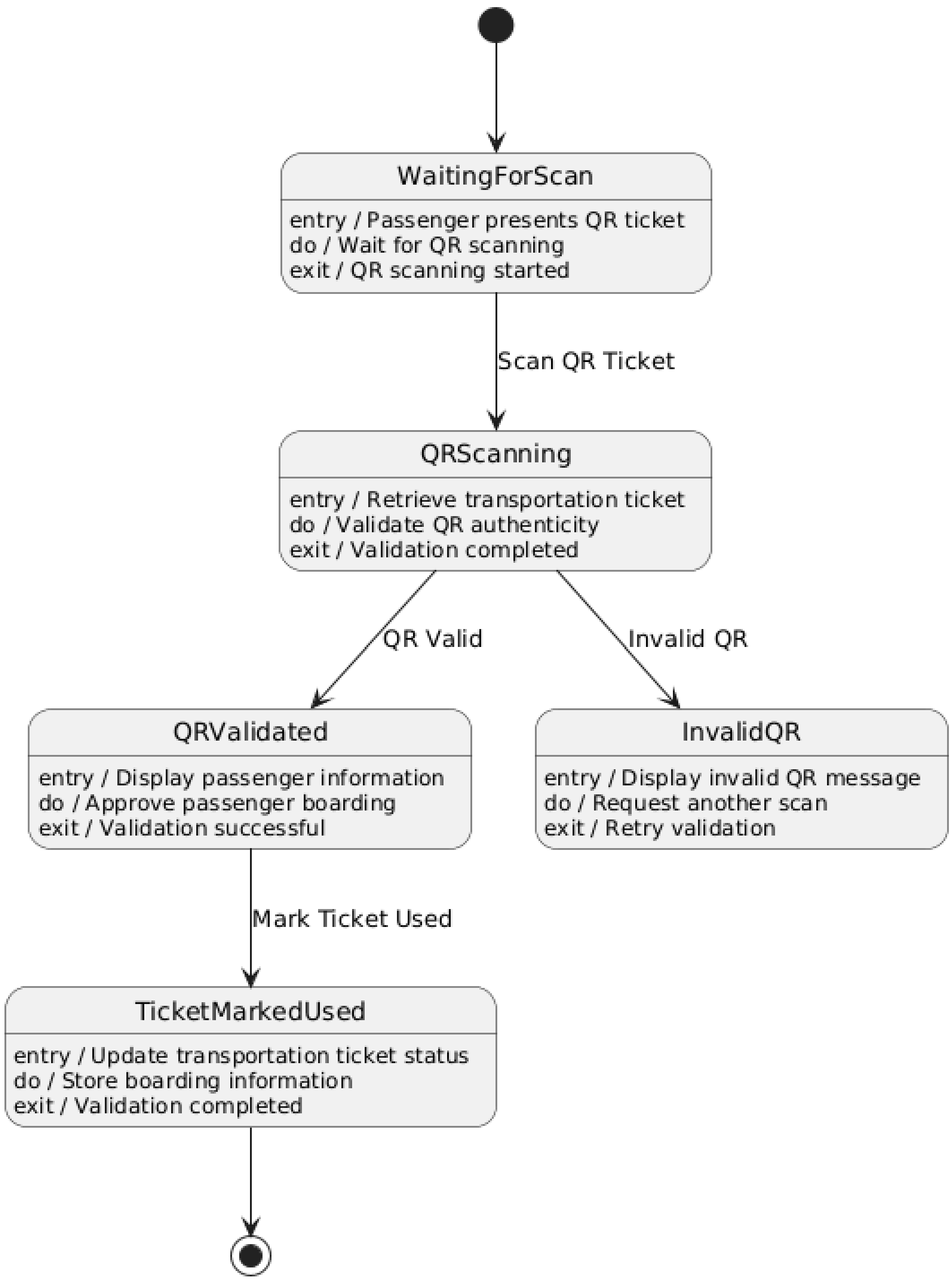
SD_01 — Reservation State Diagram



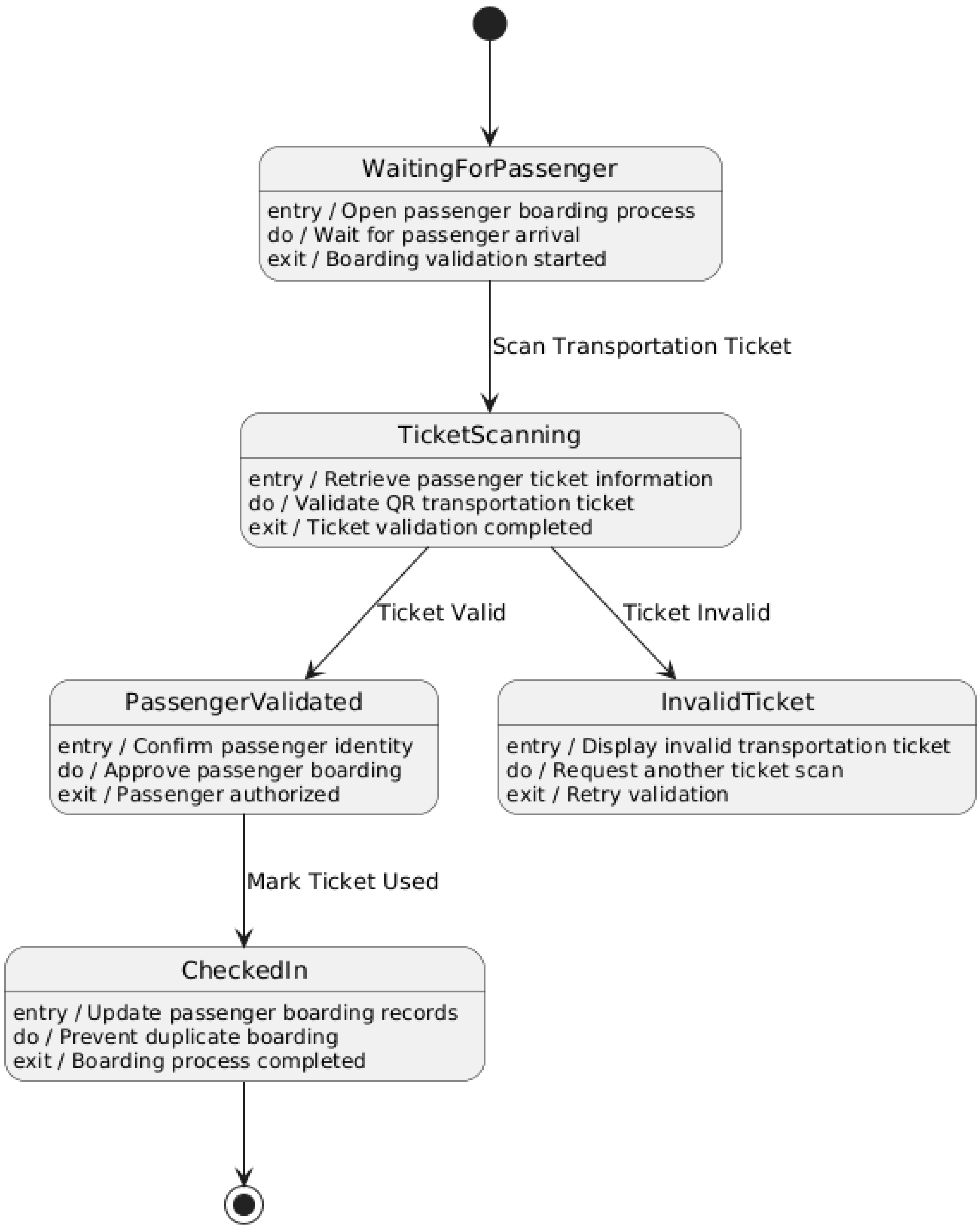
SD_02 — Ticket State Diagram



SD_03 — QR Validation State Diagram

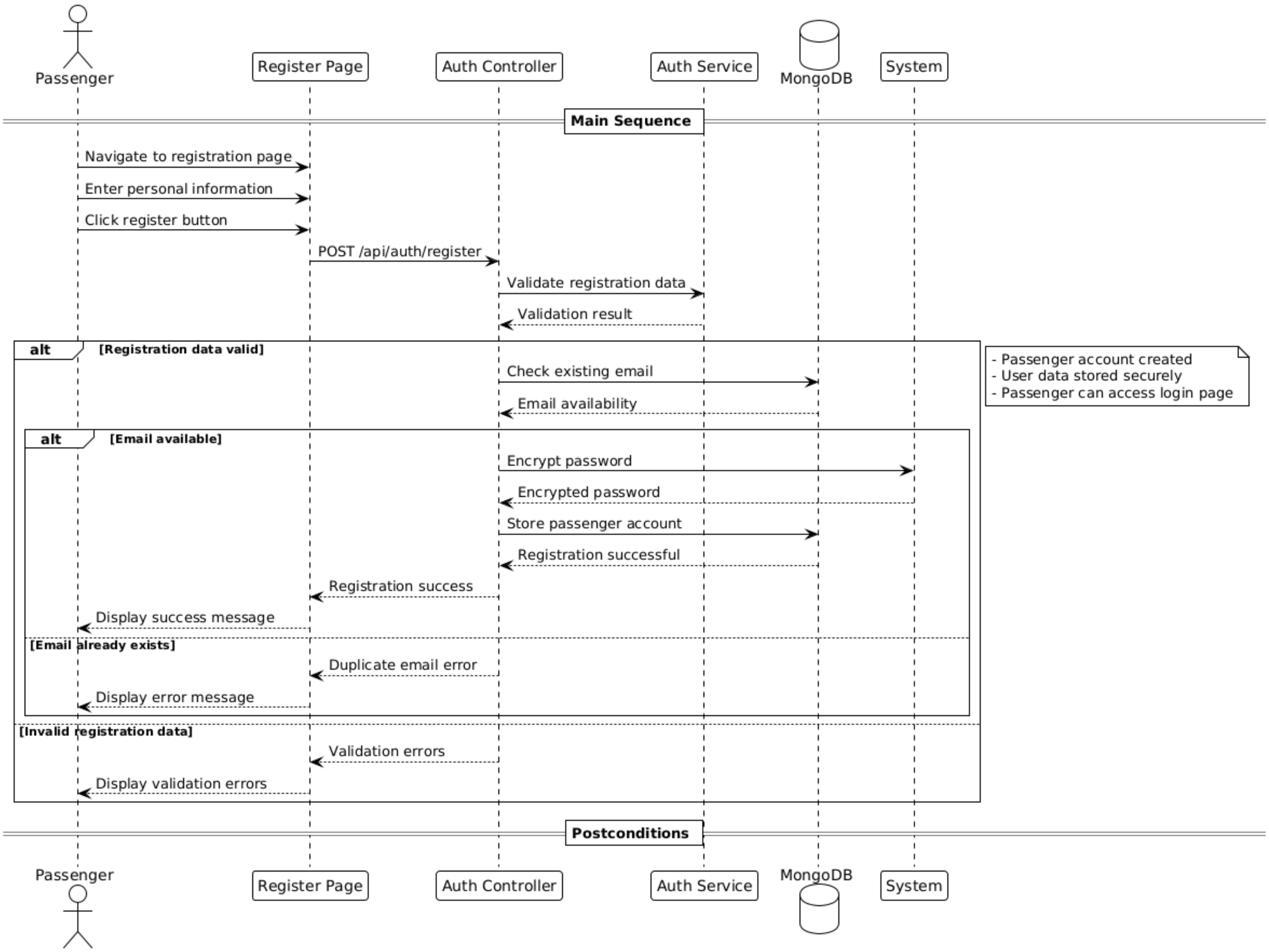


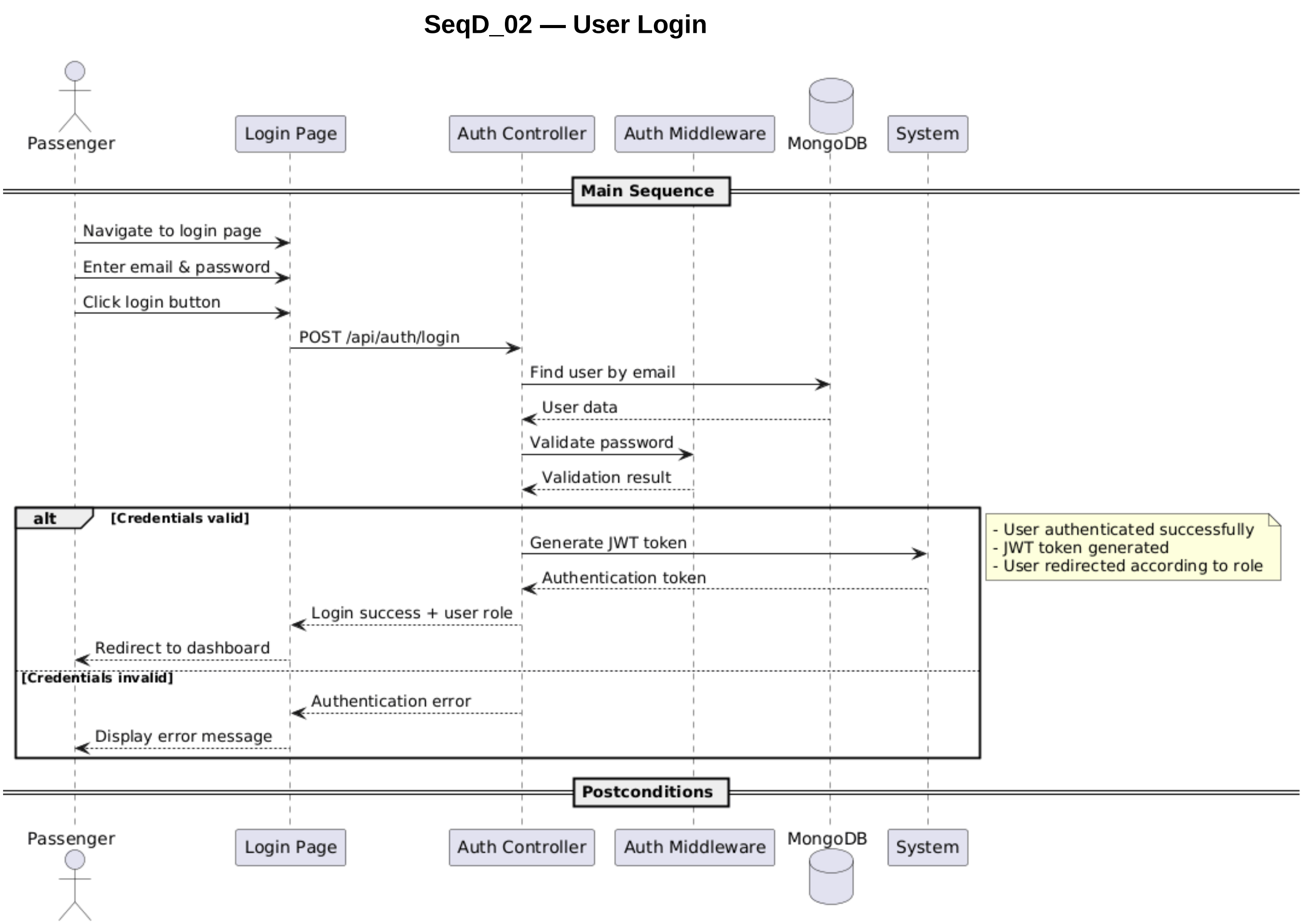
SD_04 — Check-In State Diagram



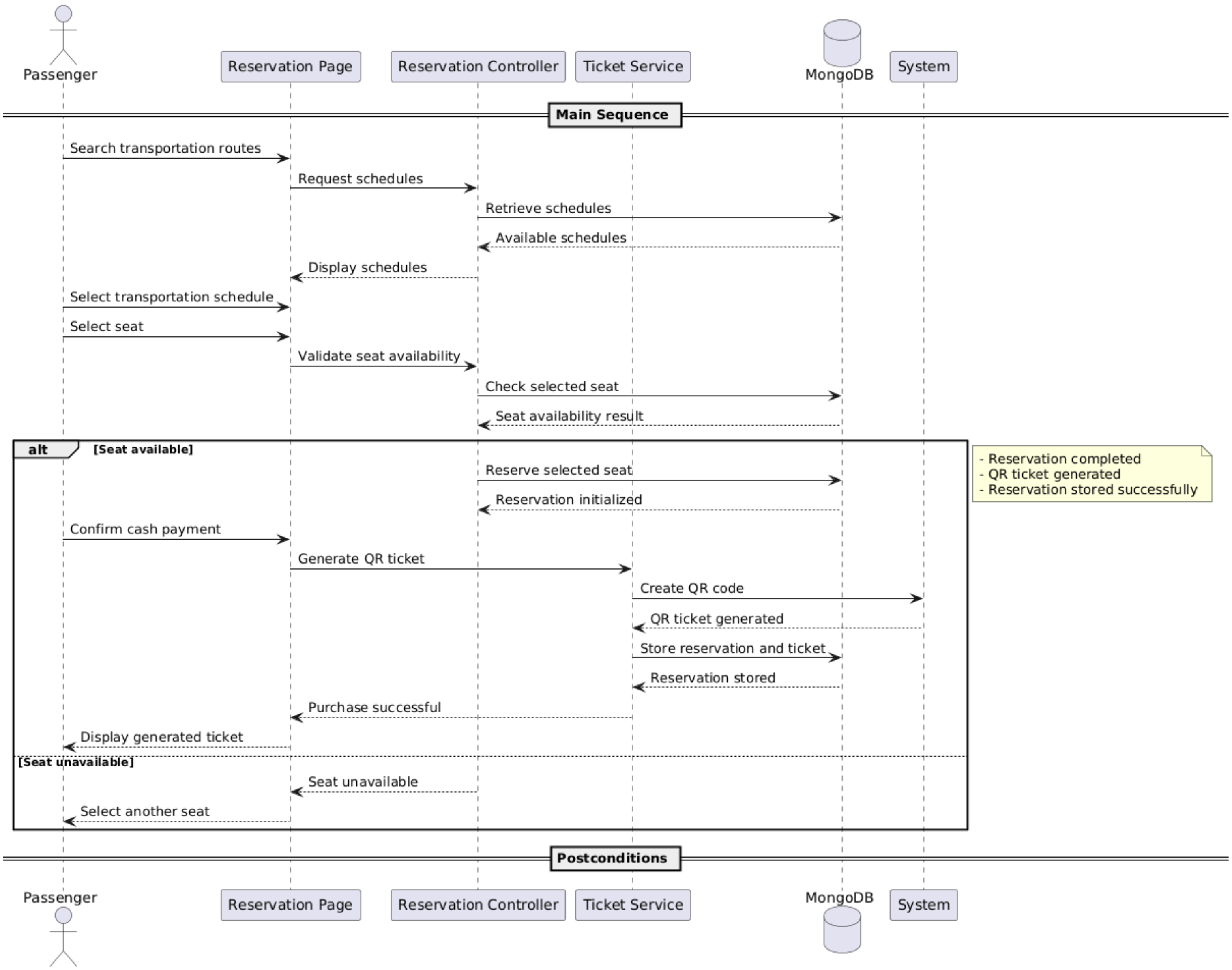
4.2.4 Sequence Diagrams

SeqD_01 — User Registration

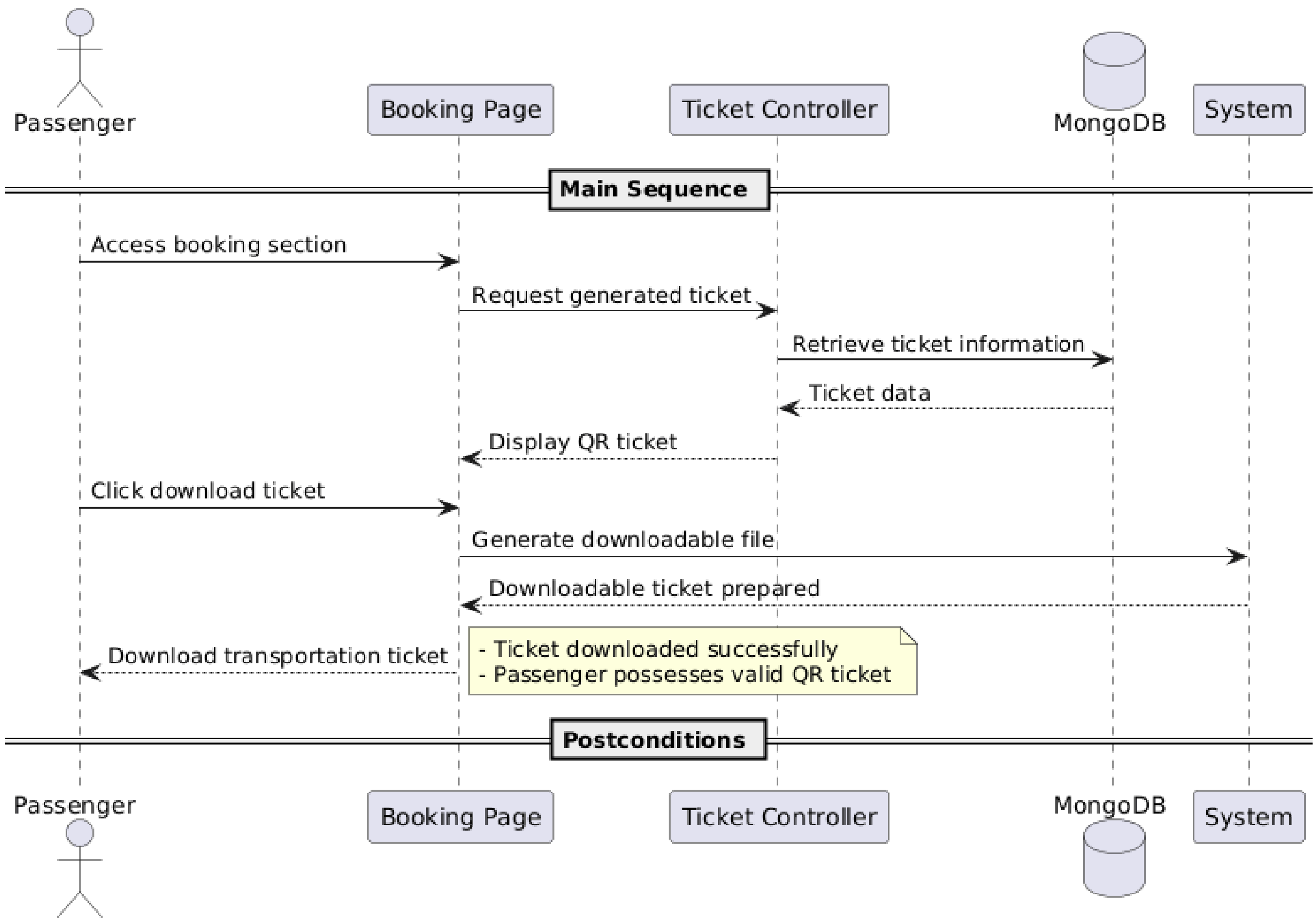



SeqD_02 — User Login
Passenger
Login Page
Auth Controller
Auth Middleware
MongoDB
System
Main Sequence
Navigate to login page
Enter email & password
Click login button
POST /api/auth/login
Find user by email
User data
Validate password
Validation result
alt
[Credentials valid]
Generate JWT token
Authentication token
Login success + user role
Redirect to dashboard
[Credentials invalid]
Authentication error
Display error message
- User authenticated successfully
- JWT token generated
- User redirected according to role
Postconditions

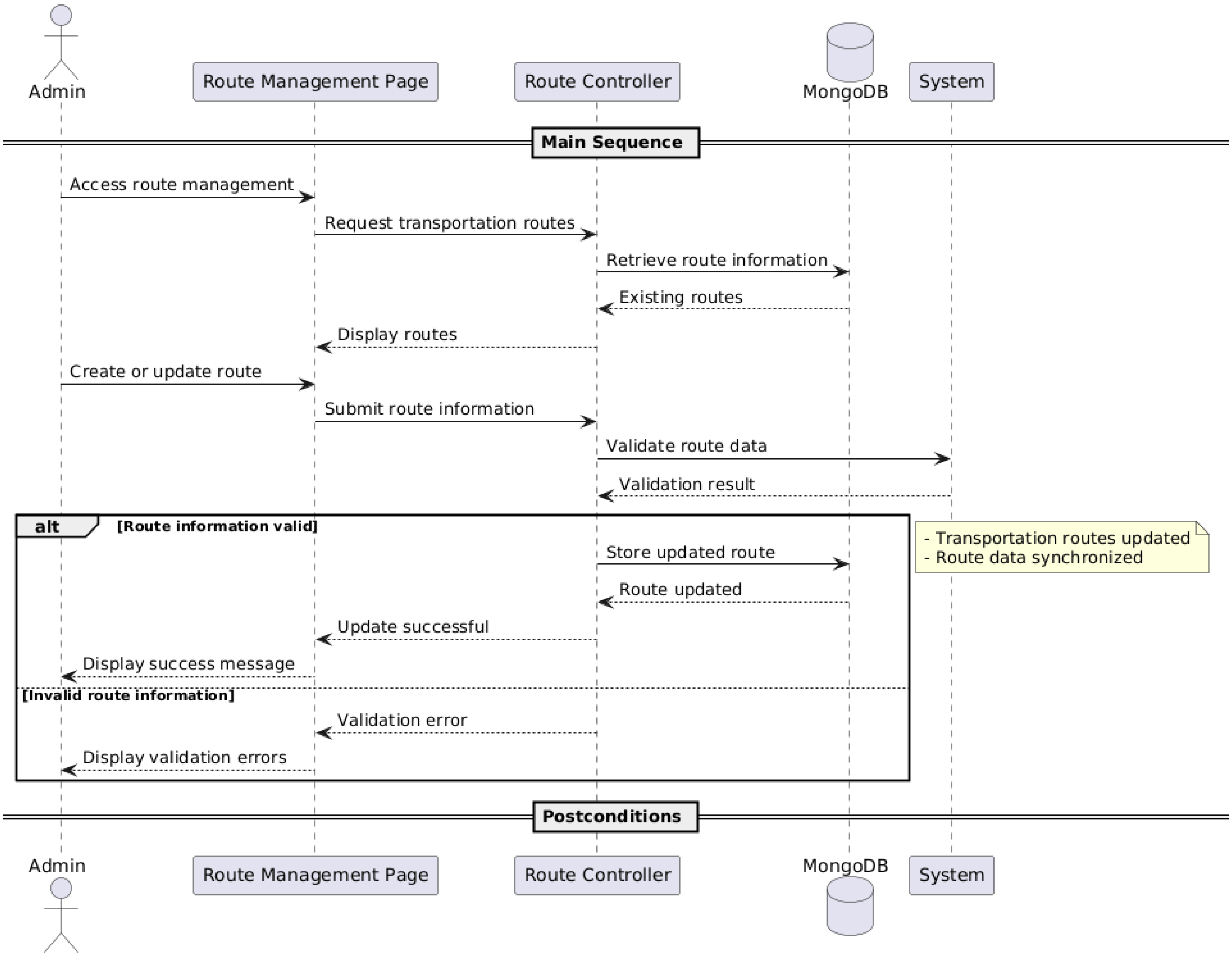
SeqD_03 — Purchase Ticket



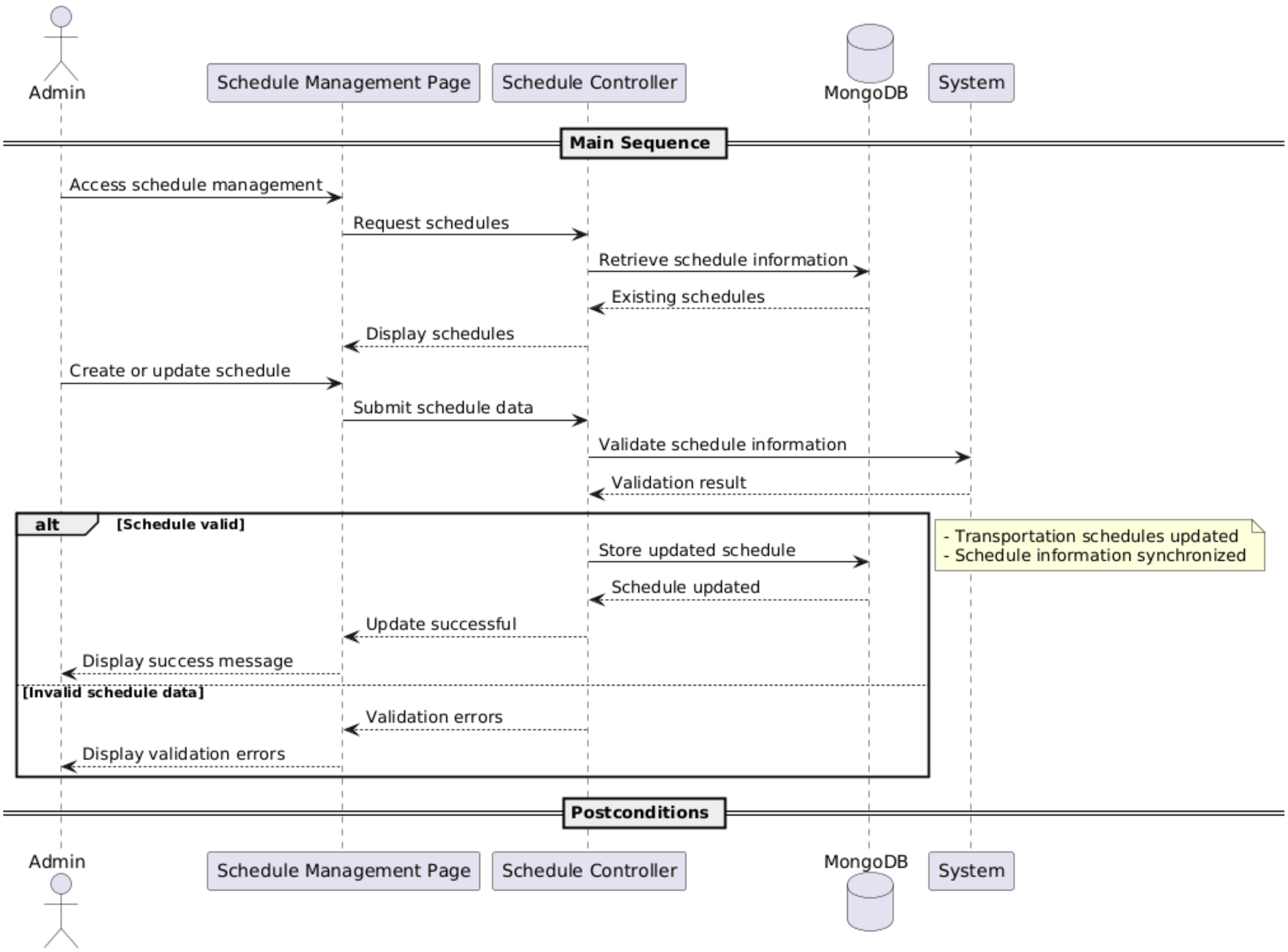
SeqD_04 — Download Ticket



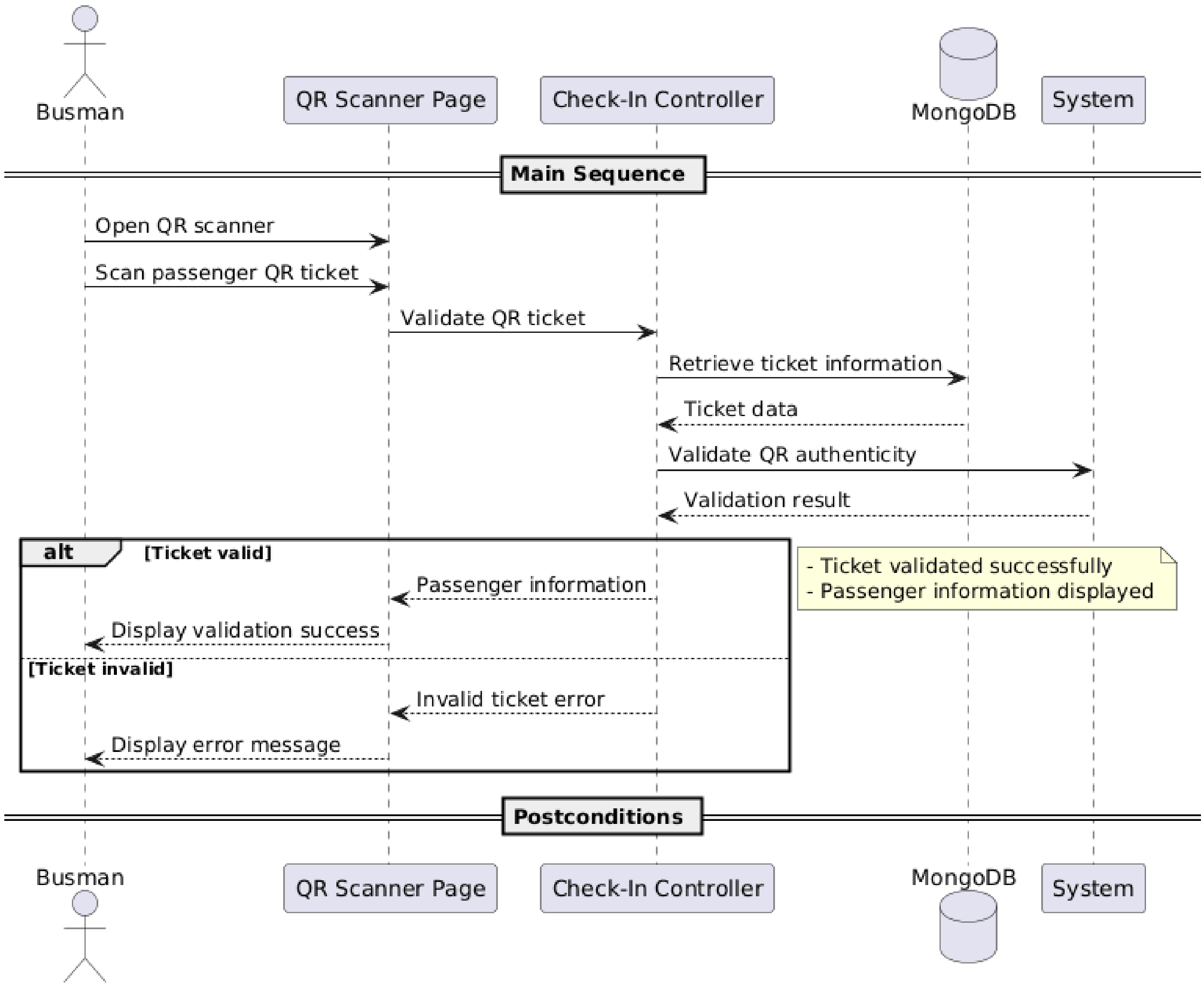
SeqD_05 — Manage Routes



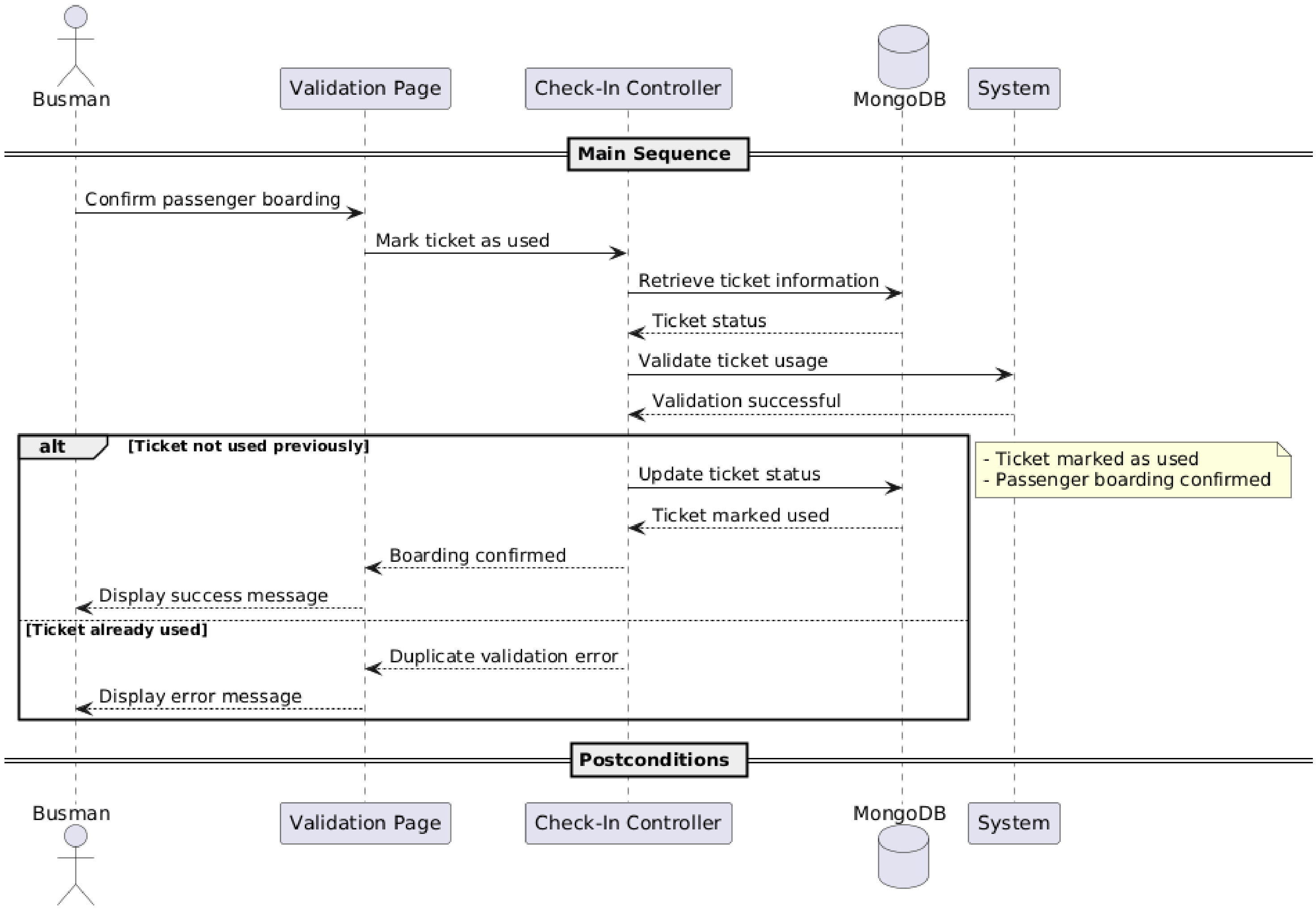
SeqD_06 — Manage Schedules

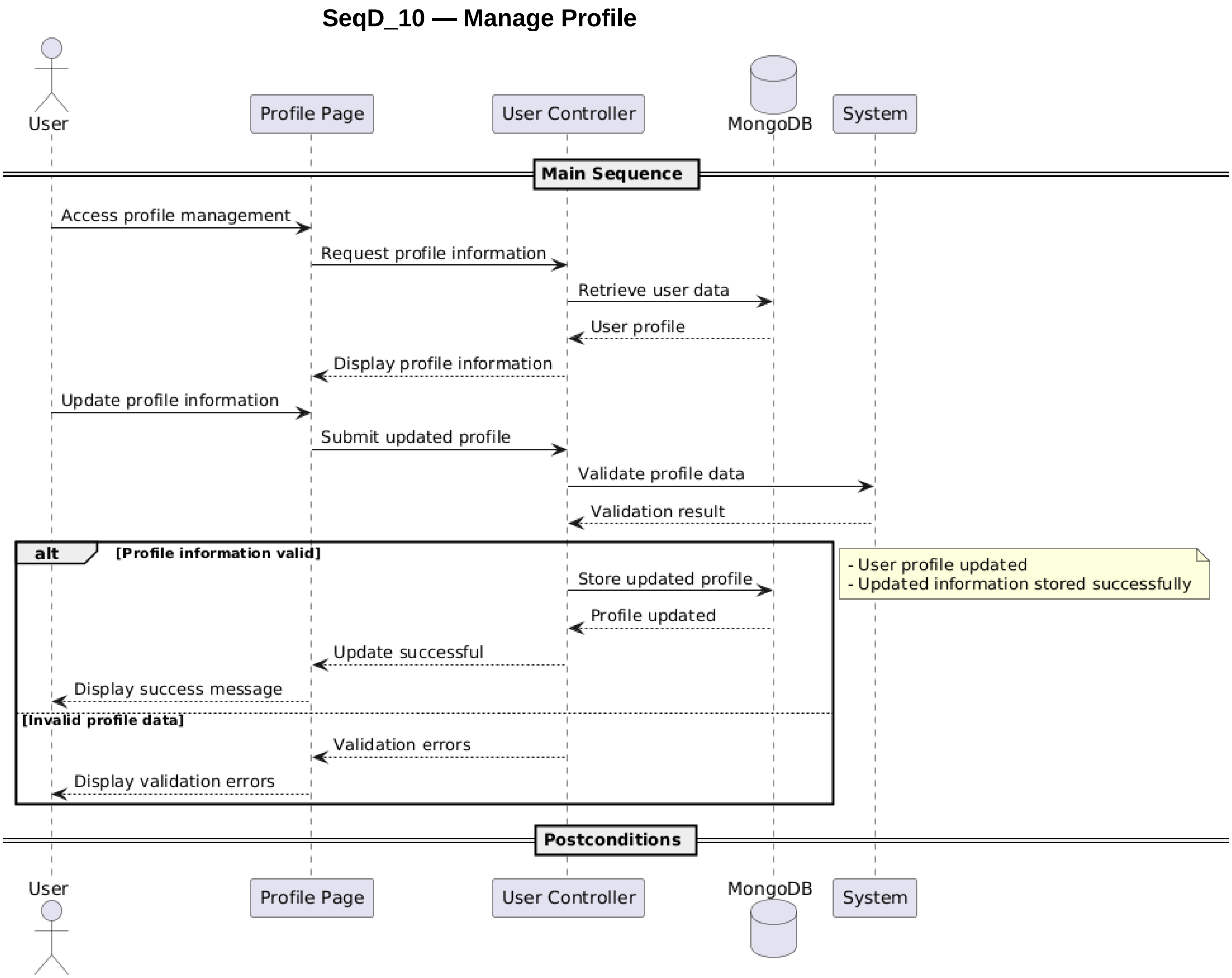


SeqD_07 — Scan Ticket

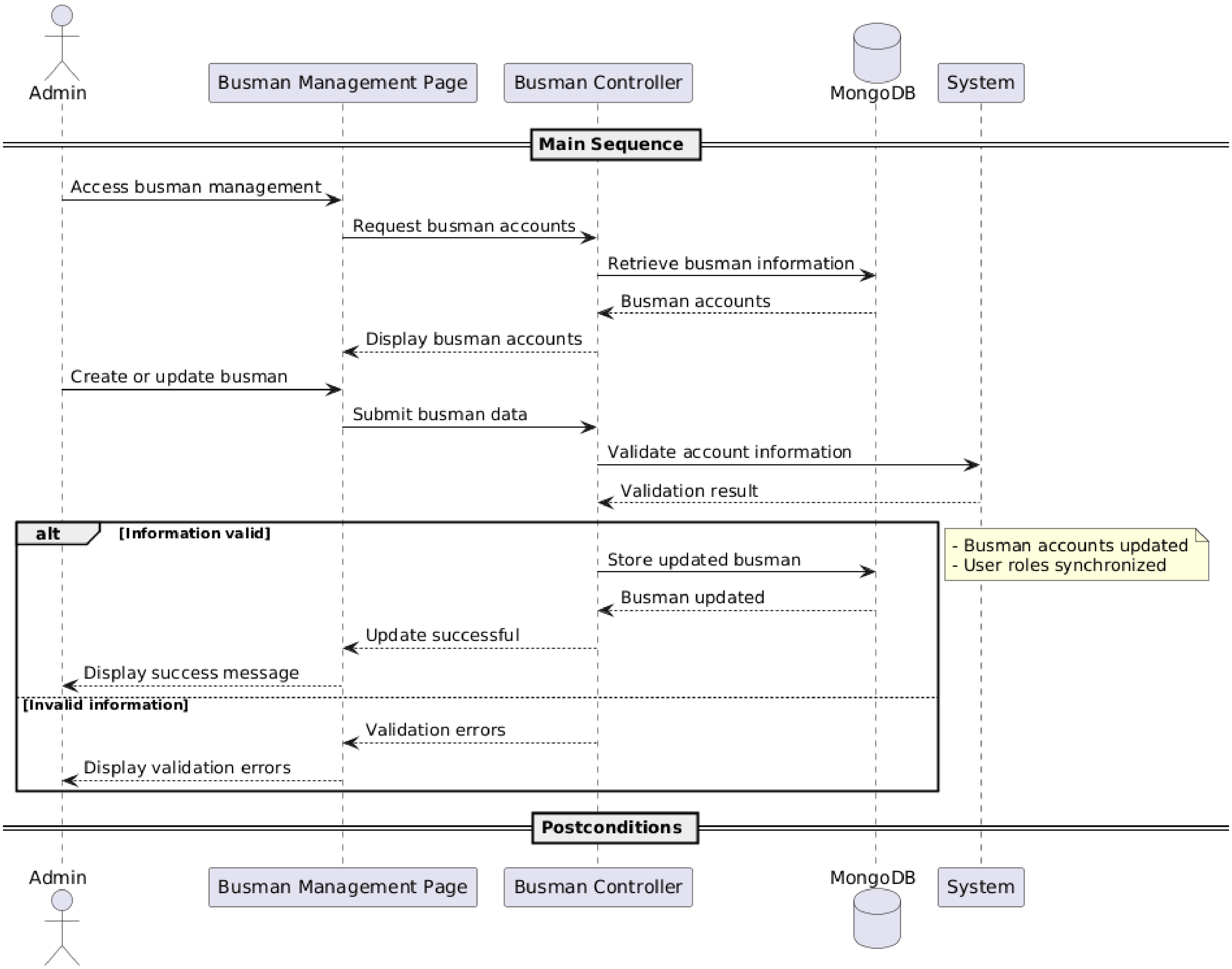


SeqD_10 — Mark Ticket as Used



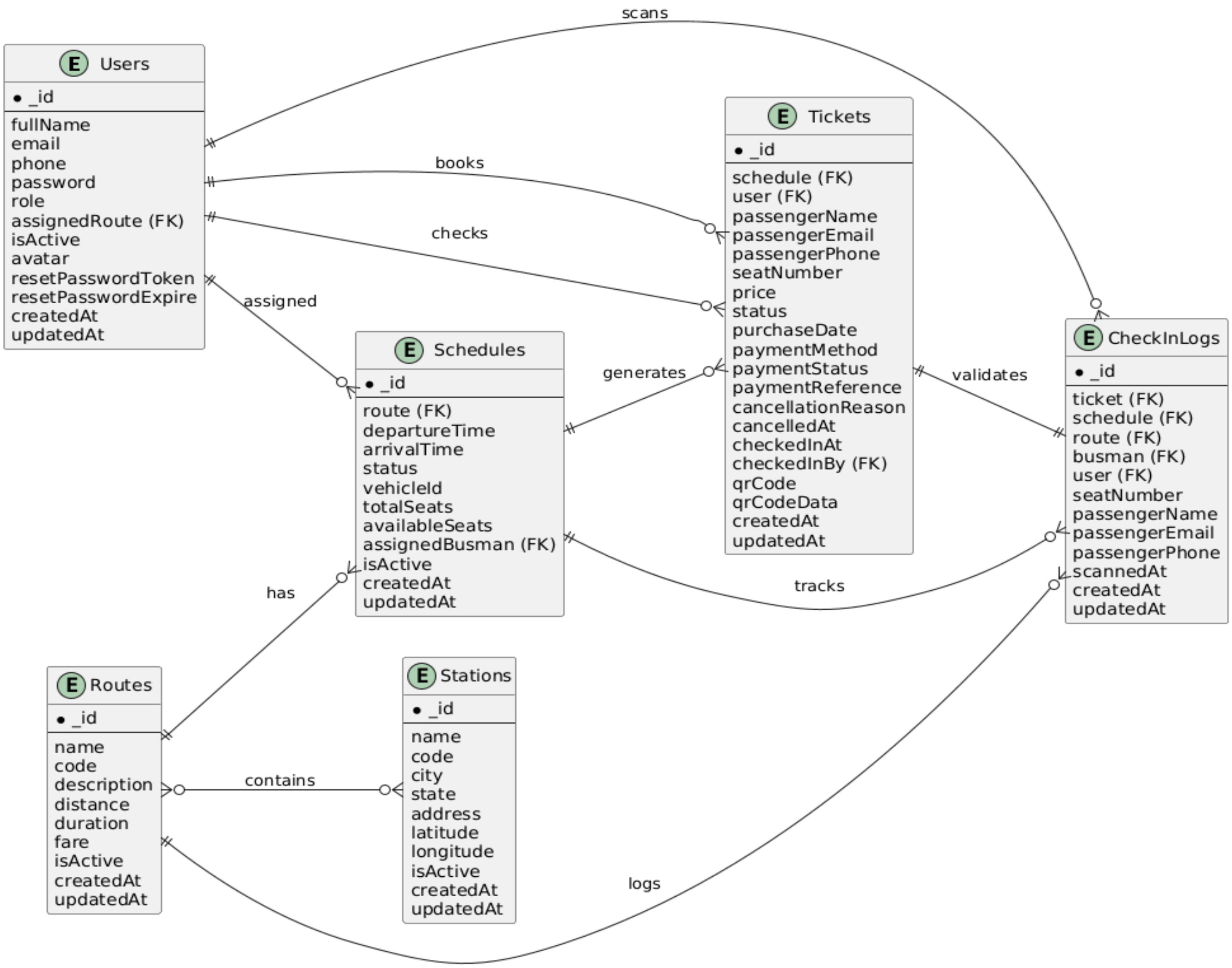

SeqD_10 — Manage Profile
User
Profile Page
User Controller
MongoDB
System
Main Sequence
Access profile management
Request profile information
Retrieve user data
User profile
Display profile information
Update profile information
Submit updated profile
Validate profile data
Validation result
alt
[Profile information valid]
Store updated profile
Profile updated
Update successful
Display success message
- User profile updated
- Updated information stored successfully
[Invalid profile data]
Validation errors
Display validation errors
Postconditions
User
Profile Page
User Controller
MongoDB
System

SeqD_11 — Manage Busman

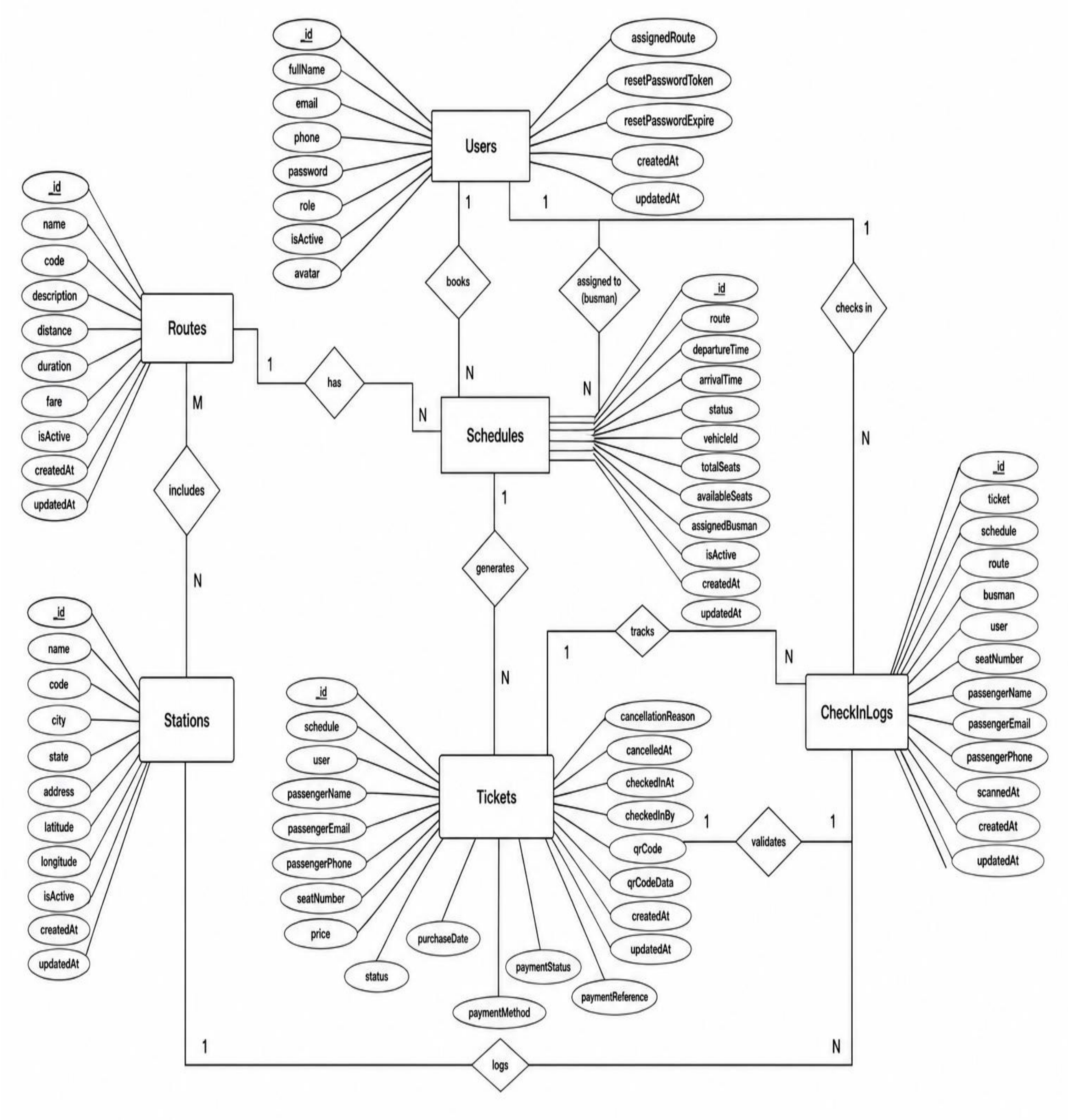


4.4 Entity Relation

4.4.1 Database Schema Design

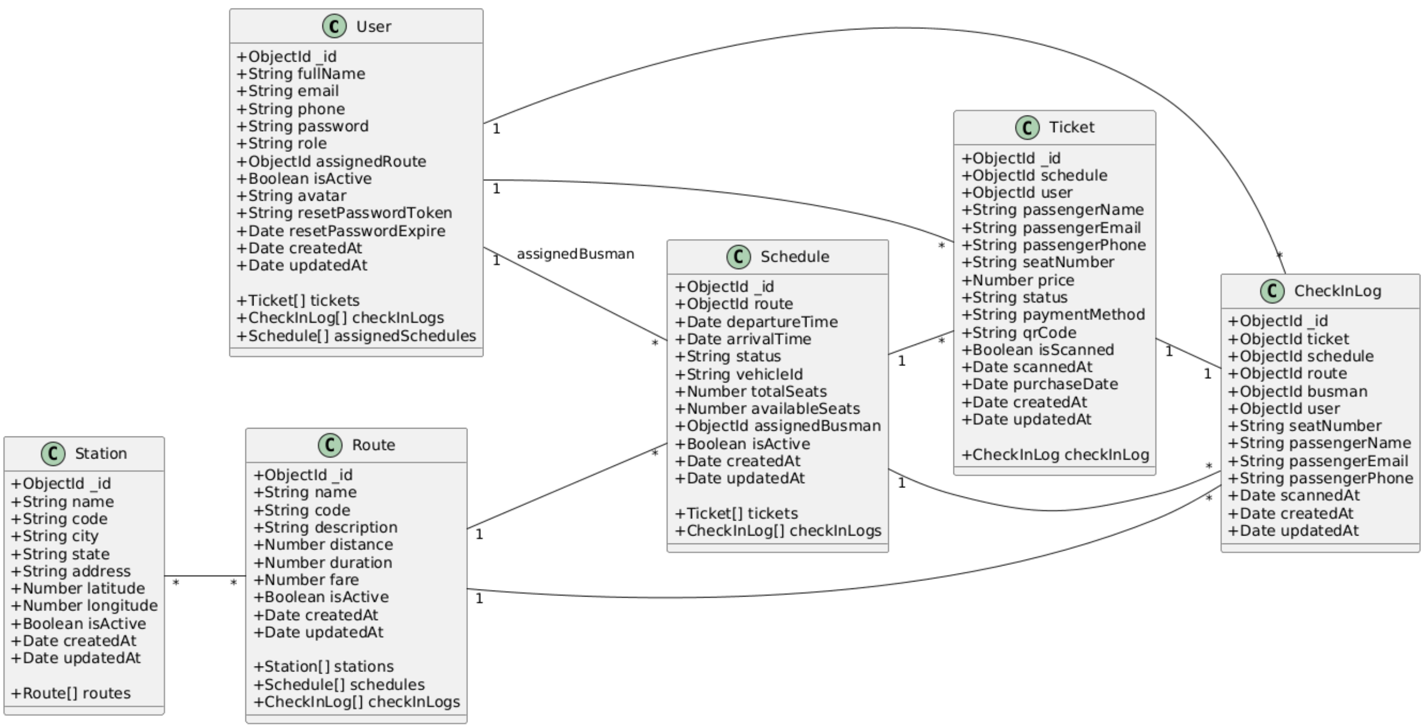


4.4.2 Entity Relation Diagram

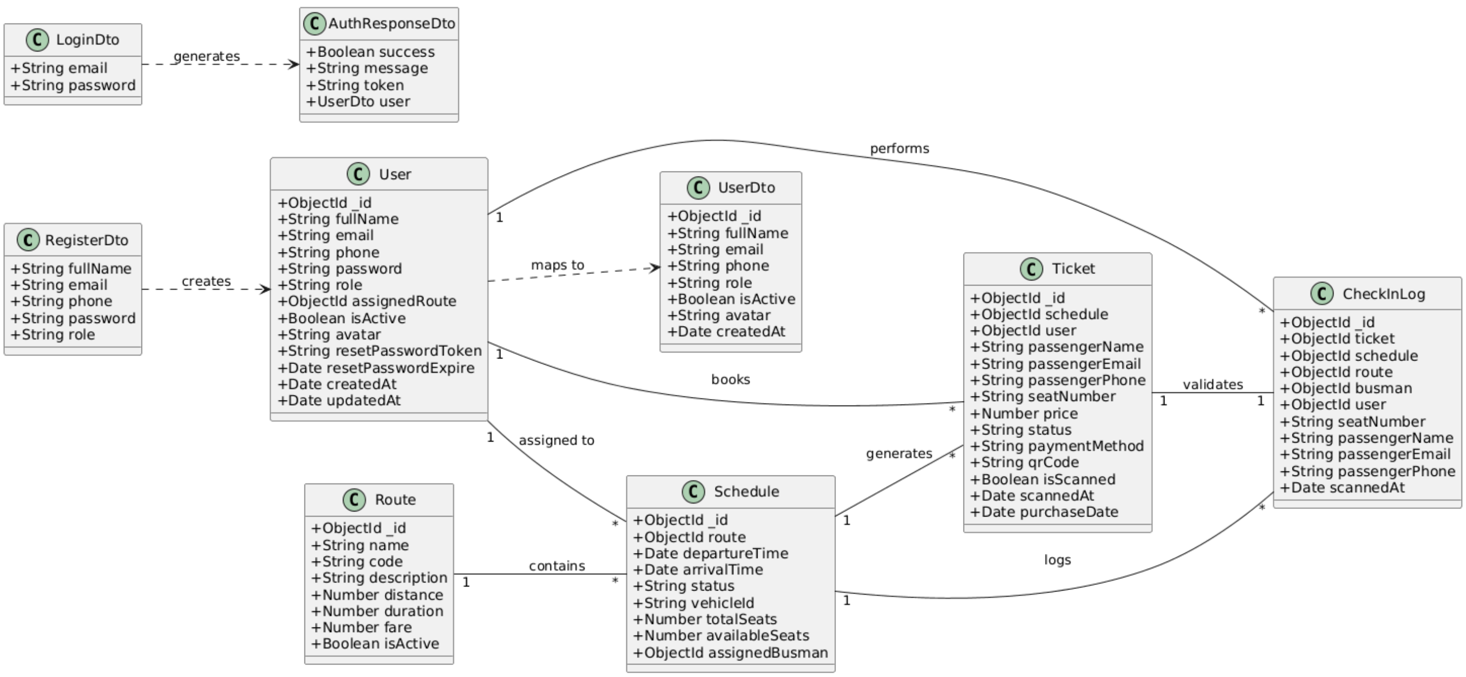


4.5 Structural Diagrams

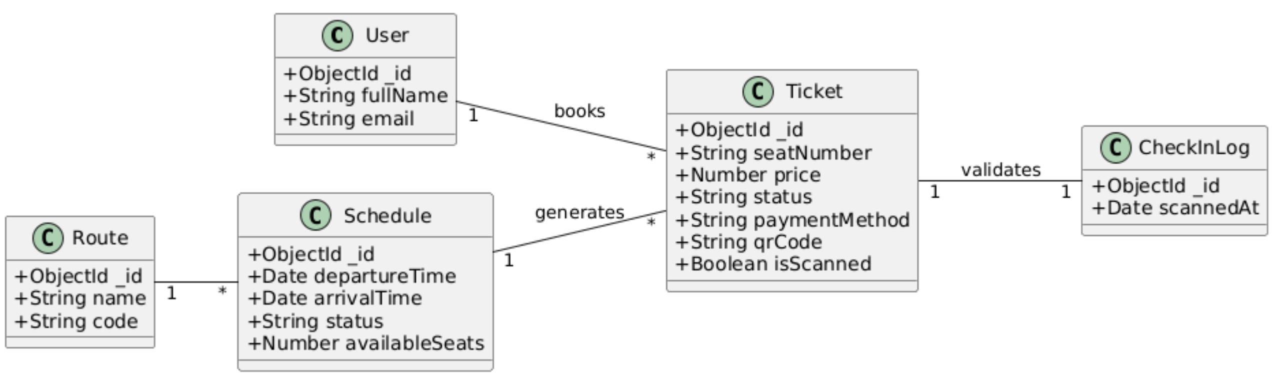
4.5.1 Class Diagram



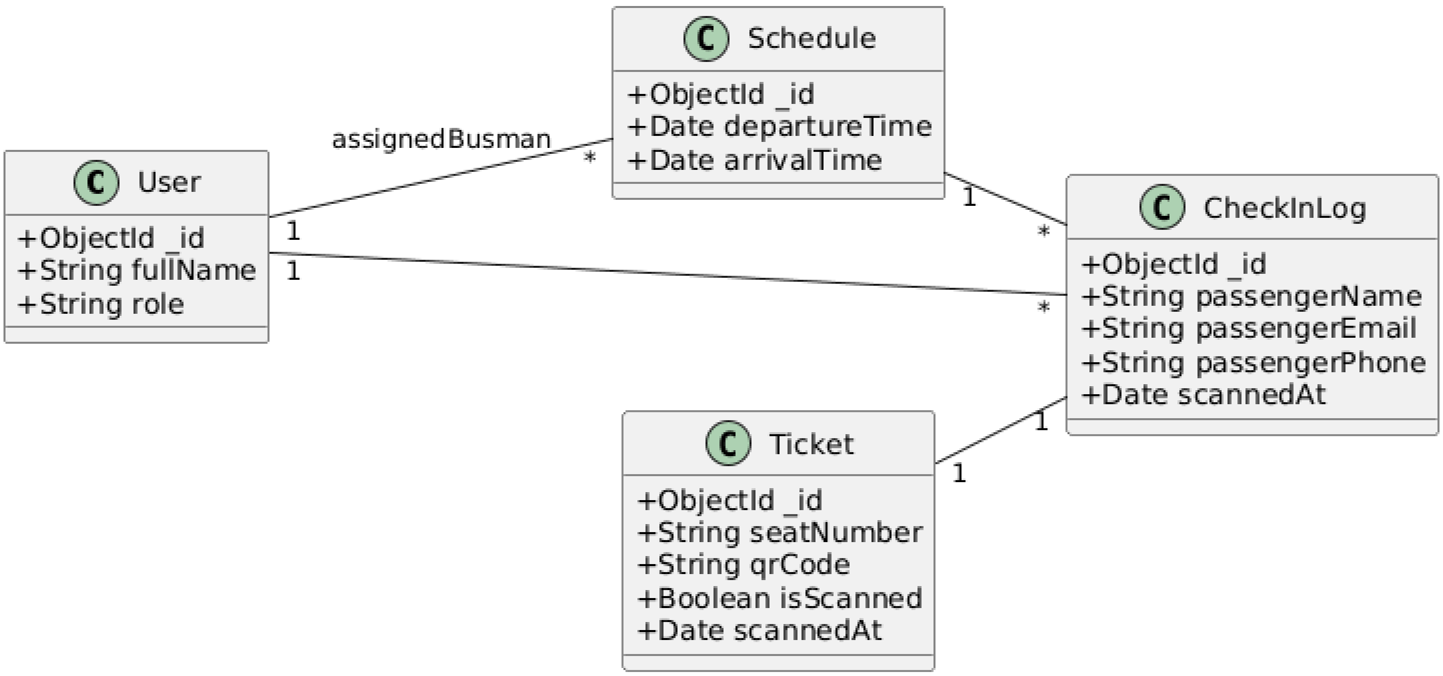
AUTHENTICATION CLASS DIAGRAM



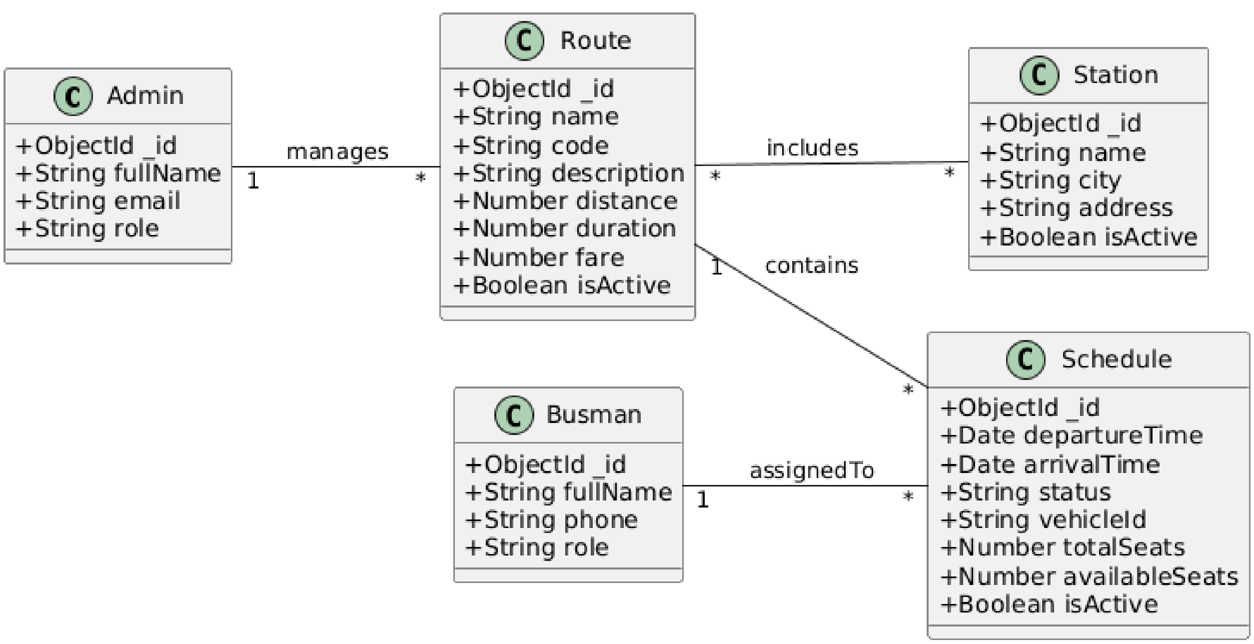
TICKET MANAGEMENT CLASS DIAGRAM



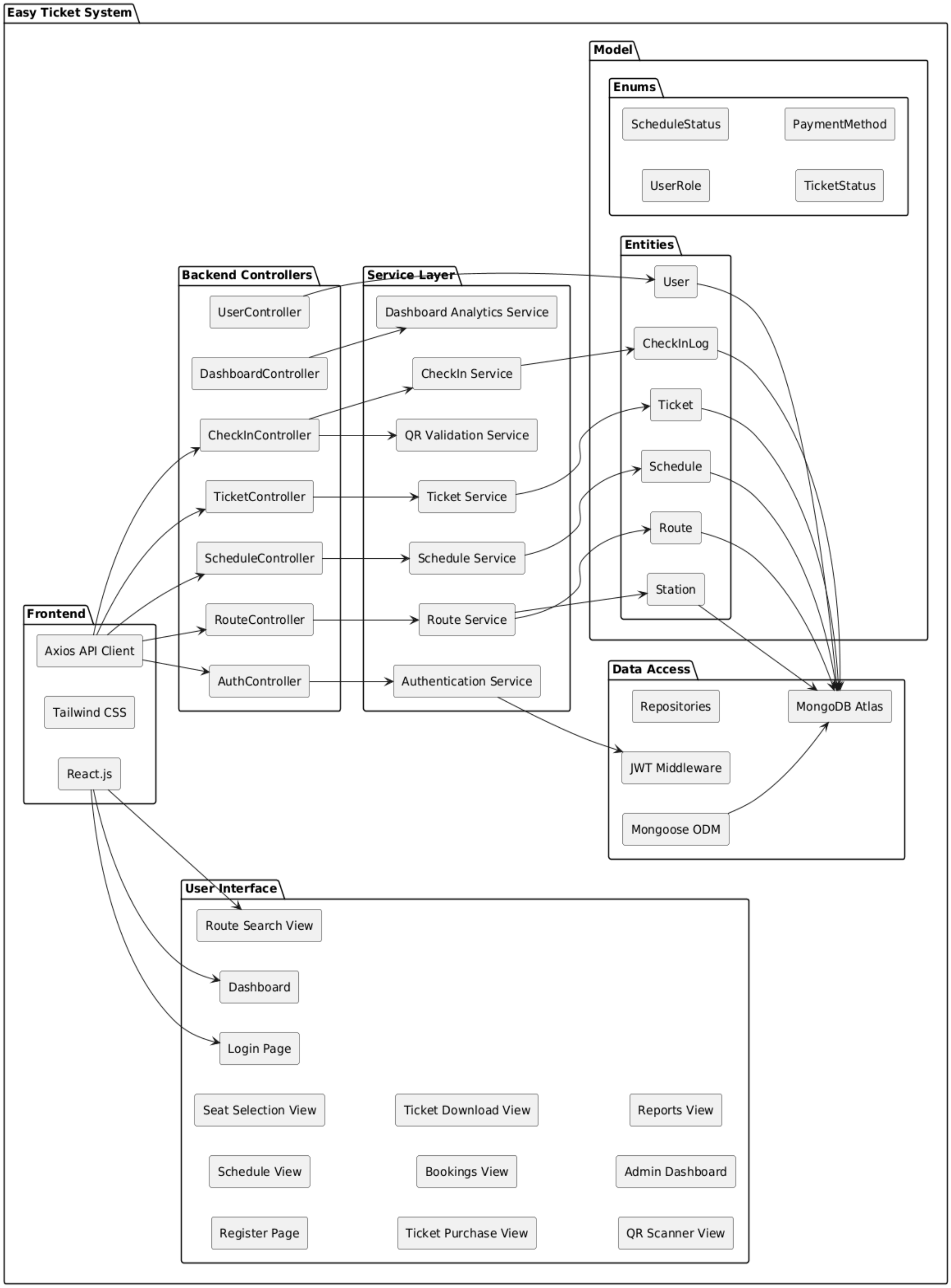
CHECK-IN VALIDATION CLASS DIAGRAM



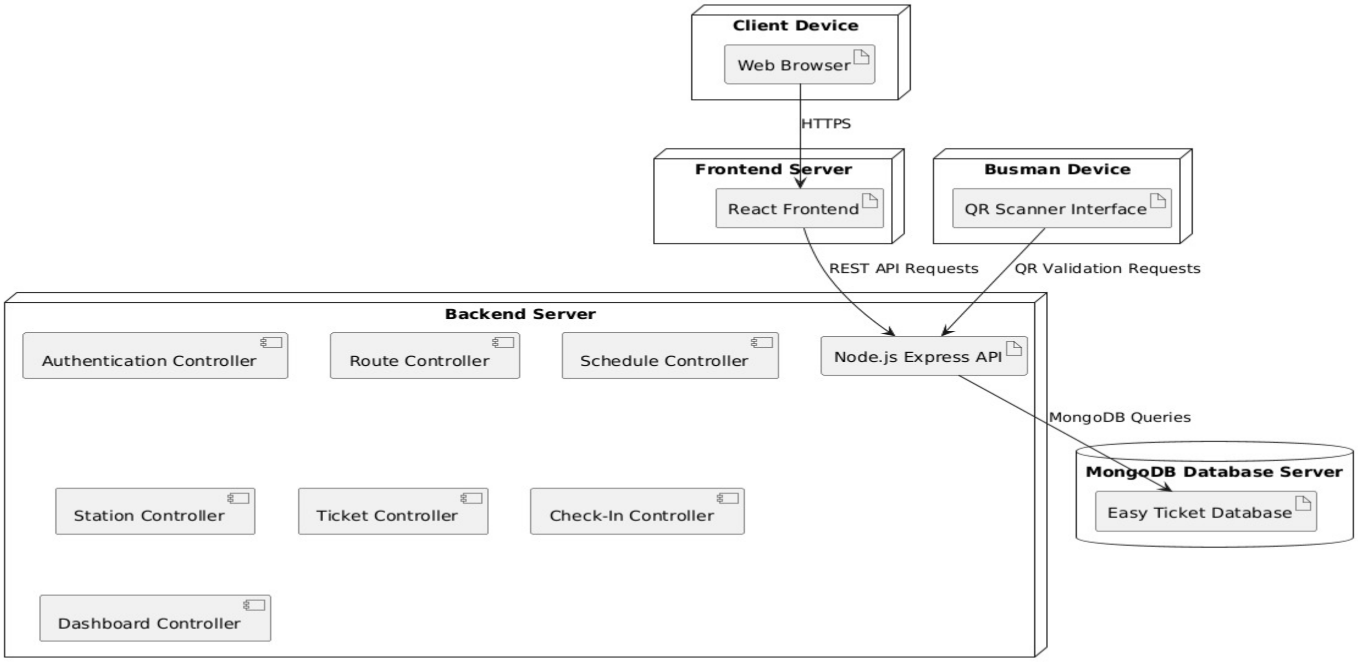
ROUTE & SCHEDULE MANAGEMENT CLASS DIAGRAM



4.5.3 Component Diagrams



4.5.4 Deployment Diagram



5. Implementation Technology

Implementation Technology

Easy Ticket System is a dynamic web-based transportation reservation and ticket management application.

The system was developed by combining modern client-side and server-side technologies to provide efficient communication between users and the application through the HTTPS protocol.

For the **Client Side**, the following technologies were used:

- JavaScript
- React.js
- HTML5
- CSS3
- Axios
- React Router DOM

The frontend application provides an interactive and responsive user interface for passengers, administrators, and busman users. Through the React-based architecture, the system ensures fast navigation, real-time updates, and efficient state management during route searching, reservation handling, and QR ticket validation processes.

For the **Server Side**, the system utilizes the following technologies:

- Node.js
- Express.js
- JSON Web Token (JWT) Authentication
- RESTful API Architecture
- Middleware-based Request Handling

The backend architecture follows a modular and scalable structure using controllers, models, middleware, and route management components. The application implements authentication and authorization mechanisms for secure role-based access control between passengers, administrators, and busman users.

The system also integrates QR-based ticket validation functionality, reservation management services, transportation schedule management, and dashboard analytics for system monitoring and operational reporting.

For the **Database**, the project uses:

- MongoDB
- Mongoose ODM

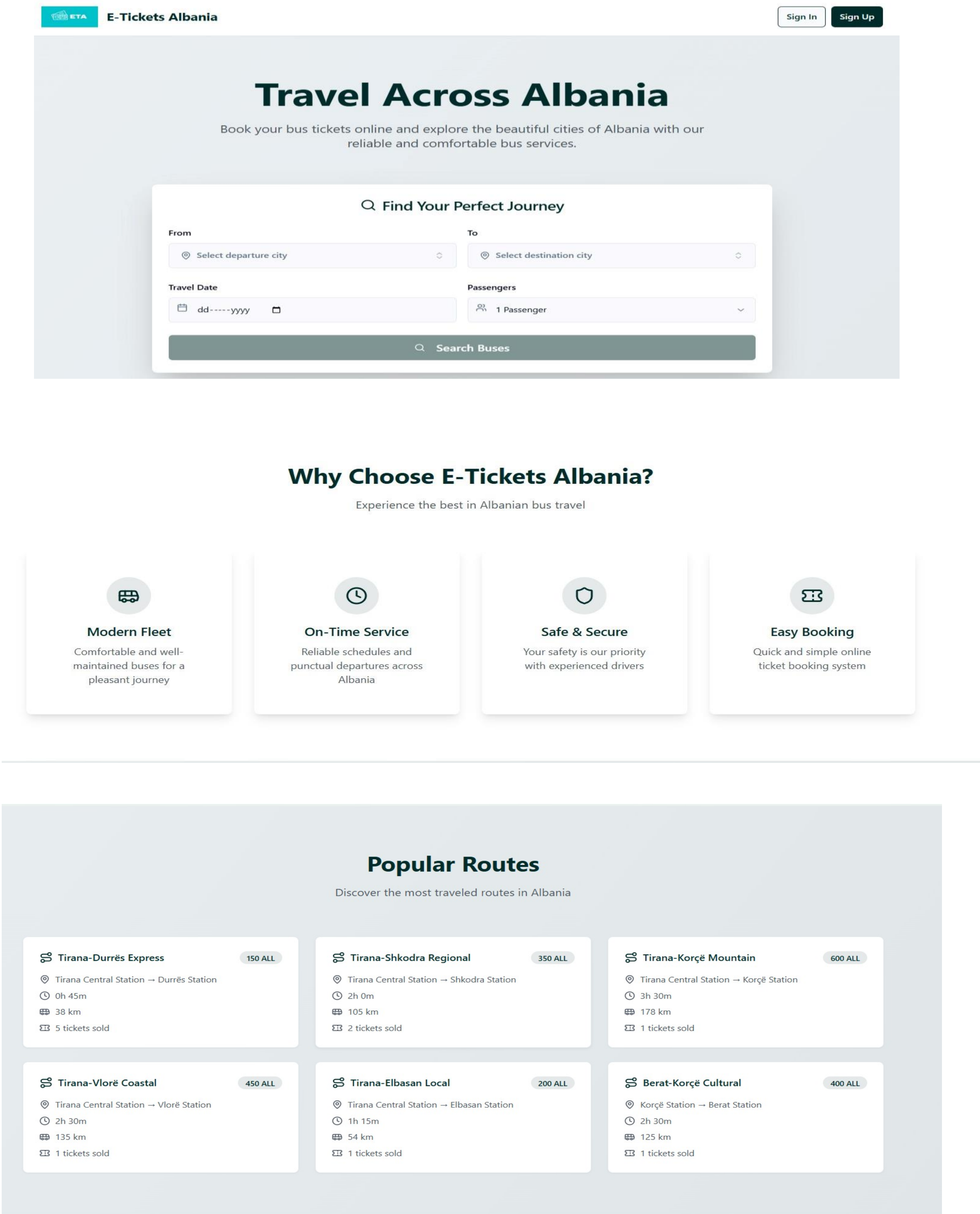
MongoDB is used as the primary NoSQL database for storing user information, transportation routes, schedules, reservations, tickets, and QR validation records. Mongoose is utilized to simplify schema modeling and database interactions within the Node.js environment.

The communication between the frontend, backend, and database layers is performed through REST API requests, ensuring efficient data exchange and system scalability.

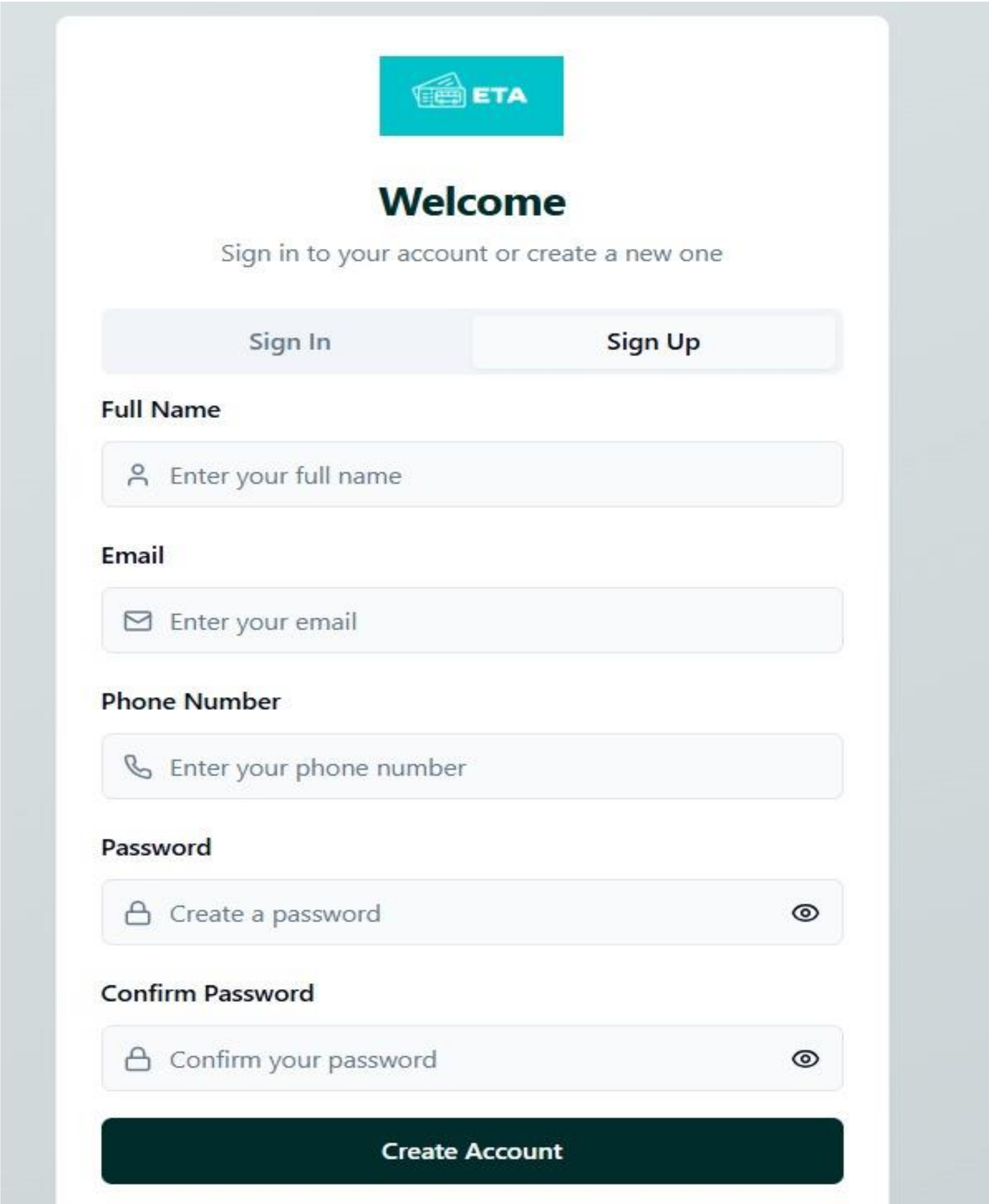
This project is also published on GitHub, where the complete development process of the system can be found, including source code, UML diagrams, software architecture documentation, implementation details, and project development progress.

Appendix - Detailed Designs

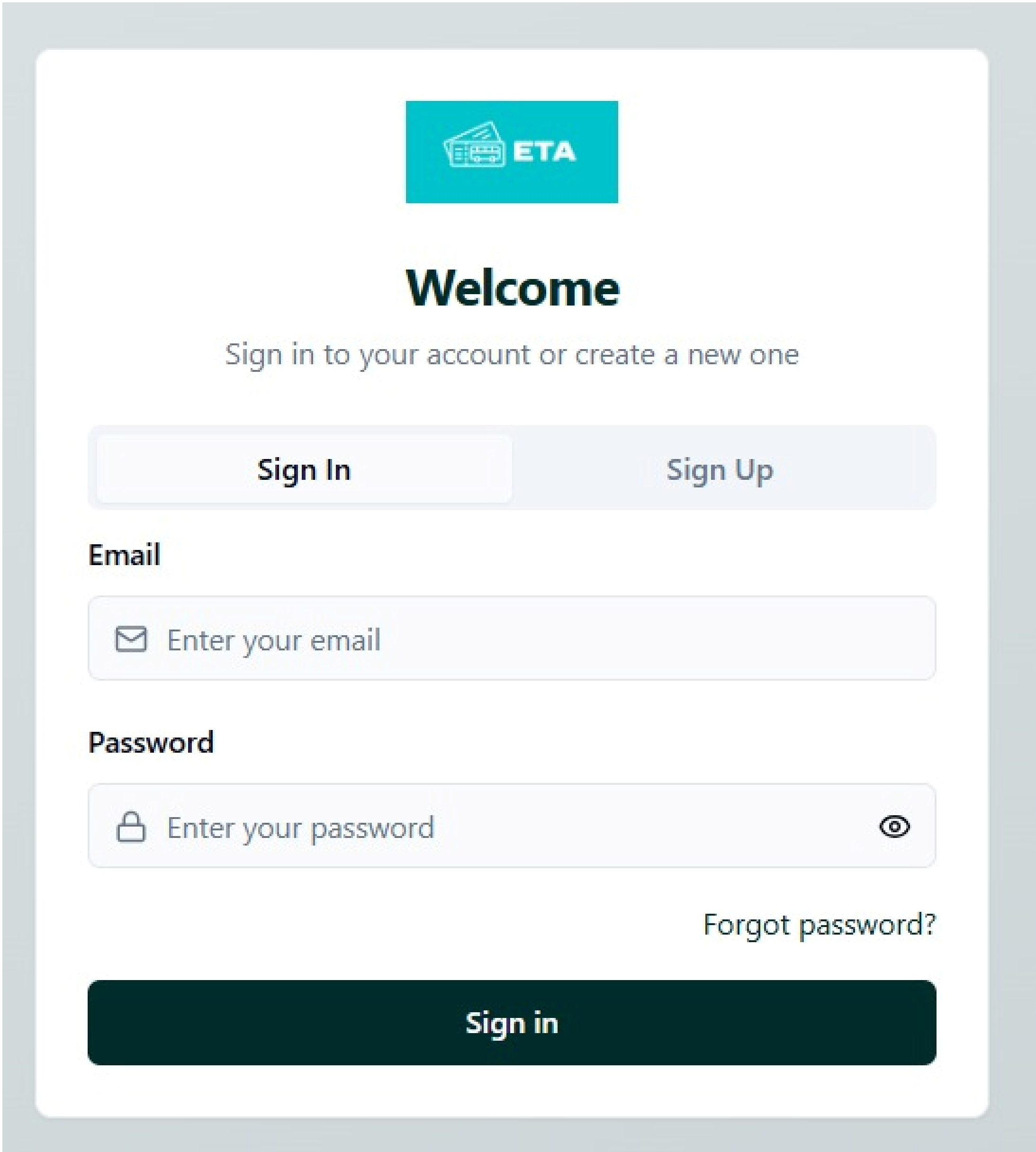
Home Page


ETA
E-Tickets Albania
Sign In
Sign Up
Travel Across Albania
Book your bus tickets online and explore the beautiful cities of Albania with our reliable and comfortable bus services.
Find Your Perfect Journey
From
Select departure city
To
Select destination city
Travel Date
dd-----yyyy
Passengers
1 Passenger
Search Buses
Why Choose E-Tickets Albania?
Experience the best in Albanian bus travel
Modern Fleet
Comfortable and well-maintained buses for a pleasant journey
On-Time Service
Reliable schedules and punctual departures across Albania
Safe & Secure
Your safety is our priority with experienced drivers
Easy Booking
Quick and simple online ticket booking system
Popular Routes
Discover the most traveled routes in Albania
Tirana-Durrës Express
150 ALL
Tirana Central Station → Durrës Station
0h 45m
38 km
5 tickets sold
Tirana-Shkodra Regional
350 ALL
Tirana Central Station → Shkodra Station
2h 0m
105 km
2 tickets sold
Tirana-Korçë Mountain
600 ALL
Tirana Central Station → Korçë Station
3h 30m
178 km
1 tickets sold
Tirana-Vlorë Coastal
450 ALL
Tirana Central Station → Vlorë Station
2h 30m
135 km
1 tickets sold
Tirana-Elbasan Local
200 ALL
Tirana Central Station → Elbasan Station
1h 15m
54 km
1 tickets sold
Berat-Korçë Cultural
400 ALL
Korçë Station → Berat Station
2h 30m
125 km
1 tickets sold

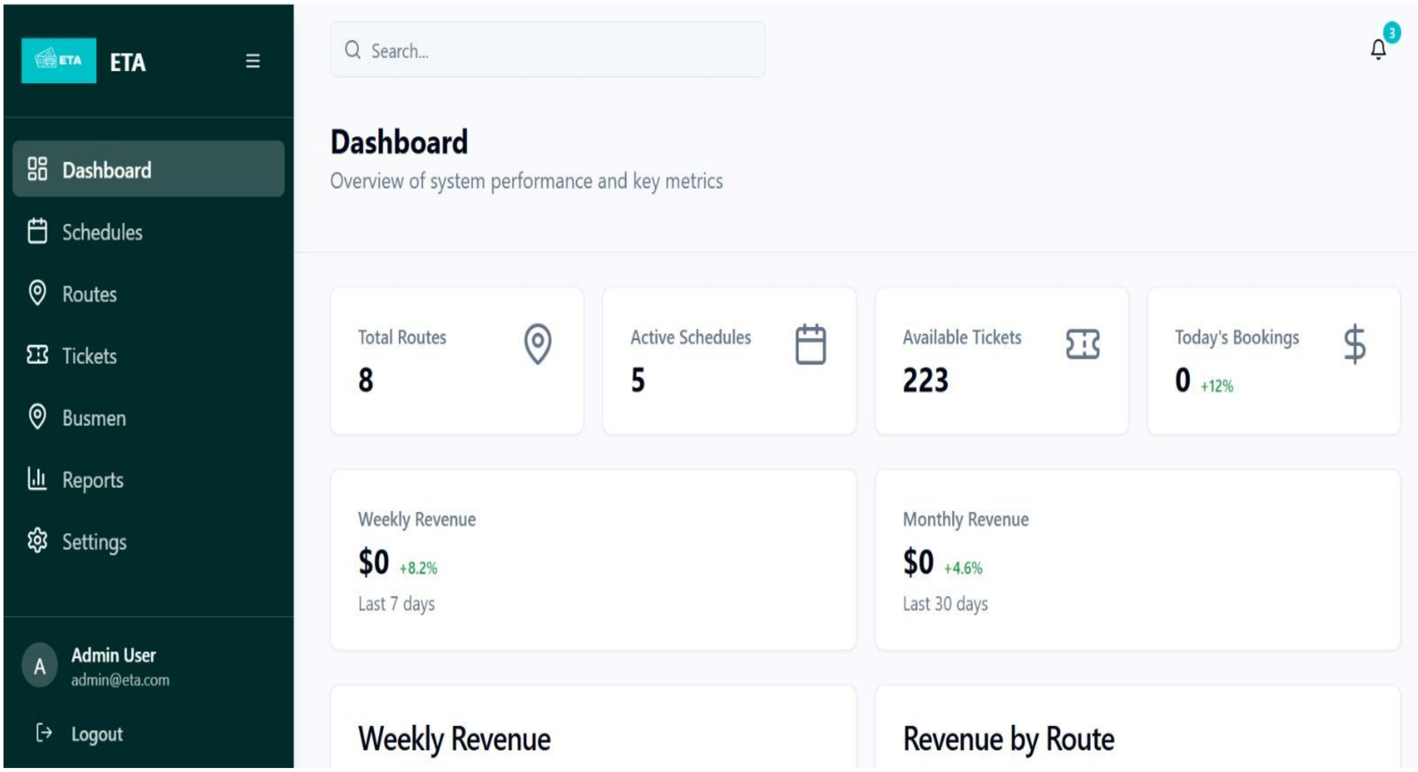
Register Form

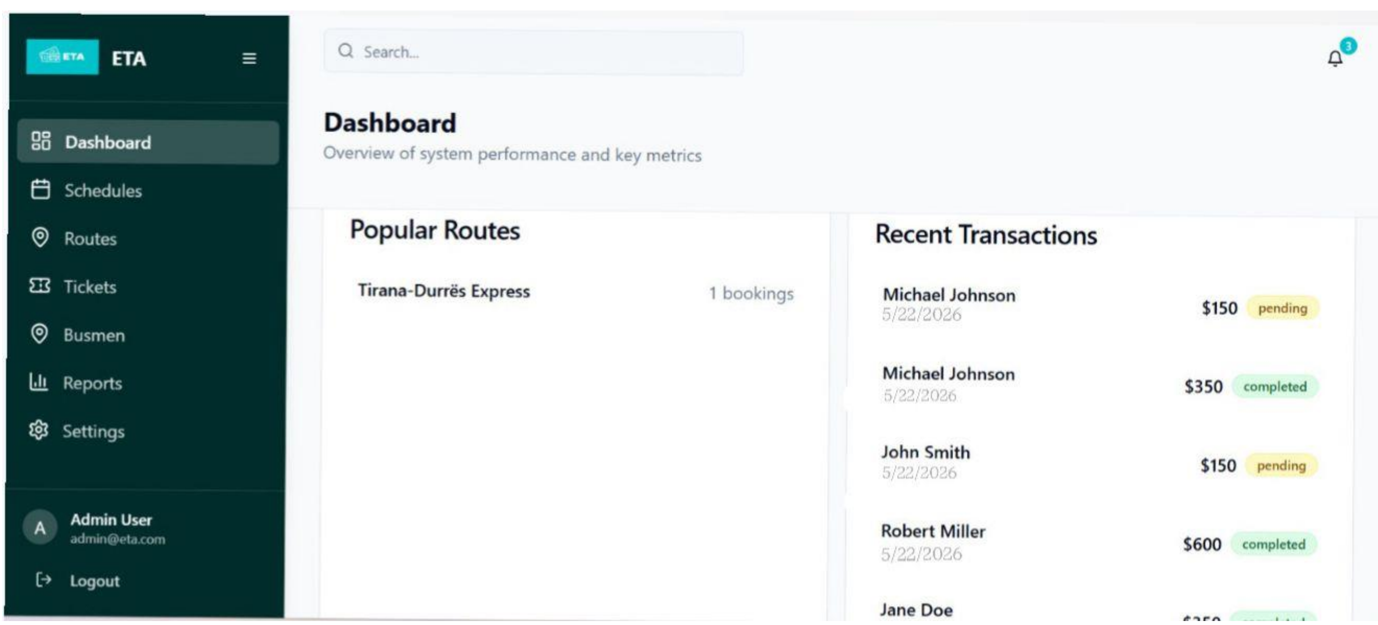


LogIn Form

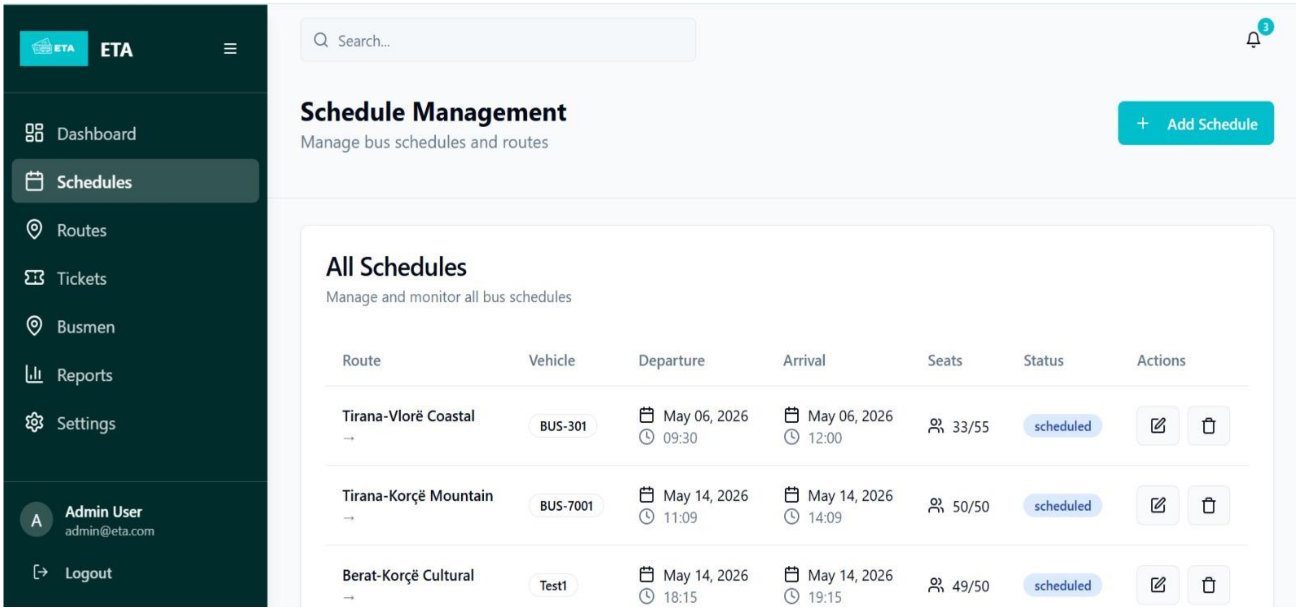

ETA
Welcome
Sign in to your account or create a new one
Sign In
Sign Up
Email
Enter your email
Password
Enter your password
Forgot password?
Sign in

Admin Dashboard

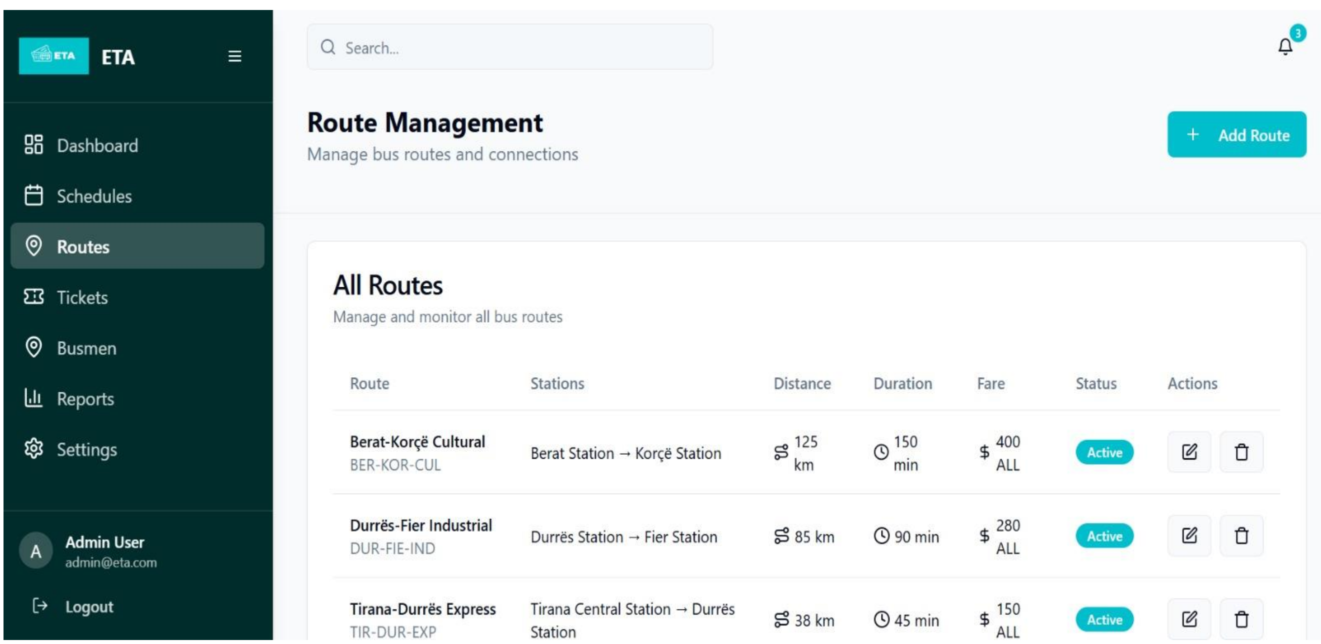




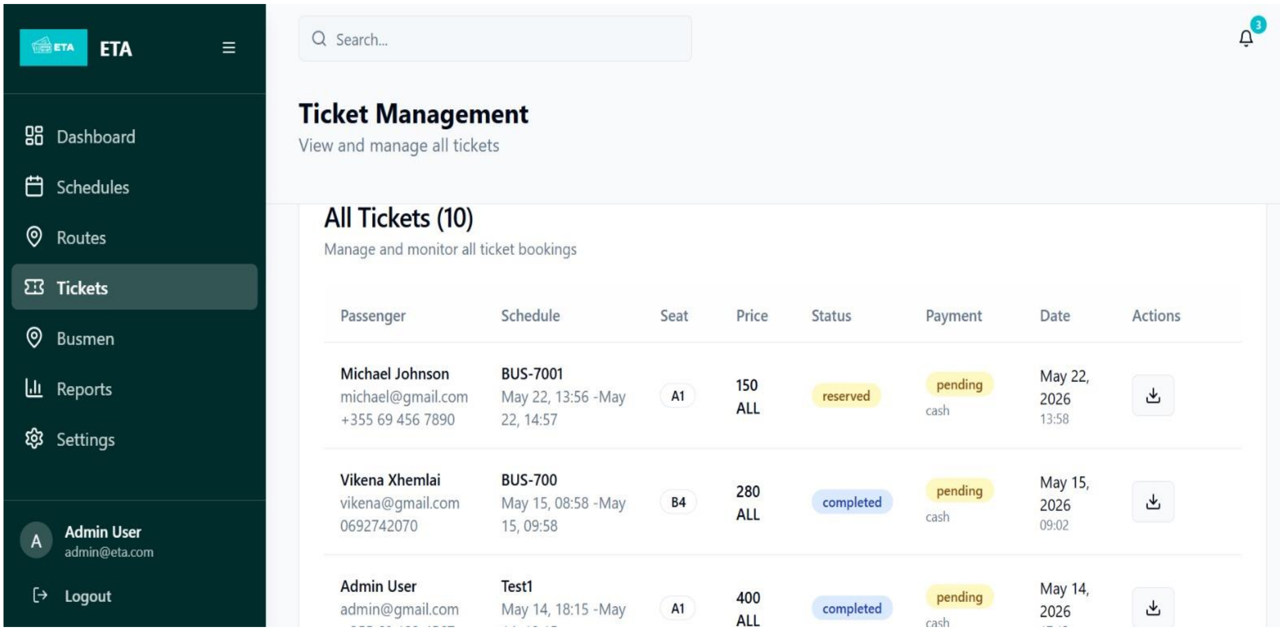
Schedule Management

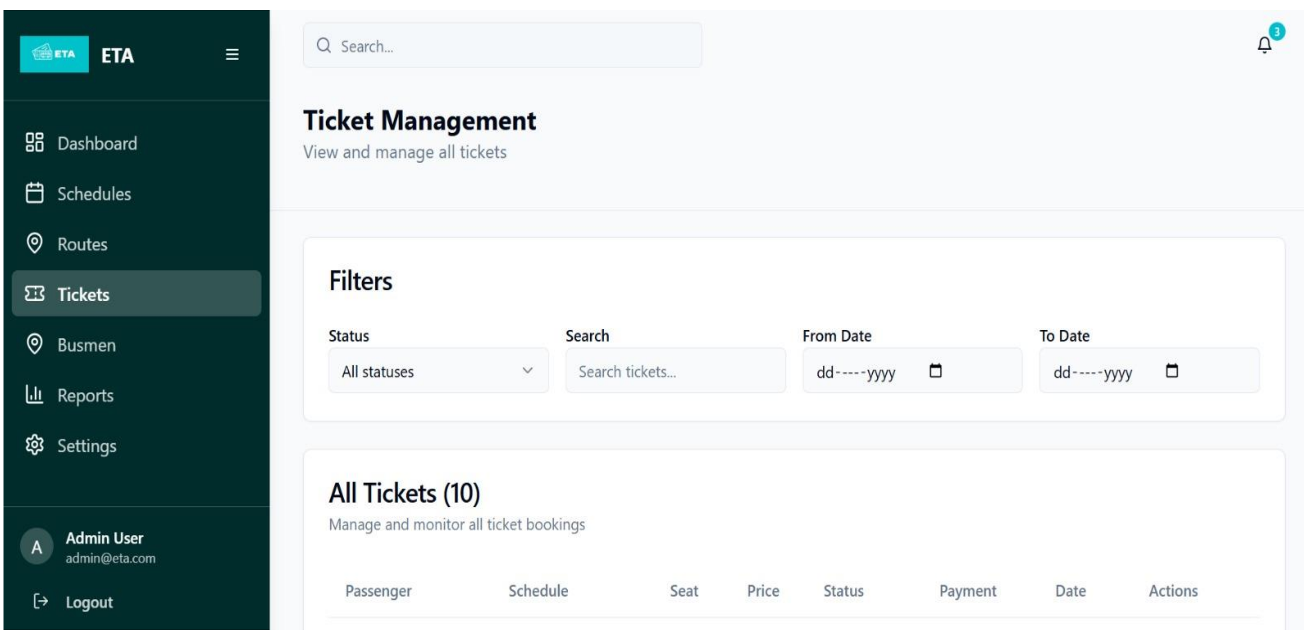


Route Management

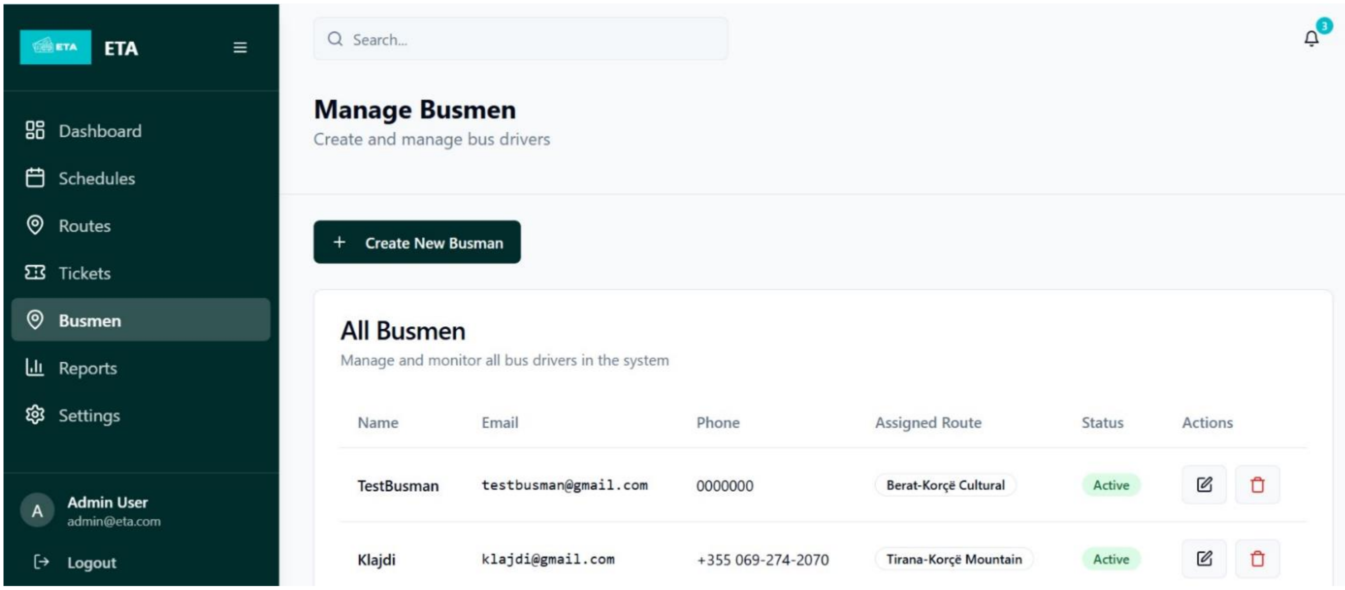


Ticket Management

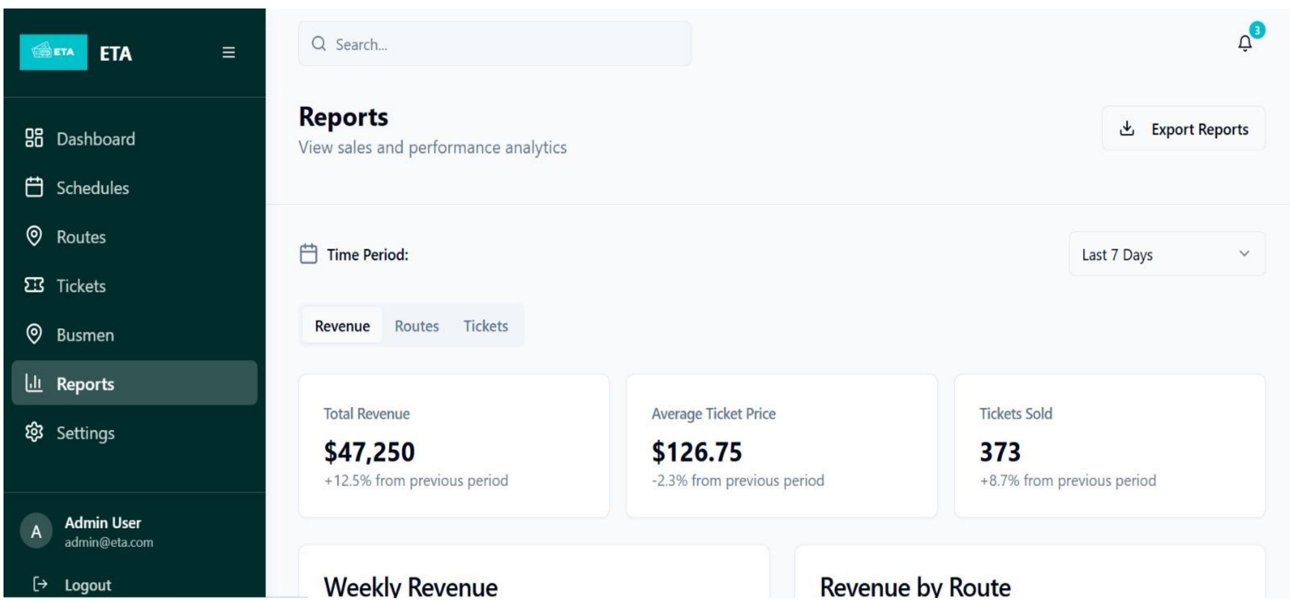




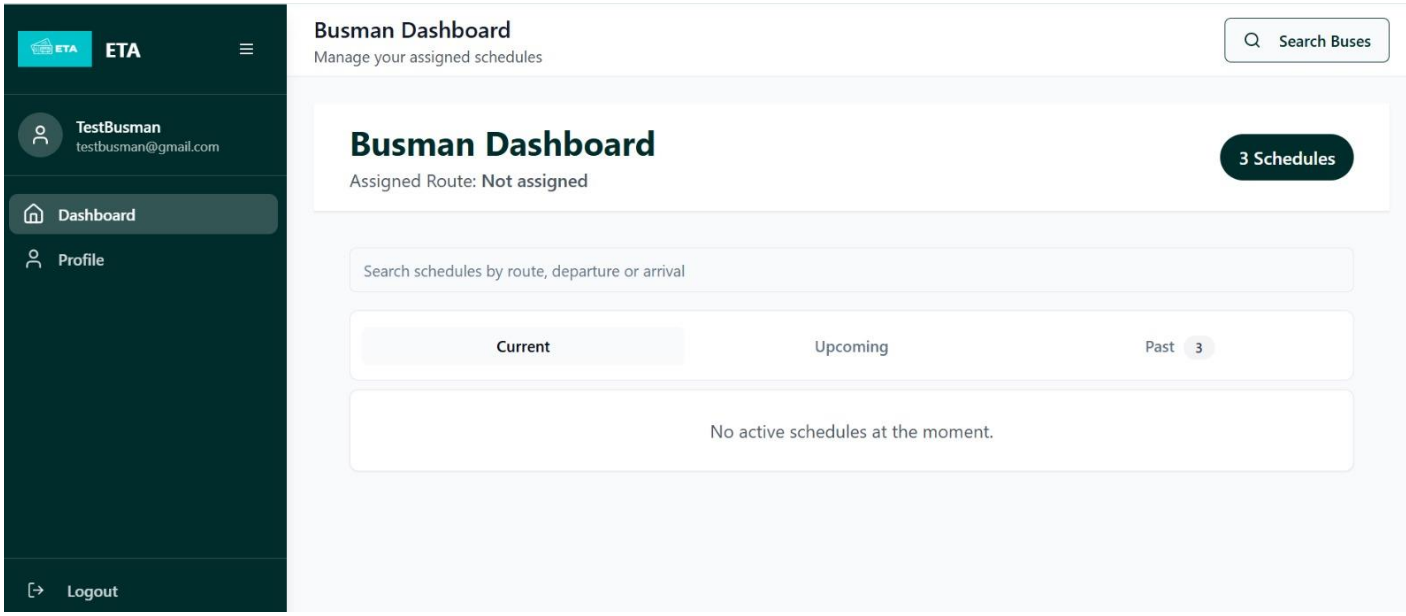
Manage Busman



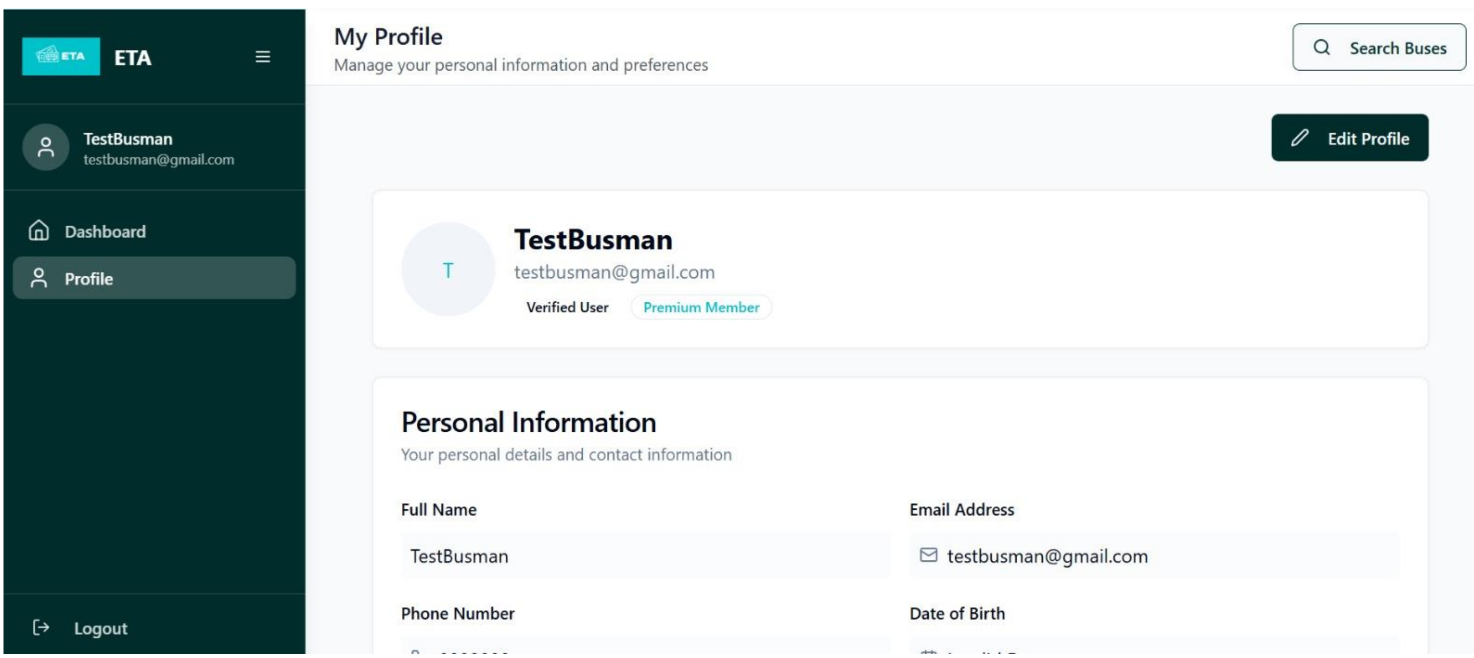
Reports



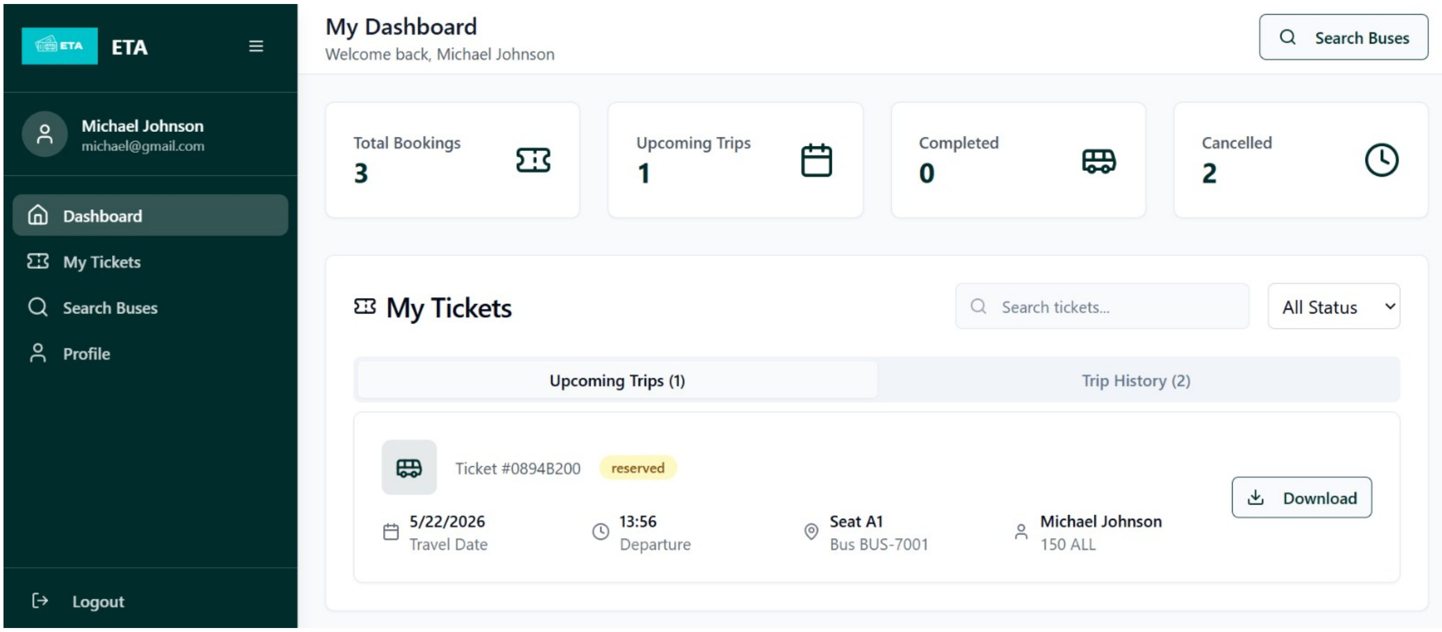
Busman Dashboard



Busman Profile



User Dashboard



Manage Profile